

TOWARD QUASICLASSICAL FUNDAMENTAL LOCAL EQUIVALENCE

GURBIR DHILLON AND IVAN LOSEV

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1. CATEGORIES AND FUNCTORS

1.1. Basics on vertex algebras. In this section we will introduce the notion of vertex algebras in a setting we need, and discuss basic examples and constructions. The base field is a characteristic 0 field denoted by $\mathbb{K}$.

1.1.1. Definitions and examples. We consider vertex algebras $V_{\hbar}$ over $\mathbb{K}[\hbar]$ such that $V_{\hbar}/(\hbar)$ is commutative that come equipped with the following structures:

- the PBW grading, $V_h = \bigoplus_{n \in \mathbb{Z}} V_h^n$, where $\hbar$ has degree 1 and V_h is graded with respect to all the products on V_h ;
- a compatible energy grading $V_h = \bigoplus_{i \in \mathbb{Z}} V_{h,i}$, where the n -th product has degree $n + 1$ and $\hbar$ has degree 0.
- a distinguished central element c of PBW degree 1 and energy degree 0 (the level, which is treated as an indeterminate).

The PBW and energy grading give rise to the actions of $\mathbb{G}_m$ (an element $t \in \mathbb{G}_m$ acts on the i -th graded component by t^i). We denote these tori by T_d (for the PBW grading; “d” is from “dilation”) and T_ℓ (for the energy grading; “ℓ” is from “loop rotation”). We write $T_{\ell,d}$ for $T_\ell \times T_d$.

By a homomorphism between two such vertex algebras we mean a $\mathbb{K}[\hbar]$ -linear $T_{\ell,d}$ -equivariant vertex algebra homomorphism (which is not required to send c to c).

Example 1.1.1. Let G be a simple algebraic group. We take the Kac-Moody algebra $\hat{\mathfrak{g}} = \mathfrak{g}((t)) \oplus \mathbb{K}c'$ associated to the standardly normalized (the square of every short coroot is 2) symmetric bilinear form κ_0 on $\mathfrak{g}$. Let $\mathfrak{h}^a = \mathfrak{h} \oplus \mathbb{K}c'$ be the affine Cartan. Consider the element $c := c' + h^\vee \hbar$, where $h^\vee$ is the dual Coxeter number (so that specializing $c := 0, \hbar = 1$ gives the critical level). For T_ℓ we take the loop rotation torus acting on $\hat{\mathfrak{g}}$ by rescaling t . Let T_d be the torus acting on $\hat{\mathfrak{g}}$ with weight 1.

The universal enveloping algebra $U(\hat{\mathfrak{g}})$ is equipped with the PBW filtration, so we can consider the Rees algebra $R_\hbar U(\hat{\mathfrak{g}})$ (its grading will be called the PBW grading: both $\hat{\mathfrak{g}}$ and $\hbar$ are in degree 1).

The *affine vertex algebra* $V_{c,\hbar}(\mathfrak{g})$ is defined as the quotient of $R_\hbar U(\hat{\mathfrak{g}})$ by the left ideal generated by the elements of the form $xf(t)$ with $x \in \mathfrak{g}$ and $f(t) \in \mathbb{K}[[t]]$.

We note that the construction above generalizes to any connected reductive algebraic group.

Example 1.1.2. Let G be a connected reductive algebraic group. To this group we assign its algebra of chiral differential operators $\text{CDO}_{c,\hbar}(G)$. As an $R_\hbar U(\hat{\mathfrak{g}})$ -module it is constructed as follows. Let $\mathcal{J}G$ denote the jet group of G (so that the algebra of regular functions $\mathbb{K}[\mathcal{J}G]$ is a commutative vertex algebra). Then $\mathbb{K}[\mathcal{J}G, \hbar]$ becomes a module over $R_\hbar U(\mathfrak{g}[[t]])$, where $xf(t) \in \mathfrak{g}[[t]]$ acts by $\hbar L_{xf(t)}$, where $L_{xf(t)}$ is the Lie derivative coming from the left translation action of $\mathcal{J}G$ on itself. Then $\text{CDO}_{c,\hbar}(G)$ is the induction of $\mathbb{K}[\mathcal{J}G, \hbar]$ from $R_\hbar U(\mathfrak{g}[[t]])$ to $R_\hbar U(\hat{\mathfrak{g}})$. It is equipped with the actions of $T_{\ell,d}$, where the action on $R_\hbar U(\hat{\mathfrak{g}})$ is as before, and the action on $\mathbb{K}[\mathcal{J}G]$ is as follows: T_d acts trivially, and T_ℓ acts via the grading by the total degree of derivations – if one fixes a closed embedding of G into $\mathbb{A}^N$ with coordinates $x^1, \dots, x^N$, then $\mathbb{K}[\mathcal{J}G]$ is generated by the elements x_j^i with $j \leq -1$, and we set $\deg x_j^i = j + 1$.

Note that $\mathbb{K}[\hbar] \subset \mathbb{K}[\mathcal{J}G, \hbar]$ as the submodule of $\mathfrak{g}[[t]]$ -invariants, so we get a natural map $V_{c,\hbar}(\mathfrak{g}) \hookrightarrow \text{CDO}_{c,\hbar}(G)$, which is $\mathbb{K}[c, \hbar]$ -linear. This map is a homomorphism as above.

1.1.2. Weak and strong equivariance. In this section we consider actions of jet groups on vertex algebras. A reference is [12, Section 3.5].

We note that $\mathbb{K}[\mathcal{J}G]$ is a Hopf algebra object in the category of vertex algebras (over $\mathbb{K}$). A weak action of $\mathcal{J}G$ on V_h is therefore defined as a $T_{\ell,d}$ -equivariant and $\mathbb{K}[c, \hbar]$ -linear vertex algebra homomorphism $V_h \rightarrow V_h \otimes \mathbb{K}[\mathcal{J}G]$ satisfying the usual axioms of a

co-action. A definition in terms of the action (rather than the co-action) is given in [12, Definition 3.5.2].

For example, the construction of $V_{c,\hbar}(\mathfrak{g})$ as a quotient of $R_\hbar U(\hat{\mathfrak{g}})$ by a left ideal stable under the $\mathcal{JG}$ -action equips $V_{c,\hbar}(\mathfrak{g})$ with a weak action of $\mathcal{JG}$ in the sense of the definition above.

Now we proceed to the definition of a strong action. Notice that a homomorphism $\varphi : V_{c,\hbar} \rightarrow V_\hbar$ makes $V_\hbar$ into $\mathfrak{g}[[t]]$ -module, where xt^n acts by $\hbar^{-1}\varphi(xt^{-1}|\emptyset\rangle)_{(n)}$. If $V_\hbar$ is equipped with a weak action of $\mathcal{JG}$, then we say that φ is a *quantum comoment map* if it is $\mathcal{JG}$ -equivariant and the two actions of $\mathfrak{g}[[t]]$ on $V_\hbar$ coincide. By a *strong action* of $\mathcal{JG}$ on $V_\hbar$ we mean a weak action together with a quantum comoment map. For example, the action of $\mathcal{JG}$ on $V_{c,\hbar}(\mathfrak{g})$ is strong, where the quantum comoment map is the identity.

Example 1.1.3. We note that $\text{CDO}_{c,\hbar}(G)$ comes with two commuting $\mathcal{JG}$ -actions, induced by the commuting left and right translation actions of $\mathcal{JG}$ on itself (we still refer to the induced actions as the left and right translation action). Both actions are weak actions in the sense above. The inclusion $V_{c,\hbar}(\mathfrak{g}) \hookrightarrow \text{CDO}_{c,\hbar}(G)$ mentioned in Example 1.1.2 is a quantum comoment map for the left translation action. According to [5], the subalgebra of invariants for the left translation action is isomorphic to $V_{c,\hbar}(\mathfrak{g})$ but the isomorphism is not $\mathbb{K}[c, \hbar]$ -linear, rather it is semi-linear with respect to the automorphism sending $\hbar$ to $\hbar$ and c to $-c$. The identification of $V_{c,\hbar}$ with this subalgebra is the quantum comoment map for the right translation action. The corresponding $\mathbb{K}[c, \hbar]$ -semilinear inclusion $V_{c,\hbar}(\mathfrak{g}) \hookrightarrow \text{CDO}_{c,\hbar}(G)$ is a quantum comoment map for the action of $\mathcal{JG}$ by right translations.

1.1.3. Universal enveloping algebra and 0-mode subquotient. To a vertex algebra $V_\hbar$ as in Section 1.1.1, we assign its universal enveloping algebra $\tilde{U}(V_\hbar)$, cf. [15, Section 4.3]. This is a complete topological $\mathbb{K}[c, \hbar]$ -algebra equipped with a pro-rational action of $T_{\ell,d}$. More precisely, there is a sequence of $T_{\ell,d}$ -stable left ideals $I_0 \supset I_1 \supset \dots$ such that $\tilde{U}(V_\hbar) = \varprojlim_n \tilde{U}(V_\hbar)/I_n$ and $T_{\ell,d}$ acts on each $\tilde{U}(V_\hbar)/I_n$ rationally.

For example, if $V_\hbar = V_{c,\hbar}(\mathfrak{g})$, then $\tilde{U}(V_\hbar)$ is the completion of $R_\hbar U(\hat{\mathfrak{g}})$ with respect to the system of left ideals $R_\hbar U(\hat{\mathfrak{g}})(t^n \mathfrak{g}[[t]])$.

The construction of $\tilde{U}(V_\hbar)$ is functorial: a homomorphism $\varphi : V_\hbar^1 \rightarrow V_\hbar^2$ yields a $T_{\ell,d}$ -equivariant topological $\mathbb{K}[\hbar]$ -algebra homomorphism $\tilde{U}(\varphi) : \tilde{U}(V_\hbar^1) \rightarrow \tilde{U}(V_\hbar^2)$.

A weak (resp., strong) action of $\mathcal{JG}$ on $V_\hbar$ induces a weak (resp., strong) action of the loop group $\mathcal{LG}$ on $\tilde{U}(V_\hbar)$. In the latter case, the quantum comoment map is $\tilde{U}(\varphi)$.

One has the notion of a module over a vertex algebra $V_\hbar$, [15, Section 5.1]. It is the same as a smooth module over $\tilde{U}(V_\hbar)$.

Finally, let us discuss a certain subquotient of $\tilde{U}(V_\hbar)$, the 0-mode algebra, e.g., from [19, Section 3.3], it is closely related to the Zhu algebra, [19, Section A.2]. Let $J_{>0}$ denote the closure of the left ideal in $\tilde{U}(V_\hbar)$ generated by the positive energy modes of energy homogeneous elements in $V_\hbar$. The subspace $A^0(V_\hbar) := [\tilde{U}(V_\hbar)/J_{>0}]^{T_\ell}$ carries the structure of a graded associative unital $\mathbb{K}[\hbar]$ -algebra. For example, for $V_\hbar = V_{c,\hbar}(\mathfrak{g})$, we have $J_{>0} = \mathcal{I}_1$, and $A^0(V_{c,\hbar}(\mathfrak{g})) = \mathbb{K}[c] \otimes R_\hbar U(\mathfrak{g})$.

1.2. Categories of equivariant smooth modules: abelian level. The categories we care about in this paper basically have the same general form. We discuss categories of this form in the present section.

1.2.1. Definition. Let G be a connected reductive group, and V_h be a vertex algebra as in Section 1.1.1 equipped with a strong action of $\mathcal{J}G$.

Let $I \subset \mathcal{J}G$ denote the Iwahori subgroup. Next we consider a pro-algebraic group F of the form $F = H \ltimes F^0$, where H is a torus and F^0 is a T_ℓ -stable subgroup of the pro-unipotent radical I^0 of I . We form the group $\tilde{F} = (T_\ell \ltimes F) \times T_d$. Note that $\tilde{F}$ acts on $\tilde{U}(V_h)$ by topological algebra automorphisms. We also have a quantum comoment map $\Phi : \mathfrak{f} \rightarrow \tilde{U}(V_h)$, an $\tilde{F}$ -equivariant map.

We want to define the corresponding category $\mathcal{C}^\heartsuit := \mathcal{C}^\heartsuit(V_h, \tilde{F}, \Phi)$. It consists of V_h -modules (equivalently, smooth $\tilde{U}(V_h)$ -modules) M also equipped with an action of $\tilde{F}$ subject to the following conditions:

- (1) The action of $\tilde{F}$ on M is *rational* meaning that every $m \in M$ is contained in a finite dimensional $\tilde{F}$ -stable subspace M_0 such that the action of $\tilde{F}$ on M_0 factors through a rational representation of a finite dimensional algebraic group quotient of $\tilde{F}$.
- (2) The V_h -module M is *weakly $\tilde{F}$ -equivariant* meaning that the action map $\tilde{U}(V_h) \times M \rightarrow M$ is $\tilde{F}$ -equivariant.
- (3) The $\tilde{U}(V_h)$ -module M is strongly F^0 -equivariant meaning that for all $x \in \mathfrak{f}^0, m \in M$, we have $\Phi(x)m = \hbar L_x m$, where in the right hand side L_x is the operator on M induced by x .

We note that M automatically acquires an action of $S(\mathfrak{h})$ via

$$(1.1) \quad ym = \Phi(y)m - \hbar L_y m, y \in \mathfrak{h}, m \in M$$

The action commutes with $\tilde{U}(V_h)$ and $T_\ell \times F$, while the operators from $\mathfrak{h}$ have degree 1 with respect to T_d . We consider the double affine Cartan space $\mathfrak{h}^{da} := \mathfrak{h} \oplus \mathbb{K}c \oplus \mathbb{K}\hbar$ and let $R := S(\mathfrak{h}^{da})$. So, $\mathcal{C}^\heartsuit$ becomes graded linear over R .

For objects $M, N \in \mathcal{C}^\heartsuit$, we write

$$(1.2) \quad \mathrm{grHom}_{\mathcal{C}}(M, N) = \bigoplus_{i \in \mathbb{Z}} \mathrm{Hom}_{\mathcal{C}}(M, N\langle i \rangle), \mathrm{grExt}_{\mathcal{C}}^1(M, N) = \bigoplus_{i \in \mathbb{Z}} \mathrm{Ext}_{\mathcal{C}}^1(M, N\langle i \rangle),$$

where $\langle i \rangle$ is the endofunctor shifting the PBW degree by i .

1.2.2. Category $\mathcal{O}$. We are going to consider an example of the general situation from Section 1.2.1.

Let G be a simple algebraic group. Take $V_h = V_{c, \hbar}(\mathfrak{g})$ so that $\tilde{U}(V_h)$ is the completed version of $R_{\hbar}U(\mathfrak{g})$, and take $F = I$ so that F^0 is the pro-unipotent radical of I . The resulting category $\mathcal{C}^\heartsuit$ is called the category $\mathcal{O}$ and is denoted by $\mathcal{O}^\heartsuit$.

We are going to introduce some objects in $\mathcal{O}^\heartsuit$.

Consider the universal homogenized Verma module $\Delta := R_{\hbar}U(\mathfrak{g})/\mathfrak{f}^0$. If we equip Δ with discrete topology, the action of $R_{\hbar}U(\mathfrak{g})$ uniquely extends to that of $\tilde{U}(V_{c, \hbar})$ making Δ a smooth $\tilde{U}(V_{c, \hbar})$ -module. Next, note that the action of $\tilde{F}$ on $R_{\hbar}U(\mathfrak{g})$ descends to the quotient Δ and this turns Δ into an object of $\mathcal{O}^\heartsuit$. We will need to twist this action by

a character. Namely, we can embed the level 0 affine weight lattice Λ^a into the character group $\mathfrak{X}(H \times T_\ell)$ as follows: the finite type weights embed into $\mathfrak{X}(H) \subset \mathfrak{X}(H \times T_\ell)$ in the default way, while the indecomposable imaginary root δ corresponds to $(0, 1) \in \mathfrak{X}(H) \oplus \mathfrak{X}(T_\ell)$. Fix $\lambda^a \in \Lambda^a$ and consider the weakly $\tilde{F}$ -equivariant $V_{c,h}(\mathfrak{g})$ -module $\Delta(\lambda^a)$, that is the twist of Δ by the character λ^a of $H \times T_{\ell,c}$. In particular, we can consider the highest weight space of $\Delta(\lambda^a)$, that is identified with $\mathbb{R}$, and the action of $\mathfrak{h} \subset R_h U(\hat{\mathfrak{g}})$ on the highest weight space is by $y \mapsto y - \hbar \langle \lambda, y \rangle$, where λ is the projection of λ^a to $\mathfrak{h}^*$.

Next, we will need projective objects in suitable subcategories of $\mathcal{O}^\vee$. Note that in $\text{Hom}_{\mathbb{Z}}(\Lambda^a, \mathbb{Q})$ we have the element ρ^a that pairs with every simple affine root by 1. For $n \in \mathbb{Z}$, define the full subcategory $\mathcal{O}_{\leq n}$ consisting of all $M = \bigoplus_{\mu^a \in \Lambda^a} M_{\mu^a}$ with $M_{\mu^a} \neq 0 \Rightarrow \langle \mu^a, \rho^a \rangle \leq n$. Pick λ^a satisfying $\langle \lambda^a, \rho^a \rangle \leq n$.

Given a free finite rank $\mathbb{R}$ -module equipped with a compatible rational action of $\tilde{F}$, we can view it as a module over $R_h U(\mathfrak{i})$, where $\mathfrak{i} = \text{Lie}(I)$: the action of $\mathfrak{i}^0$ comes from the strong equivariance condition (3), while the action of $\mathfrak{h}^{da}$ is recovered from the $\mathbb{R}$ -module structure and (1.1). Consider the $\tilde{F}$ -module R_{λ^a} with the usual grading (equivalently, T_d -action) and $H \times T_\ell$ -action by λ^a (which then gets inflated to $T_\ell \ltimes I$). This module has a projective cover in the category of $\mathbb{R}$ -modules with a compatible rational I -action and weights of height $\leq i - \text{ht}(\lambda^a)$: we tensor R_{λ^a} with $U(\mathfrak{i}^0)$ and then mod out by the sum of weight spaces, where the height is $> i$. Denote this cover by $\mathbb{R}_{\lambda^a}^P$. Set

$$P_{\lambda^a, \leq n} := R_h U(\hat{\mathfrak{g}}) \otimes_{R_h U(\mathfrak{i})} \mathbb{R}_{\lambda^a}^P.$$

In particular, this construction has the following consequence.

Lemma 1.2.1. *Any finitely generated (over $R_h U(\hat{\mathfrak{g}})$) module $M \in \mathcal{O}^\vee$ admits an epimorphism over a finitely generated module that is graded free over $\mathbb{R}$.*

Proof. Indeed, notice that $M \in \mathcal{O}_{\leq n}^\vee$ for a suitable n and that M admits an epimorphism from $\bigoplus_{i=1}^r P_{\lambda_i^a, \leq n}$, where λ_i^a 's are the weights of generators of M . $\square$

1.2.3. Base change. We return to the general setup of Section 1.2.1 and observe that thanks to (1.1), every object in $\mathcal{C}^\vee$ is automatically a module over $\tilde{U}(V_h) \hat{\otimes}_{\mathbb{K}[c, \hbar]} \mathbb{R}$, where $\hat{\otimes}$ stands for the completed tensor product.

Now let $\mathbb{R}'$ be a graded Noetherian $\mathbb{R}$ -algebra. We can form the topological $\mathbb{R}'$ -algebra $\tilde{U}(V_h) \hat{\otimes}_{\mathbb{K}[c, \hbar]} \mathbb{R}'$ and consider the category $\mathcal{C}_{\mathbb{R}'}^\vee$ of strongly F^0 - and weakly $T_{\ell,d} \times H$ -equivariant smooth $\tilde{U}(V_h) \hat{\otimes}_{\mathbb{K}[c, \hbar]} \mathbb{R}'$ -modules so that (1.1) holds, where in the left hand side we view y as an element of $\mathbb{R}'$.

Note that we have a right exact functor $\mathbb{R}' \otimes_{\mathbb{R}} \bullet : \mathcal{C}^\vee \rightarrow \mathcal{C}_{\mathbb{R}'}^\vee$. For an object $M \in \mathcal{C}^\vee$, we write $M_{\mathbb{R}'}$ for $\mathbb{R}' \otimes_{\mathbb{R}} M$.

Let $\mathbb{F}_0$ denote the full graded localization of a graded quotient $\mathbb{R}_0$ of $\mathbb{R}$. Suppose $\mathbb{R}_0$ is a domain and suppose $\hbar$ is nonzero in $\mathbb{R}_0$. Set $\mathbf{f} = \mathbb{F}_0^{T_d}$ so that $\mathbb{F}_0 = \mathbf{f}[\hbar^{\pm 1}]$. The category $\mathcal{C}_{\mathbb{F}_0}^\vee$ is equivalent to the following category $\mathcal{C}_{\mathbf{f}}^\vee$: we consider the vertex algebra $V_{\mathbf{f}} := V_h \otimes_{\mathbb{K}[c, \hbar]} \mathbf{f}$ over $\mathbf{f}$ (without the PBW grading). Note that $\mathfrak{h}$ comes with a linear map to $\mathbf{f}$, the composition $\mathfrak{h} \rightarrow \mathbb{F}_0 \xrightarrow{\hbar^{-1}} \mathbf{f}$, so it makes sense to talk about twisted F -equivariant $V_{\mathbf{f}}$ -modules with respect to that map $\mathfrak{h} \rightarrow \mathbf{f}$. The category $\mathcal{C}_{\mathbf{f}}^\vee$ by definition, consists of twisted F -equivariant and weakly T_ℓ -equivariant $V_{\mathbf{f}}$ -modules. We have an equivalence $\mathcal{C}_{\mathbb{F}_0}^\vee \xrightarrow{\sim} \mathcal{C}_{\mathbf{f}}^\vee$ given by taking the T_d -invariants.

1.3. Reduction functor. Our goal for now is to construct a functor

$$\text{Red} : \mathcal{O} \rightarrow \mathbb{K}[c, \hbar] \text{-Mod}^{T_{\ell,d}}.$$

1.3.1. BRST complex. To this end, we construct a certain complex from a module in $\mathcal{O}$.

The construction is a homogenized version of the minus version of the reduction functor from [2]. Namely, consider the Lie algebra $\mathfrak{n}^-((t))$ and its subalgebras $\mathfrak{n}^-[[t]]$ and $t^{-1}\mathfrak{n}^-[t^{-1}]$, the former is complete and the latter is discrete. Also, consider the spaces $\mathfrak{n}^{-,*}((t)) = \mathfrak{n}^{-,*}[[t]] \oplus t^{-1}\mathfrak{n}^{-,*}[t^{-1}]$. Note that there are $\mathbb{K}$ -valued perfect pairings between discrete and complete spaces

$$\mathfrak{n}^-[[t]] \times t^{-1}\mathfrak{n}^{-,*}[t^{-1}] \rightarrow \mathbb{C}, \mathfrak{n}^{-,*}[[t]] \times t^{-1}\mathfrak{n}^-[t^{-1}] \rightarrow \mathbb{C},$$

that are compositions of $\mathbb{K}((t))$ -valued pairings and taking the residue. These pairings give a perfect pairing of the Tate vector spaces $\mathfrak{n}^-((t)) \times \mathfrak{n}^{-,*}((t)) \rightarrow \mathbb{C}$ and hence an orthogonal form on $\mathfrak{n}^-((t)) \oplus \mathfrak{n}^{-,*}((t))$. Form the homogenized Clifford algebra Cl (“homogenized” means that for $u, v \in \mathfrak{n}^-((t)) \oplus \mathfrak{n}^{-,*}((t))$ the relation is $uv + vu = \hbar(u, v)$). Then we can form its Fock space representation $\mathfrak{F}$ with generator $|0\rangle$ and relations $x|0\rangle = 0$ for $x \in t\mathfrak{n}^-[[t]] \oplus \mathfrak{n}^{-,*}[[t]]$. We equip $\mathfrak{F}$ with the unique grading, $\mathfrak{F} = \bigoplus_{i \in \mathbb{Z}} \mathfrak{F}^i$, where $|0\rangle$ is in degree 0, the operators from $\mathfrak{n}^-((t))$ have degree -1 , and the operators from $\mathfrak{n}^{-,*}((t))$ have degree 1. This is to be referred to as the *cohomological grading*. Also, Cl and $\mathfrak{F}$ are naturally graded (as an algebra and its module) by the root lattice of $\hat{\mathfrak{g}}$ and also by the PBW grading. For example, the element $x^\alpha t^n \in \mathfrak{n}^{-,*}((t))$ has Λ^a -degree $\alpha + n\delta$ and PBW degree 0.

Let M be a module from $\mathcal{O}$. We are going to equip the cohomologically graded space $M \otimes_{\mathbb{K}[\hbar]} \mathfrak{F}$ with a differential, a homogenized version of what is denoted by Q_- in [2, Section 6.5]. Consider the root basis $x_{-\alpha} := e_{-\alpha}$ in $\mathfrak{n}^-$, where α runs over the positive roots. Let x^α denote the dual basis in $\mathfrak{n}^{-,*}$. Consider the following fields for Cl

$$x_{-\alpha}(z) := \sum_{n \in \mathbb{Z}} (x_{-\alpha} t^n) z^{-n-1}, x^\alpha(z) := \sum_{n \in \mathbb{Z}} (x^\alpha t^n) z^{-n}.$$

For positive roots α, β, γ , let $c_{\alpha, \beta}^\gamma := \langle x^\gamma, [x_{-\alpha}, x_{-\beta}] \rangle$ denote the structure constants of $\mathfrak{n}^-$.

Consider the non-degenerate character ψ of $\mathfrak{n}^-$ and extend it to a loop invariant character of $\mathfrak{n}^-((t))$: it sends an element at^k with $a \in \mathfrak{n}^-$ to $\delta_{n,0}\psi(a)$.

Now, consider the following formal current into $\mathfrak{n}^-((t)) \otimes \text{Cl} \oplus \text{Cl}$:

$$Q_-(z) := \sum_{\alpha} e_{-\alpha}(z) \otimes x^\alpha(z) + \sum_{\alpha, \beta, \gamma} c_{\alpha, \beta}^\gamma : x^\alpha(z) x^\beta(z) x_{-\gamma}(z) : - \sum_{\alpha} \psi(e_{-\alpha}) x^\alpha.$$

Let Q_- denote the coefficient of z^{-1} in the field $Q_-(z)$. We use the notation $\text{res } Q_-(z)$ for this term.

Lemma 1.3.1. *For any module $M \in \mathcal{O}$, we have a well-defined operator $\hbar^{-1}Q_-$ on $M \otimes \otimes_{\mathbb{K}[\hbar]} \mathfrak{F}$. Its square is 0.*

Proof. Note that

$$Q_- = \sum_{\alpha, n} (e_{-\alpha} t^n) \otimes (x^\alpha t^{-n}) + \text{res} \sum_{\alpha, \beta, \gamma} c_{\alpha, \beta}^\gamma : x^\alpha(z) x^\beta(z) x_{-\gamma}(z) : - \sum_{\alpha} \psi(e_{-\alpha}) x^\alpha.$$

To show that the operator is well-defined we will show that

- (1) $\text{res} \sum_{\alpha, \beta, \gamma} c_{\alpha, \beta}^{\gamma} : x^{\alpha}(z)x^{\beta}(z)x_{-\gamma}(z) :$ acts on the free $\mathbb{K}[\hbar]$ -module $\mathfrak{F}$ by an operator divisible by $\hbar$,
- (2) and for each α and n , we have one of the following:
 - $e_{-\alpha}t^n \in \mathfrak{i}^0 := \text{Lie}(I^0)$,
 - or $x^{\alpha}t^n$ acts on $\mathcal{F}$ by an operator divisible by $\hbar$.

Condition (1) means that the 2nd summand of Q_- divided by $\hbar$ is a well-defined operator. The similar claim for the 1st summand follows from the observations that

- $\hbar^{-1}y$ with $y \in \mathfrak{i}^0$ is a well-defined operator on any module from the category $\mathcal{O}_{\mathbf{R}}$ by the strong equivariance for the I^0 -action,
- and that ψ vanishes on $\mathfrak{i}^0 \cap \mathfrak{n}^-((t)) = t\mathfrak{n}^-[[t]]$.

(1) holds because in each summand in $\text{res} \sum_{\alpha, \beta, \gamma} c_{\alpha, \beta}^{\gamma} : x^{\alpha}(z)x^{\beta}(z)x_{-\gamma}(z) :$ one of the factors $x^{\alpha}t^n, x^{\beta}t^m, x_{-\gamma}t^k$ is an annihilation operator for $\mathcal{F}$, so is divisible by $\hbar$.

(2) holds by inspection. So, indeed, $\hbar^{-1}Q_-$ is a well-defined operator.

To show that the square of this operator is 0 we use two observations. First, note that it is enough to check this when M is finitely generated. Now recall that every finitely generated module in $\mathcal{O}_{\mathbf{R}}$ is a quotient of a module free over $\mathbb{K}[\hbar]$, see Lemma 1.2.1. So it is enough to assume that M is free over $\mathbb{K}[\hbar]$. In this case it is enough to prove the claim after inverting $\hbar$. Here it is equivalent to the claim the version of Q_- introduced in [2, Section 6.5] squares to 0, which is checked in loc. cit., see [2, Equation (297)]. $\square$

So, we can view $\hbar^{-1}Q_-$ as a differential on the cohomologically graded module $M \otimes_{\mathbb{K}[\hbar]} \mathcal{F}$. We define $\text{Red}(M)$ as the 0th cohomology of this complex. It carries a T_{ℓ} -action by the construction. It also carries the commuting modified T_d -action: we can view $\check{\rho} \in \mathfrak{h}$ as a homomorphism $\mathbb{G}_m \rightarrow \text{Ad}(G)$ and we consider the action on $\hat{\mathfrak{g}}$ given by $t.x = t\check{\rho}(t)x$. With these actions, the differential $\hbar^{-1}Q_-$ is $T_{\ell, d}$ -equivariant.

1.3.2. Exactness.

Proposition 1.3.2. *Let M be an arbitrary object in $\mathcal{O}$. We have $H^i(M \otimes_{\mathbb{K}[\hbar]} \mathcal{F}, \hbar^{-1}Q_-) = 0$ for $i \neq 0$. The functor $M \mapsto \text{Red}(M) = H^0(M \otimes_{\mathbb{K}[\hbar]} \mathcal{F}, \hbar^{-1}Q_-)$ is exact. Moreover, if every energy graded component of M is finitely generated over $R_{\hbar}U(\mathfrak{g})[c] \otimes S(\mathfrak{h})$, then each energy graded component in $\text{Red}(M)$ is finitely generated over $\mathbf{R} = S(\mathfrak{h})[c, \hbar]$.*

Proof. The proof is in five steps.

Step 1. We first handle the case when $\hbar$ annihilates M , equivalently, M lies in $\mathcal{O}_{\mathbf{R}_0}$, where $\mathbf{R}_0 = \mathbf{R}/(\hbar)$. Our approach is inspired by the proof of [6, Theorem 7.1].

The objects of $\mathcal{O}_{\mathbf{R}_0}$ are $T_{\ell, d} \ltimes I$ -equivariant $S(\hat{\mathfrak{g}}/\mathfrak{i}^0)$ -modules. The algebra $S(\hat{\mathfrak{g}}/\mathfrak{i}^0)$ can be thought of as the algebra of polynomial functions on the space of λ -connections $\mathbb{K}d + (\mathfrak{n}^{\perp}t^{-1} + \mathfrak{g}^*[[t]])dt$.

At $\hbar = 0$, the algebra Cl becomes a suitably completed version of the exterior algebra of $\mathfrak{n}^-((t)) \oplus \mathfrak{n}^{-,*}((t))$. We can consider $M \otimes \mathfrak{F}/\hbar\mathfrak{F}$ as a bigraded vector space:

$$M \otimes \mathfrak{F}/\hbar\mathfrak{F} = \bigoplus_{i \leq 0 \leq j} M \otimes \Lambda^{-i}(\mathfrak{n}^-[t^{-1}]) \otimes \Lambda^j(t^{-1}\mathfrak{n}^{-,*}[t^{-1}]).$$

The differential $\hbar^{-1}Q_-$ decomposes as the sum $d^1 + d^2$, where d^1 (resp., d^2) sends the graded piece (i, j) to $(i + 1, j)$ (resp., $(i, j + 1)$). Namely, d^1 equals

$$\sum_{\alpha, n \leq 0} (e_{-\alpha} t^n) \otimes \iota_{x^\alpha} t^{-n} - \sum_{\alpha} \psi(e_{-\alpha}) \iota_{x^\alpha},$$

where $\iota_?$ stands for the contraction operation $\Lambda^{-i}(\mathfrak{n}^-[t^{-1}]) \rightarrow \Lambda^{-i-1}(\mathfrak{n}^-[t^{-1}])$. Note that d^1 is the tensor product of the Koszul differential on $M \otimes \Lambda(\mathfrak{n}^-[t^{-1}])$ and the identity endomorphism of $\Lambda^*(t^{-1}\mathfrak{n}^{*,*}[t^{-1}])$.

Note that since the cohomology commutes with filtered colimits, it is enough to consider the case when M is finitely generated over $S(\hat{\mathfrak{g}}/\mathfrak{i}^0)$. In this case the energy degrees in M are bounded from above. It follows that in each energy graded component of $\bigoplus_{i \leq 0 \leq j} M \otimes \Lambda^{-i}(\mathfrak{n}^-[t^{-1}]) \otimes \Lambda^j(t^{-1}\mathfrak{n}^{*,*}[t^{-1}])$, the i degree is bounded from below, which implies that the filtration on each cohomology group of $(M \otimes \mathfrak{F}/\hbar\mathfrak{F}), \hbar^{-1}Q^{st})$ induced by the filtration on the complex by the degree in $\mathfrak{n}^-[t^{-1}]$ is separated. We will show that for the corresponding spectral sequence $E_2^{p,q}$ vanishes unless $p = q = 0$, which will finish this step.

We claim that $E_1^{p,q}$ equals 0 for $p > 0$ and

$$E_1^{0,q} = ((M \otimes \mathbb{C}_\psi)/\mathfrak{n}^-[t^{-1}](M \otimes \mathbb{C}_\psi)) \otimes \Lambda^q(t^{-1}\mathfrak{n}^{*,*}[t^{-1}]),$$

where $\mathbb{C}_\psi$ is the 1-dimensional $\mathfrak{n}^-[t^{-1}]$ -module, where the action is by ψ . For this we need to show that the higher homology of $\mathfrak{n}^-[t^{-1}]$ in $M \otimes \mathbb{C}_\psi$ vanishes. Geometrically, taking this homology is the same as taking the derived restriction of M to

$$\mathbb{K}d + ((\psi + \mathfrak{h}^*)t^{-1} + (\mathfrak{n}^-)^\perp[[t]]) dt \subset \mathbb{K}d + (\mathfrak{n}^\perp t^{-1} + \mathfrak{g}^*[[t]]) dt.$$

By choosing a standard principal $\mathfrak{sl}_2$ -triple (e, h, f) (where $h = 2\check{\rho}$) and identifying $\mathfrak{g}$ with $\mathfrak{g}^*$ we identify these loci with

$$\mathbb{K}d + ((e + \mathfrak{h})t^{-1} + \mathfrak{b}^-[[t]]) dt \subset \mathbb{K}d + (\mathfrak{b}t^{-1} + \mathfrak{g}[[t]]) dt.$$

A standard argument shows that the action morphism

$$\mathcal{J}_k B \times [\mathbb{K}d + ((e + \mathfrak{h}^*)t^{-1} + \mathfrak{b}^- \otimes \mathbb{K}[t]/(t^k)) dt] \rightarrow \mathbb{K}d + (\mathfrak{b}t^{-1} + \mathfrak{g} \otimes \mathbb{K}[t]/(t^k)) dt$$

is flat for all k , where $\mathcal{J}_k B$ is the k th jet group of B . By passing to the limit, we see that

$$\mathcal{J}B \times [\mathbb{K}d + ((e + \mathfrak{h})t^{-1} + \mathfrak{b}^- \otimes \mathbb{K}[[t]]) dt] \rightarrow \mathbb{K}d + (\mathfrak{b}t^{-1} + \mathfrak{g} \otimes \mathbb{K}[[t]]) dt$$

is flat. Since all objects in $\mathcal{O}_{R_0}$ are I - and hence $\mathcal{J}B$ -equivariant, the restriction to $\mathbb{K}d + ((\psi + \mathfrak{h}^*)t^{-1} + (\mathfrak{n}^-)^\perp[[t]]) dt$ is a t -exact functor, and, in particular, we get the claim in the beginning of the paragraph.

The differential $d_2 : E_1^{0,q} \rightarrow E_1^{0,q+1}$ is nothing else but the Chevalley-Eilenberg differential. So, we are computing the cohomology for the action of $t\mathfrak{n}^-[[t]]$ on

$$(1.3) \quad ((M \otimes \mathbb{C}_\psi)/\mathfrak{n}^-[t^{-1}](M \otimes \mathbb{C}_\psi)).$$

We note that the action morphism

$$\mathcal{J}^1 N^- \times [\mathbb{K}d + ((e + \mathfrak{h}^*)t^{-1} + \mathfrak{z}_\mathfrak{g}(f)[[t]]) dt] \rightarrow \mathbb{K}d + ((e + \mathfrak{h})t^{-1} + \mathfrak{b}^-[[t]]) dt,$$

where $\mathcal{J}^1 N^-$ is the kernel of $\mathcal{J}N^- \rightarrow N^-$, is an isomorphism. Set

$$(1.4) \quad S'_{\mathfrak{h}^a} := \mathbb{K}d + ((e + \mathfrak{h})t^{-1} + \mathfrak{z}_\mathfrak{g}(f)[[t]]) dt.$$

Since M , and hence its restriction to $\mathbb{K}d + ((e + \mathfrak{h})t^{-1} + \mathfrak{b}^-[[t]]) dt$, is $\mathcal{J}^1 N^-$ -equivariant, we see that (1.3) decomposes into the tensor product $M|_{S'_{\mathfrak{h}^a}} \otimes \mathbb{K}[\mathcal{J}^1 N^-]$ as a module over $tn^-[[t]]$.

Recall that for a unipotent group, U , we have $\dim H^i(\mathfrak{u}, \mathbb{K}[U]) = \delta_{i,0}$. The group $\mathcal{J}^1 N^-$ is pro-unipotent, i.e., it is the inverse limit of unipotent groups $U_i, i \geq 1$ (under epimorphisms). We note that the Chevalley-Eilenberg complex for $\mathcal{J}^1 N^-$ is a filtered colimit (in fact, a union) of the Chevalley-Eilenberg complexes for the groups U_i . Taking the cohomology commutes with taking the filtered colimits so the higher cohomology of (1.3) vanishes.

This implies the claim of this step for objects $M \in \mathcal{O}_{R_0}$.

Step 2. We apply the functor $\mathbf{Red}$ to $\Delta(\lambda^a)$. Using an argument repeating [1, Section 5] we get the following:

- (1) $\mathbf{Red}(\Delta(\lambda^a))$ is concentrated in cohomological degree 0.
- (2) The energy graded components of the 0-th cohomology of $\mathbf{Red}(\Delta(\lambda^a))$ are finitely generated graded R -modules.

Step 3. The claims of the proposition for general M (i.e., those not annihilated by $\hbar$) reduces to the case when M is finitely generated, again because taking the cohomology commutes with filtered colimits. We remark that since M is finitely generated, each energy graded component of M is finitely generated over $R_{\hbar}U(\mathfrak{g})[c]$. In what follows we restrict to objects M of $\mathcal{O}$ with energy graded components finitely generated over $R_{\hbar}U(\mathfrak{g})[c]$ (this class of objects is closed under kernels because $R_{\hbar}U(\mathfrak{g})[c]$ is Noetherian, while the class of finitely generated modules is not).

Let r be the highest energy degree appearing in M . We claim that there is a complex

$$(1.5) \quad 0 \rightarrow N_k \rightarrow N_{k-1} \rightarrow \dots \rightarrow N_0 \rightarrow M \rightarrow 0$$

in $\mathcal{O}$ such that:

- (i) Each N_i admits a finite filtration by Verma modules and the energy degrees in N_i are $\leq r$,
- (ii) The cohomology of (1.5) are in energy degrees $< r$.

Indeed, every energy graded component of an object in $\mathcal{O}$ is an object in the analogously defined category for $R_{\hbar}U(\mathfrak{g})[c]$ and the Borel subgroup $B \subset G$, to be denoted by $\mathcal{O}_{fin}$. If M is finitely generated over $R_{\hbar}U(\mathfrak{g})[c]$, then all energy graded components are finitely generated over $R_{\hbar}U(\mathfrak{g})[c]$. For a resolution $0 \rightarrow N'_i \rightarrow \dots \rightarrow N'_0$ of M_r by Verma filtered objects, we can take $N_i := \text{Ind}_{\mathfrak{g}}^{\hat{\mathfrak{g}}} N'_i$ (with highest energy degrees set to r). Since the highest energy degree in M is r , the homomorphism $N'_0 \rightarrow M_r$ lifts to $N_0 \rightarrow M_r$. Moreover, since the composition $N'_1 \rightarrow N'_0 \rightarrow M_r$ is 0, the composition $N_1 \rightarrow N_0 \rightarrow M$ is 0 as well. We have constructed a complex $0 \rightarrow N_k \rightarrow \dots \rightarrow N_0 \rightarrow M$ from $0 \rightarrow N'_k \rightarrow \dots \rightarrow N'_0 \rightarrow M_r \rightarrow 0$ with properties (i) and (ii). So it remains to construct the latter complex, which will be done in the next step.

Step 4. Take a finitely generated object $M' \in \mathcal{O}_{fin}$. We want to show that it has a finite resolution by Verma filtered objects. For this we need two claims:

- (a) $R_{\hbar}U(\mathfrak{b}^-)[c]$ has finite homological dimension.
- (b) an object in $\mathcal{O}_{fin}$ has a Verma filtration if and only if it is free finite rank Λ^a -graded module over $R_{\hbar}U(\mathfrak{b}^-)[c]$.

To show (a), one equips $R_h U(\mathfrak{b}^-)[c]$ with a PBW filtration, where $\mathfrak{b}^-$ is in degree 1, and $\hbar, c$ are in degree 0. The associated graded is the polynomial algebra $S(\mathfrak{b}^-)[\hbar, c]$, so it has finite homological dimension. (a) follows.

The only if direction in (b) follows because every Verma module is a free $R_h U(\mathfrak{b}^-)[c]$ -module. Conversely, suppose $M' \in \mathcal{O}_{fin}$ is free finite rank graded module over $R_h U(\mathfrak{b}^-)[c]$. To establish a Verma filtration, we argue by induction on the number of generators. To get the induction step, we choose a basis $m_1, \dots, m_k$ of homogeneous weight vectors. Assume m_k has maximal weight. Then m_k is B -semiinvariant and $R_h U(\mathfrak{b}^-)[c]m_k$ is a subobject in M' (as an object in $\mathcal{O}_{fin}$) isomorphic to a Verma module. So we can quotient it out and continue by induction.

Once we know (a) and (b), we just need to observe that M' admits an epimorphism from a Verma filtered object, cf. the proof of Lemma 1.2.1, and the kernel is still finitely generated because $R_h U(\mathfrak{g})[c]$ is Noetherian.

Step 5. Now we finish the proof. From Step 2 combined with Step 3 we deduce that every energy graded component of $\text{Red}(M)$ is quasi-isomorphic to a bounded complex of finitely generated graded R -modules. Let $C^\bullet$ be one of these complexes. From Step 1 we see that the cohomology of $\mathbb{K} \otimes_{\mathbb{K}[\hbar]}^L C^\bullet$ are concentrated in cohomological degree 0. It follows that the cohomology of $C^\bullet$ itself are in cohomological degree 0 finishing the proof of the exactness. The proof of finite generation of energy graded components follows from Steps 2 and 3. $\square$

Remark 1.3.3. As in the proof of [3, Theorem 4.4.3], a comment after equation (44) there, Step 1 shows that for $M \in \mathcal{O}_{R_0}^\heartsuit$, we actually have a natural isomorphism

$$(1.6) \quad \text{Red}(M) \cong ((M \otimes \mathbb{C}_\psi) / \mathfrak{n}^-[t^{-1}](M \otimes \mathbb{C}_\psi))^{\mathcal{J}^{1N^-}}$$

Also, by Step 1, we have

$$(1.7) \quad \text{Red}(M) \cong M|_{S'_{\mathfrak{h}_a}},$$

where $S'_{\mathfrak{h}_a}$ is defined by (1.4).

1.4. Affine W-algebras. In this section we will recall two other vertex algebras important for this work: the usual and equivariant affine W-algebras. We follow Arakawa's papers, [2, 3, 4]. We note that the setting of these papers is a bit different from ours: the level c is specialized to an arbitrary number, and $\hbar$ is set to be 1.

1.4.1. Plus version of Drinfeld-Sokolov reduction. Let $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{h} \oplus \mathfrak{n}^+$ be the triangular decomposition. Choose root vectors $e_\alpha \in \mathfrak{g}$, where α runs through the root system. We consider the character $\chi : \mathfrak{n}^+((t)) \rightarrow \mathbb{K}$ given by $e_\alpha \otimes f(t) \mapsto 0$ if α is not simple and $e_\alpha \otimes f(t) \mapsto \text{Res}_{t=0}(f(t)dt)$ if α is simple. We note that this map is not $T_{\ell,d}$ -equivariant, but instead it becomes equivariant if we modify the action as follows. We write we write $t \mapsto \check{\rho}(t)$ for the corresponding homomorphism $\mathbb{G}_m \rightarrow \text{Ad}(G)$. Let $t \mapsto t_\ell, t \mapsto t_d$ denote actions of $\mathbb{G}_m = T_\ell$ and T_d on $\hat{\mathfrak{g}}$. The modified actions we need are $t \mapsto t_d \check{\rho}(t)^{-1}$ and $t \mapsto t_\ell \check{\rho}(t)$.

In our context by the Kazhdan-Lusztig category we mean the category of $T_{\ell,d} \ltimes \mathcal{JG}$ -equivariant smooth $R_h U(\hat{\mathfrak{g}})$ -modules. Denote this category by KL . To $M \in \text{KL}$ one assigns the plus version of the BRST complex (of $\mathbb{K}[c, \hbar]$ -modules). As the space is the exterior tensor product of M and a certain Fock space $\mathfrak{F}_+$, similar to but slightly different from

one in Section 1.3.1 (one needs to homogenize the BRST complex with differential Q_+ as, say, in [2, Section 6.4], while Section 1.3.1 homogenizes Q_- from [2, Section 6.5]).

We would like to point out that the complex is equivariant for the modified action of $T_{\ell,d}$, so the cohomology carries this action.

The following result is a homogenized deformed version of [3, Theorem 4.5.7] (a similar result was obtained in [16]). Here “homogenized” refers to the presence of $\hbar$ and “deformed” refers to c being an indeterminate.

Proposition 1.4.1. *The cohomology of the BRST complex vanishes outside of degree 0. The functor of taking the 0th cohomology (from $\mathbf{KL}$ to the category of (weakly) $T_{\ell,d}$ -equivariant $\mathbb{K}[c, \hbar]$ -modules) is exact.*

Sketch of proof. We note that while an approach in [3] is based on considering filtrations and associated graded spaces, they are quite different from what we use in the present paper, so an argument needs to be changed. On the other hand, there is an argument essentially repeating the proof of Proposition 1.3.2. Let us explain necessary modifications:

In Step 1, one shows that for $M \in \mathbf{KL}$ annihilated by $\hbar$, $\text{Red}_+(M)$ is the restriction of M to the following version of the slice:

$$(1.8) \quad \mathbb{K}d + f + \mathfrak{z}_{\mathfrak{g}}(e)[[t]],$$

where $(e, \hbar = 2\check{\rho}, f)$ is an $\mathfrak{sl}_2$ -triple. This restriction functor is exact. In Steps 2-4 one deals with the homogenized version of Weyl modules getting that for every finitely generated object in the Kazhdan-Lusztig category the energy graded components of the cohomology is finitely generated over $\mathbb{K}[c, \hbar]$. The result follows like in Step 5 of the proof of Proposition 1.3.2. $\square$

We denote the 0th cohomology of the BRST complex for M by $\text{Red}_+(M)$.

Now suppose that $V_{\hbar}$ is a vertex algebra equipped with a strong $\mathcal{JG}$ -action whose quantum comoment map φ is $\mathbb{K}[c, \hbar]$ -linear. The two examples we care about are the $\mathcal{JG}$ -action on $V_{c, \hbar}(\mathfrak{g})$ and the left translation $\mathcal{JG}$ -action on $\text{CDO}_{c, \hbar}(G)$, see Section 1.1.2. One can show that the BRST complex of $V_{\hbar}$ is a dg-vertex algebra, and so $\text{Red}_+(V_{\hbar})$ is a vertex algebra of the kind considered in Section 1.1.1. And $\text{Red}_+(M)$ is a module over $\text{Red}_+(V_{\hbar})$.

Definition 1.4.2. The vertex algebra $\text{Red}_+(V_{c, \hbar}(\mathfrak{g}))$ is called the *affine W -algebra*, we denote it by $\mathfrak{W}_{c, \hbar}$ (or $\mathfrak{W}_{c, \hbar}(\mathfrak{g})$ when we need to indicate the dependence on $\mathfrak{g}$). The vertex algebra $\text{Red}_+(\text{CDO}_{c, \hbar}(G))$ (considered in [4]) is called the *equivariant affine W -algebra*, it will be denoted by $\mathfrak{EW}_{c, \hbar}$.

We note that we still have a strong action of $\mathcal{JG}$ on $\mathfrak{EW}_{c, \hbar}$ with a $\mathbb{K}[c, \hbar]$ -semilinear quantum comoment map. Since Red_+ is an exact functor, we have a vertex algebra inclusion from $\mathfrak{W}_{c, \hbar}$ to the $\mathcal{JG}$ -invariant part of $\mathfrak{EW}_{c, \hbar}$. In fact, this inclusion is an isomorphism: the $\mathcal{JG}$ -invariants in $V_{\hbar}$ is the equalizer of two homomorphisms (the co-action and the embedding as the first factor) $V_{\hbar} \rightarrow V_{\hbar} \otimes \mathbb{K}[\mathcal{JG}]$, the equalizers are intertwined by the exactness of Red_+ .

1.4.2. Structure of affine W -algebra. The goal of this section is to give a basis of $\mathfrak{W}_{c, \hbar}$ as a bigraded $\mathbb{K}[c, \hbar]$ -module, as well as a topological basis of $\mathcal{W}_{c, \hbar} := \widetilde{U}(\mathfrak{W}_{c, \hbar})$. We also

describe the quotients of $\mathfrak{W}_{c,\hbar}$ and $\mathcal{W}_{c,\hbar}$ by $(\hbar)$ and $(c, \hbar)$ and the 0-mode subquotient $A^0(\mathfrak{W}_{c,\hbar})$.

The following is a homogenized and deformed version of [2, Proposition 4.12.1]. Let $r := \text{rk } \mathfrak{g}$ and let $d_1, \dots, d_r$ denote the degree of free homogeneous generators of $\mathbb{K}[\mathfrak{g}]^G$.

Proposition 1.4.3. *There are elements $x^i \in \mathfrak{W}_{c,\hbar}, i = 1, \dots, r$ of PBW degree d_i and energy degree $-d_i$ such that the elements $x_{(n_1)}^{i_1} \dots x_{(n_k)}^{i_k} |\emptyset\rangle$, where $k > 0, n_1 \leq n_2 \leq \dots n_k < 0$ and if $n_j = n_{j+1}$, then $i_j < i_{j+1}$, form a $\mathbb{K}[c, \hbar]$ -basis in $\mathfrak{W}_{c,\hbar}$.*

From $x^i \in \mathfrak{W}_{c,\hbar}$ we get elements $x_j^i \in \mathcal{W}_{c,\hbar}$ with $j \in \mathbb{Z}$, where our convention is that the energy degree of x_j^i is j . According to [2, Section 3.14], we get the following.

Corollary 1.4.4. *The elements $x_{n_1}^{i_1} \dots x_{n_k}^{i_k}$ with $n_1 \leq n_2 \leq \dots \leq n_k$ and if $n_j = n_{j+1}$, then $i_j < i_{j+1}$ form a topological $\mathbb{K}[c, \hbar]$ -basis in $\mathcal{W}_{c,\hbar}$ meaning that an arbitrary element of $\mathcal{W}_{c,\hbar}$ is uniquely written as an infinite linear combination of these monomials, where for each $N \in \mathbb{Z}$ only finitely many summands have $n_k < N$.*

Now we proceed to the description of quotients of $\mathfrak{W}_{c,\hbar}$ and $\mathcal{W}_{c,\hbar}$ by the ideals $(\hbar)$. Note that (1.8) is a one-parameter deformation of $\mathcal{J}S$ over $\mathbb{A}^1$, where c is a coordinate on $\mathbb{A}^1$. Denote it by $\mathcal{J}_c S$. We also have the corresponding deformation of the loop space to be denoted $\mathcal{L}_c S$.

Lemma 1.4.5. $\mathfrak{W}_{c,\hbar}/(\hbar) = \mathbb{K}[\mathcal{J}_c S]$ and $\mathcal{W}_{c,\hbar}/(\hbar) = \mathbb{K}[\mathcal{L}_c S]$.

Finally, let us discuss the 0-mode subquotient. The following is proved in the same way as [2, Theorem 4.16.3]. Let $\mathcal{Z}(\mathfrak{g})$ denote the center of $U(\mathfrak{g})$.

Lemma 1.4.6. *There is a T_d -equivariant $\mathbb{K}[c, \hbar]$ -algebra isomorphism*

$$R_{\hbar} \mathcal{Z}(\mathfrak{g})[c] \xrightarrow{\sim} A^0(\mathfrak{W}_{c,\hbar}).$$

Remark 1.4.7. In what follows we will twist this identification by the automorphism of $\mathfrak{g}$ that is given on $\mathfrak{h}$ by $-w_0$.

1.5. Representation theory of affine W-algebras. Here we study the category $\mathbf{C}$: the full subcategory of $\mathcal{C}(\mathcal{W}_{c,\hbar}, H \times T_{\ell,d})$ as in Section 1.2.1 consisting of all objects with trivial action of H , which is the same as the category of weakly $T_{\ell,d}$ -equivariant smooth $\mathcal{W}_{c,\hbar} \hat{\otimes} S(\mathfrak{h})$ -modules.

Remark 1.5.1. We would like to point out some autoequivalences of $\mathbf{C}$. Let φ be an automorphism of $\mathfrak{h}^{da} = \mathfrak{h}^a \oplus \mathbb{K}\hbar$ fixing $c, \hbar$. Then φ gives rise to an autoequivalence of $\mathbf{C}$ that sends an object M to the same bigraded $\mathcal{W}$ -module, but with $\mathbf{R}$ -action twisted by φ . We denote this equivalence by $M \mapsto {}^\varphi M$.

1.5.1. Verma modules. First, we are going to assign a $T_{\ell,d}$ -equivariant $R_{\hbar} \mathcal{Z}(\mathfrak{g}) \otimes S(\mathfrak{h}^a)$ -module structure on $\mathbf{R} = S(\mathfrak{h}^{da})$. Recall that we have fixed the normalized W -invariant form κ_0 on $\mathfrak{h}$. In particular, it gives a W -equivariant isomorphism $\mathfrak{h} \rightarrow \mathfrak{h}^*$ to be denoted also by κ_0 . Let $\text{HC} : R_{\hbar} \mathcal{Z}(\mathfrak{g}) \hookrightarrow S(\mathfrak{h})[\hbar] = \mathbb{K}[\mathfrak{h}^*, \hbar]$ denote the Harish-Chandra isomorphism (whose image consists of the W -invariants for the $\hbar\rho$ -shifted action, i.e. for $w \in W, \lambda \in \mathfrak{h}^*$ we have $w \bullet^{\hbar} \lambda = w(\lambda + \hbar\rho) - \hbar\rho$). Let $\sigma_{\lambda} : \mathbf{R} \rightarrow \mathbf{R}$ be the $\mathbb{K}[c, \hbar]$ -linear automorphism given on $\mu \in \mathfrak{h}^*$ by $\mu \mapsto \mu - c\kappa_0(\check{\rho}) + \hbar\lambda$.

Definition 1.5.2. Let γ_λ (where $\lambda^a = (\lambda, n)$) denote the homomorphism

$$R_\hbar \mathcal{Z}(\mathfrak{g})[c] \rightarrow \mathbb{R}$$

gives as follows:

- It sends c to c viewed as an element of $\mathbb{R}$,
- On $R_\hbar \mathcal{Z}(\mathfrak{g})$ it is given by $\sigma_\lambda \circ \text{HC}$, where we view HC as a homomorphism to $\mathbb{R}$ via the tautological embedding of $S(\mathfrak{h})[\hbar]$ to $\mathbb{R}$.

In particular, notice that γ_0 identifies $\mathbb{K}[c] \otimes R_\hbar \mathcal{Z}(\mathfrak{g})$ with the W -invariants in $\mathbb{R}$ for the following action:

$$(1.9) \quad w \bullet^{c, \hbar} \lambda = w(\lambda + \hbar \rho - c\kappa_0(\check{\rho})) - \hbar \rho + c\kappa_0(\check{\rho}).$$

Now we proceed to defining the module $\underline{\Delta}(\lambda^a)$, where $\lambda^a = \lambda + n\delta$. Recall the left ideal $J_{>0} \subset \mathcal{W}_{c, \hbar}$, see Section 1.1.3. By definition, $\underline{\Delta}(\lambda^a)$ is the tensor product

$$[\mathcal{W}_{c, \hbar} / J_{>0}] \otimes_{A^0(\mathfrak{W}_{c, \hbar})} \mathbb{R},$$

where $A^0(\mathfrak{W}_{c, \hbar})$ is identified with $R_\hbar \mathcal{Z}(\mathfrak{g})[c]$ as in Remark 1.4.7, and the latter acts on $\mathbb{R}$ via γ_λ ; the generator $1 \otimes 1$ is in energy degree n and PBW degree 0. In other words,

$$(1.10) \quad \underline{\Delta}(\lambda^a) = {}^{t_\lambda} \underline{\Delta}(n\delta).$$

By the construction of $\underline{\Delta}(\lambda^a)$, it has the following property. Note that for any $M \in \mathbb{C}^\vee$, $A^0(\mathfrak{W}_{c, \hbar})$ acts on $M^{J_{>0}}$, the annihilator of $J_{>0}$ in M . Let $M = \bigoplus_{n \in \mathbb{Z}} M_n$ be the energy grading. Then we have a natural isomorphism

$$(1.11) \quad \text{grHom}_{\mathbb{C}}(\underline{\Delta}(\lambda + n\delta), M) \xrightarrow{\sim} \{m \in M_n^{J_{>0}} \mid am = \gamma_\lambda(a)m, \forall a \in R_\hbar U(\mathfrak{g})[c]\}.$$

We now proceed to a description of $\underline{\Delta}_{\mathbb{R}_0}(\lambda^a)$. Set $S_c^+ := \mathbb{K}d + t^{-1}(e + \mathfrak{z}_{\mathfrak{g}}(f)[[t]])dt \subset \mathcal{L}_c S$, this is the contracting locus for the T_ℓ -action. Taking the residue at 0 defines a projection $S_c^+ \rightarrow S \cong \mathfrak{h}^*/W$. We set

$$(1.12) \quad S_{\mathfrak{h}^a}^+ := S_c^+ \times_{\mathfrak{h}^*/W} \mathfrak{h}^*,$$

this is now a scheme over $(\mathfrak{h}^a)^*$ with a natural map to $\mathcal{L}_c S$, the composition of the projection $S_{\mathfrak{h}^a}^+ \rightarrow S_c^+$ that is the quotient by the action of W and the inclusion $S_c^+ \hookrightarrow \mathcal{L}_c S$.

We note that this description of $\underline{\Delta}(\lambda^a)$ combined with Lemma 1.4.5 implies the following isomorphism

$$(1.13) \quad \mathbb{K} \otimes_{\mathbb{K}[\hbar]} \underline{\Delta}(\lambda^a) \cong \mathbb{K}[S_{\mathfrak{h}^a}^+] \otimes \mathbb{C}_{n,0},$$

where $\mathbb{C}_{n,0}$ is the 1-dimensional $T_{\ell, d}$ -module, where T_ℓ acts via $t \mapsto t^n$ and T_d acts trivially.

1.5.2. Functor Red revisited. In Section 1.3 we have constructed an exact functor

$$\text{Red} : \mathcal{O}^\vee \rightarrow \mathbb{K}[c, \hbar]\text{-Mod}^{T_{\ell, d}}.$$

We now explain how to upgrade this functor to

$$\text{Red} : \mathcal{O}^\vee \rightarrow \mathbb{C}^\vee$$

by establishing a natural action of $\mathcal{W}_{c, \hbar}$ on $\text{Red}(M)$ for $M \in \mathcal{O}^\vee$ (the action of $S(\mathfrak{h})$ is induced from that on M by functoriality).

Our construction is a homogenized version of that in [2, Section 6.5].

We will need details on this action. On the BRST complex for M we have an action of the BRST complex for $\tilde{U}(V_{c, \hbar}(\mathfrak{g}))$ defined as the topological dg-algebra $\tilde{U}(V_{c, \hbar}(\mathfrak{g})) \hat{\otimes}_{\mathbb{K}[\hbar]} \text{Cl}$

with differential $\hbar^{-1}[Q_-, \cdot]$. The 0-th cohomology of this dg-algebra naturally acts on the only nonzero cohomology of $\text{Red}(M)$.

Now consider an element of $\mathcal{L}G$ defined as follows. We fix a lift $\dot{w}_0$ of w_0 in the Weyl group. Then we have the automorphism $\eta := \text{Ad}(\dot{w}_0 t^{-\rho^\vee})$ of $\hat{\mathfrak{g}}$ and $\mathcal{L}G$. Note that $v(e_i \otimes t^{-1}) = f_{i^*} \otimes 1$, where $i \mapsto i^*$ is the self-bijection of the simple roots induced by $-w_0$. So v maps $\mathcal{L}N^+$ to $\mathcal{L}N^-$ and $\psi \circ \eta = \chi$. Also η^{-1} induces an isomorphism between Cl and its counterpart Cl_+ that is used to define Red_+ and sends Q_- to Q_+ . In particular, the 0-th cohomology of

$$(1.14) \quad (\tilde{U}(V_{c,\hbar}(\mathfrak{g})) \hat{\otimes}_{\mathbb{K}[\hbar]} \text{Cl}_+, \hbar^{-1}[Q_+, \cdot])$$

acts on $\text{Red}(M)$ via η . As explained in [2, Section 3.18], see, in particular, [2, Equation (120)], $\tilde{U}(\mathfrak{W}_{c,\hbar})$ admits a natural homomorphism to the 0-th cohomology of (1.14) and so $\text{Red}(M)$ acquires an action of $\mathcal{W}_{c,\hbar}$. In this particular case the map from $\mathcal{W}_{c,\hbar}$ to the 0-th cohomology is an isomorphism, which is proved similarly to [2, Theorem 3.19.1].

1.5.3. Red at $\hbar = 0$. We are now going to specialize the constructions of the previous section to the case $\hbar = 0$.

The dg-algebra $(\tilde{U}(V_{c,\hbar}(\mathfrak{g})) \hat{\otimes}_{\mathbb{K}[\hbar]} \text{Cl}/\hbar \text{Cl}, \hbar^{-1}[Q_-, \cdot])$ is quasi-isomorphic to the Hamiltonian reduction of $\mathbb{K}[\mathcal{L}_c \mathfrak{g}^*]$ by the action of $\mathcal{L}N^-$ with character ψ to be denoted by $\mathbb{K}[\mathcal{L}_c \mathfrak{g}^*] //_{\psi} \mathcal{L}N^-$. This Hamiltonian reduction is identified with $\mathbb{K}[\mathcal{L}_c S]$, where $S = e + \mathfrak{z}_{\mathfrak{g}}(f)$. The Hamiltonian reduction $\mathbb{K}[\mathcal{L}_c G] //_{\psi} \mathcal{L}N^-$ naturally acts on the right hand side of (1.6), and this is the action that we introduced in Section 1.5.2.

Now we interpret this action in terms of the realization of the Hamiltonian reduction as $\mathbb{K}[\mathcal{L}_c S]$. The construction in Step 1 of the proof of Proposition 1.3.2 shows that $\mathbb{K}[S'_{\mathfrak{h}^a}]$ is identified as an algebra with

$$(1.15) \quad ([S(\hat{\mathfrak{g}}/\mathfrak{i}^0) \otimes \mathbb{C}_{\psi}] / [S(\hat{\mathfrak{g}}/\mathfrak{i}^0) \otimes \mathbb{C}_{\psi}] \mathfrak{n}^- [t^{-1}])^{\mathcal{J}^1 N^-}.$$

(1.15) admits a natural homomorphism from $\mathbb{K}[\mathcal{L}_c \mathfrak{g}^*] //_{\psi} \mathcal{L}N^- = \mathbb{K}[\mathcal{L}_c S]$. We are going to present the corresponding morphism $S'_{\mathfrak{h}^a} \rightarrow \mathcal{L}_c S$ as the following composition.

Lemma 1.5.3. *The morphism $S'_{\mathfrak{h}^a} \rightarrow \mathcal{L}_c S$ decomposes as the composition of an isomorphism $S'_{\mathfrak{h}^a} \xrightarrow{\sim} S_{\mathfrak{h}^a}^+$ of schemes over $\mathfrak{h}^{a,*}$ and the morphism $S_{\mathfrak{h}^a}^+ \rightarrow \mathcal{L}_c S$ from Section 1.5.1.*

Proof. Since the action of T_{ℓ} on $S'_{\mathfrak{h}^a}$ is contracting, the scheme theoretic image of $S'_{\mathfrak{h}^a}$ in $\mathcal{L}_c S$ is contained in S_c^+ . Under the identification of $S'_{\mathfrak{h}^a}$ with the spectrum of (1.15), the composition $S'_{\mathfrak{h}^a} \rightarrow S_c \rightarrow \mathfrak{h}^*/W$ can be described as follows: we first map a point of $S'_{\mathfrak{h}^a}$ to its residue, an element of $e + \mathfrak{h}$, and then this element is mapped to $\mathfrak{h}^*/W$ via $e + \mathfrak{h} \rightarrow (e + \mathfrak{b}^-)/N^- \cong \mathfrak{h}^*/W$. This description shows that the compositions $S'_{\mathfrak{h}^a} \rightarrow S_c^+ \rightarrow \mathfrak{h}^*/W$ and $S'_{\mathfrak{h}^a} \rightarrow \mathfrak{h}^{a,*} \rightarrow \mathfrak{h}^*/W$ coincide, yielding a morphism $S'_{\mathfrak{h}^a} \rightarrow S_{\mathfrak{h}^a}^+$. It remains to show it is an isomorphism. We note that this morphism is equivariant for the T_d -action. Since it contracts both schemes to a point, and both schemes are flat over $\mathfrak{h}^{a,*}$, it is sufficient to prove that we get an isomorphism over $0 \in \mathfrak{h}^a$. Here the fibers of both schemes are equal to $(et^{-1} + \mathfrak{z}_{\mathfrak{g}}(f)[[t]])dt$, and the morphism in question is the identity. $\square$

In what follows we view the restriction of Red to $\mathcal{O}_{R_0}^\heartsuit$ as a functor $\mathcal{O}_{R_0}^\heartsuit \rightarrow \mathbb{K}[S_{\mathfrak{h}^a}^+]\text{-Mod}^{T_{\ell,d}}$ by using the isomorphism $S'_{\mathfrak{h}^a} \xrightarrow{\sim} S_{\mathfrak{h}^a}^+$. For example, for $\lambda^a = (\lambda, n)$, we have an isomorphism

$$(1.16) \quad \text{Red}(\Delta_{R_0}(\lambda^a)) \cong \mathbb{K}[S_{\mathfrak{h}^a}^+] \otimes \mathbb{C}_{n,0}.$$

1.5.4. $\text{Red}(\Delta(\lambda^a))$. Our goal in this section is to prove the following result.

Proposition 1.5.4. *For all λ^a , we have $\text{Red}(\Delta(\lambda^a)) \xrightarrow{\sim} \underline{\Delta}(\lambda^a)$.*

Proof. Let $\mathbb{F}_0, \mathbf{f}$ have the same meaning as in Section 1.2.3. Consider the equivalent categories $\mathcal{C}_{\mathbb{F}_0}$ and $\mathcal{C}_{\mathbf{f}}$.

By Proposition 1.3.2 and (1.16), we know the following:

- (1) Every energy graded component of $\text{Red}(\Delta(\lambda^a))$ is a free finite rank R -module.
- (2) $\mathbb{K} \otimes_{\mathbb{K}[h]} \text{Red}(\Delta(\lambda^a)) \cong \mathbb{K}[S_{\mathfrak{h}^a}^+] \otimes \mathbb{C}_{n,0}$.

Combining (2) with (1.13), we see that

$$\mathbb{K} \otimes_{\mathbb{K}[h]} \text{Red}(\Delta(\lambda^a)) \xrightarrow{\sim} \mathbb{K} \otimes_{\mathbb{K}[h]} \underline{\Delta}(\lambda^a).$$

Combining this with (1) we see that $\text{Red}(\Delta(\lambda^a))$ is generated by its highest (i.e., n -th) energy graded component. It remains to prove that the actions of $R_h \mathcal{Z}(\mathfrak{g})[c]$ on the n -th graded components of $\underline{\Delta}(\lambda^a)$, $\text{Red}(\Delta(\lambda^a))$ coincide. Since both are free over R , it is enough to check the coincidence of actions after localizing to the full graded fraction ring $\mathbb{F}$ of R . The objects $\text{Red}(\Delta_{\mathbb{F}}(\lambda^a))$ and $\underline{\Delta}_{\mathbb{F}}(\lambda^a)$ are isomorphic by (an easy special case of) [2, Theorem 7.5.1]. $\square$

1.5.5. *Images of certain objects under Red.* Now let $\mathbb{F}_0$ be a full graded fraction ring of a graded domain quotient R_0 of R , and let $\mathbf{f}$ be the T_d -invariants in $\mathbb{F}_0$, a field. Assume that $\hbar \neq 0$ in R_0 . Thanks to the equivalence $\mathcal{C}_{\mathbb{F}_0}^\heartsuit \cong \mathcal{C}_{\mathbf{f}}^\heartsuit$ results on the behavior of Red from [2] apply. In particular, we have the unique energy graded irreducible quotients $L_{\mathbb{F}_0}(\lambda^a)$ of $\Delta_{\mathbb{F}_0}(\lambda^a)$ and $\underline{L}_{\mathbb{F}_0}(\lambda^a)$ of $\underline{\Delta}_{\mathbb{F}_0}(\lambda^a)$.

The following is [2, Theorem 7.6.3], note the difference in labeling, see Remark 1.4.7.

Proposition 1.5.5. *We have $\text{Red}(L_{\mathbb{F}_0}(\lambda^a)) \cong \underline{L}_{\mathbb{F}_0}(\lambda^a)$ if $\langle \lambda + \hbar\rho, \check{\alpha} \rangle - n\hbar$ is nonzero in $\mathbb{F}_0$ for all $n > 0$ and all positive finite roots $\check{\alpha}$. Otherwise, $\text{Red}(L_{\mathbb{F}_0}(\lambda^a)) = 0$.*

One can also compute the images of dual Verma modules $\nabla_{\mathbb{F}_0}(\lambda^a)$ (=the images of Verma under the usual duality functor on the category $\mathcal{O}$), which make sense in $\mathcal{O}_{\mathbb{F}_0}$. It also makes sense to define dual Verma modules for $\mathcal{W}_{\mathbf{f}}$, see [2, Section 5.5]. Let us recall the construction.

By $\mathcal{O}_{fin, \mathbb{F}_0}^{\mathcal{W}}$ we denote the full subcategory of $\mathcal{C}_{\mathbb{F}_0}$ consisting of all modules M with energy grading $N = \bigoplus_{i \in \mathbb{Z}} M_i$ such that each M_i is finite generated (automatically free) over $\mathbb{F}_0$ and $M_i = 0$ for all $i \gg 0$. The category $\mathcal{O}_{fin, \mathbb{F}_0}^{\mathcal{W}}$ comes with a contravariant involutive endo-functor D such that the ranks of M_i and $D(M_i)$ coincide. Moreover, we have the following claim which is a part of [2, Theorem 5.5.4].

Lemma 1.5.6. *We have $\text{Red}(\nabla_{\mathbb{F}_0}(\lambda)) \cong D(\underline{\Delta}(-w_0\lambda^a))$.*

1.6. Feigin-Frenkel duality. One way to realize $\mathfrak{W}_{c,\hbar}$ is via the so called free field realization embedding, [13]. The target for the embedding is the affine vertex algebra $V_{c,\hbar}(\mathfrak{h})$. One reference for this embedding in the homogenized deformed setting is [12, Section 4]. Recall that the relation for operators xt^i in this algebra is

$$[xt^i, yt^j] = \delta_{i+j,0} i \hbar \kappa_0(x, y).$$

The embedding $\text{ffr}_{\mathfrak{g}} : \mathfrak{W}_{c,\hbar} \hookrightarrow V_{c,\hbar}(\mathfrak{h})$ is $\mathbb{K}[c, \hbar]$ -linear.

We would like to explain some further properties of the embedding $\text{ffr}_{\mathfrak{g}}$. First, it intertwines the PBW gradings as well as the energy gradings. Next, $\mathfrak{W}_{c,\hbar}$ comes with a preferred pseudo-conformal vector: this is a vector whose field is of the form $L(z) := \sum_{i \in \mathbb{Z}} L_i z^{-i-2}$ and the operator $[L_0, ?]$ acts on the n -th graded component for the energy grading by $\hbar c n$. In the case of $\mathfrak{sl}_2$ this vector actually generates the vertex algebra $\mathfrak{W}_{\hbar}$. The image of the field $L(z)$ under ffr can be described as follows. Let $r := \text{rk } \mathfrak{g}$ and let $x_1, \dots, x_n$ and $x^1, \dots, x^n$ be bases of $\mathfrak{h}$ that are dual with respect to the form $\kappa_0|_{\mathfrak{h}}$. For an element $x \in \mathfrak{h}$, we set $x(z) := \sum_{i \in \mathbb{Z}} (x^i) z^{-i-1}$. We consider the elements $\check{\rho} \in \mathfrak{h}$ and $\rho \in \mathfrak{h}^*$, so that $\kappa_0^{-1}(\rho) \in \mathfrak{h}$. Then, cf. [14, Proposition 6.2.2],

$$(1.17) \quad \text{ffr}_{\mathfrak{g}}(L(z)) = \frac{1}{2} : x_i(z) x^i(z) : + \partial_z [c \check{\rho}(z) - \hbar \kappa_0^{-1}(\rho)(z)].$$

Here $:? :$ stands for the normally ordered product of fields, and κ_0 is viewed as $\kappa_0|_{\mathfrak{h}} : \mathfrak{h} \xrightarrow{\sim} \mathfrak{h}^*$.

A remarkable feature of the free field realization embedding is the Feigin-Frenkel duality between the affine W-algebras $\mathfrak{W}_{\hbar}(\mathfrak{g})$ and $\mathfrak{W}_{\check{\hbar}}(\check{\mathfrak{g}})$ (where we write $\check{\hbar}$ for the Rees parameter on the Langlands dual side). Consider the isomorphism

$$\iota : V_{c,\hbar}(\mathfrak{h}) \xrightarrow{\sim} V_{\check{c},\check{\hbar}}(\check{\mathfrak{h}}) : xt^i \mapsto -\kappa_0(x)t^i, c \mapsto \check{\hbar}, \hbar \mapsto \check{c}, \forall x \in \mathfrak{h}.$$

Note, in particular, that this isomorphism sends $\text{ffr}_{\mathfrak{g}}(L(z))$ to $\text{ffr}_{\check{\mathfrak{g}}}(\check{L}(z))$.

Proposition 1.6.1. *The maps*

$$\text{ffr}_{\mathfrak{g}} : \mathfrak{W}_{\hbar}(\mathfrak{g}) \rightarrow V_{\hbar}(\mathfrak{h}), \text{ffr}_{\check{\mathfrak{g}}} : \mathfrak{W}_{\check{\hbar}}(\check{\mathfrak{g}}) \rightarrow V_{\check{\hbar}}(\check{\mathfrak{h}})$$

are embeddings, and ι intertwines their images.

Proof. Recall that $\text{ffr}_{\mathfrak{g}}$ is energy graded. The energy graded components in $\mathfrak{W}_{\hbar}(\mathfrak{g})$ and $V_{\hbar}(\mathfrak{h})$ are finitely generated free graded $\mathbb{K}[c, \hbar]$ -modules. So the proof of injectivity reduces to checking the injectivity of the map specialized to $c = \hbar = 0$, which is done, for example in Step 1 of the proof of [12, Proposition 10.1.2].

To prove the claim about the intersection of images we argue as follows. We consider the free field realization for the affine Lie algebras: $\text{ffr}'_{\mathfrak{g}} : V_{c,\hbar}(\mathfrak{g}) \rightarrow \text{CDO}_{\hbar}(N^+) \otimes_{\mathbb{K}[\hbar]} V_{c,\hbar}(\mathfrak{h})$.

Note that we can define the plus reduction functor on strongly $\mathcal{JN}$ -equivariant modules, we do not need the whole $\mathcal{JG}$ -equivariance. Then $\text{ffr}_{\mathfrak{g}} = \text{Red}_+(\text{ffr}'_{\mathfrak{g}})$. According to [12, Proposition 4.3.1], one can factorize $\text{ffr}'_{\mathfrak{g}}$ as follows. We write L for a minimal standard Levi L_i and $P_i := LN$. Let N' be the unipotent radical of P and $N_L := N \cap L$. We can factorize $\text{CDO}_{\hbar}(N)$ as $\text{CDO}_{\hbar}(N') \otimes_{\mathbb{K}[\hbar]} \text{CDO}_{\hbar}(N_L)$. Then

$$(1.18) \quad \text{ffr}'_{\mathfrak{g}} = (\text{id}_{N'} \otimes \text{ffr}'_L) \circ \varphi,$$

where φ is a vertex algebra homomorphism $V_{c,\hbar}(\mathfrak{g}) \rightarrow \text{CDO}_{\hbar}(N') \otimes_{\mathbb{K}[\hbar]} V_{c,\hbar}(\mathfrak{l}_i)$. Let Red_+^L be the plus reduction functor for L .

We claim that

$$(1.19) \quad \text{Red}_+(\text{CDO}_h(N') \otimes_{\mathbb{K}[\hbar]} V_{c,h}(\mathfrak{l})) \xrightarrow{\sim} \text{Red}_+^L(V_{c,h}(\mathfrak{l})), \quad \text{Red}_+(\text{id}_{N'} \otimes \text{ffr}'_l) = \text{Red}_+^L(\text{ffr}'_l).$$

To establish (1.19), we embed the plus version of the BRST complex for $V_{c,h}(\mathfrak{l})$ to that for $\text{CDO}_h(N') \otimes_{\mathbb{K}[\hbar]} V_{c,h}(\mathfrak{l})$ by $x \mapsto |0\rangle \otimes x$. It is easy to see that we get a morphism of complexes. (1.19) will follow once we show that this embedding is a quasi-isomorphism. It is not difficult to see that the PBW gradings on the energy graded components of the cohomology of the BRST complex associated to are bounded from below. So it is enough to prove the quasi-isomorphism claim after taking the quotient by $\hbar$. Since the energy graded components of both complexes are finite complexes, it is enough to show the quasi-isomorphisms on E_2 -sheets of spectral sequences as in Step 1 of the proof of Proposition 1.3.2. There the claim about the quasi-isomorphism follows by reduction in stages.

Combining (1.18) with (1.19) we see that $\text{im ffr}_{\mathfrak{g}} \subset \text{im ffr}_{\mathfrak{l}_i}$ for all i . This proves

$$\text{im ffr}_{\mathfrak{g}} \subset \bigcap_{i=1}^{\text{rk } \mathfrak{g}} \text{im ffr}_{\mathfrak{l}_i}.$$

The argument that the above containment is an equality reduces to the specialization at $c = \hbar = 0$, cf. the proof of [12, Proposition 10.1.2], which is then established in the proof of that proposition. $\square$

So we get an isomorphism $\iota : \mathfrak{W}_{c,h}(\mathfrak{g}) \xrightarrow{\sim} \mathfrak{W}_{\check{c},\check{h}}(\check{\mathfrak{g}})$ compatible with the ffr maps.

We will need the homomorphisms between the 0-mode subquotients induced by the homomorphisms $\text{ffr}_{\mathfrak{g}}$. Let $\text{ffr}_{\mathfrak{g}}^0$ denote the homomorphism

$$R_{\hbar} \mathcal{Z}(\mathfrak{g})[c] = A^0(\mathfrak{W}_{\hbar}) \rightarrow A^0(V_{c,h}(\mathfrak{h})) = S(\mathfrak{h}^{da}) = \mathbf{R}.$$

induced by $\text{ffr}_{\mathfrak{g}}$.

Lemma 1.6.2. *The homomorphism $\text{ffr}_{\mathfrak{g}}^0$ coincides with γ_0 from Section 1.5.1.*

We can identify $\mathbf{R}$ with $\check{\mathbf{R}}$ via

$$(1.20) \quad \iota : c \mapsto \check{c}, \check{h} \mapsto \check{h}, x \mapsto -\kappa_0(x), \forall x \in \mathfrak{h}.$$

Then we deduce the following corollary of Lemma 1.6.2.

Corollary 1.6.3. *We have the following commutative diagram:*

$$(1.21) \quad \begin{array}{ccc} R_{\hbar} \mathcal{Z}(\mathfrak{g})[c] & \xrightarrow{\sim} & R_{\hbar} \mathcal{Z}(\check{\mathfrak{g}})[\check{c}] \\ \downarrow \text{ffr}_{\mathfrak{g}}^0 & & \downarrow \text{ffr}_{\check{\mathfrak{g}}}^0 \\ \mathbf{R} & \xrightarrow{\iota} & \check{\mathbf{R}} \end{array}$$

where the top horizontal arrow is the isomorphism of the 0-mode algebras induced by the Feigin-Frenkel duality isomorphism. In particular, the Feigin-Frenkel duality intertwines the Verma modules $\underline{\Delta}_{\mathbf{R}}(0) \cong \underline{\Delta}_{\check{\mathbf{R}}}(0)$.

1.7. Main results of the paper. Let G be a (split) connected simple algebraic group over $\mathbb{K}$ and $\check{G}$ be the Langlands dual group of G (of adjoint type). We use the invariant form κ_0 on $\mathfrak{h}$ to identify $\mathfrak{h}$ with $\check{\mathfrak{h}} = \mathfrak{h}^*$. We extend this identification $\mathfrak{h}^{da} = \mathfrak{h} \oplus \mathbb{K}c \oplus \mathbb{K}\check{h} \xrightarrow{\sim} \check{\mathfrak{h}}^{da} = \check{\mathfrak{h}} \oplus \mathbb{K}\check{c} \oplus \mathbb{K}\check{h}$ via $c \mapsto \check{h}$ and $\check{h} \mapsto \check{c}$.

One category we care about is $\mathcal{O}$. On the other hand, we consider the equivariant W-algebra $\mathcal{EW}_{\check{c}, \check{h}}$ for $\check{G}$ (i.e., the universal enveloping algebra for $\mathfrak{EW}_{\check{c}, \check{h}}(\check{\mathfrak{g}})$) and the category $\check{\mathcal{F}}$ associated to $\mathcal{EW}_{\check{c}, \check{h}}$ and the group $\check{F} := T_{\ell, d} \ltimes \check{I}$ as in Section 1.2.1. This category is graded linear over $S(\check{\mathfrak{h}}^{da})$ and hence over $\mathbb{R}$ thanks to the identification (1.20).

Our goal in this paper is to construct a right exact functor $\Xi : \mathcal{O}^\heartsuit \rightarrow \check{\mathcal{F}}^\heartsuit$ satisfying certain compatibility conditions that are outlined below in this section.

Remark 1.7.1. We expect that Ξ extends to $\mathcal{O} \rightarrow \check{\mathcal{F}}$ and is an equivalence, or at least is close to being an equivalence. On the other hand, Ξ itself is not expected to be t-exact.

1.7.1. Images of Vermas. First, we would like to explain where Ξ sends modules $\Delta(\lambda^a)$ for $\lambda \in \Lambda^a, n \in \mathbb{Z}$. We write λ^a as $\lambda + m\delta$ with λ in the weight lattice for G , equivalently, the coroot lattice for $\check{G}$. To λ we assign a free monodromic Wakimoto sheaf that we view as a strongly equivariant (for $\check{I}^0 \times \check{I}^0$) pro-object in the category of D-modules on the loop group $\mathcal{LG}$ (with some additional structures). An infinite-dimensional (for $\check{I} \backslash \mathcal{LG} / \check{I}$ instead of $\check{B} \backslash \mathcal{G} / \check{B}$) analog of the construction in [10, Section 3] allows to produce from this Wakimoto sheaf (with a suitable equivariance twist) a homogenized D-module on the Kac-Moody group $\hat{\check{G}}$ to be denoted by $\mathbf{Wak}_\lambda$. The module $\mathbf{Wak}_\lambda$ by the construction is $\check{I}^0 \times \check{I}^0$ -equivariant, weakly $H \times H \times T_{\ell, d}$ -equivariant, and moreover the left and right actions of $\check{R}$ on $\mathbf{Wak}_\lambda$ differ by t_λ .

We write $D_{\check{h}}(\hat{\check{G}})$ for the algebra of homogenized differential operators on $\hat{\check{G}}$. The left and right actions of $\hat{\check{G}}$ on itself induce quantum comoment maps $R_{\check{h}}U(\hat{\check{\mathfrak{g}}}) \rightarrow D_{\check{h}}(\hat{\check{G}})$ to be denoted by Φ_ℓ and Φ_r (for the left and right actions, respectively). In particular, $\Phi_\ell^* \mathbf{Wak}_\lambda$ can be viewed as an object in $\check{\mathcal{O}}$. Consider the object $\text{Red}(\mathbf{Wak}_\lambda)$. The left action of $\check{I}$ and the corresponding quantum comoment map Φ_ℓ turn $\text{Red}(\mathbf{Wak}_\lambda)$ into an object of $\check{\mathcal{F}}$. Suitable normalizing the $T_{\ell, d}$ -action, we get an object in $\check{\mathcal{F}}$ to be denoted by $\check{\Delta}^F(\lambda^a)$.

Here is the first property of Ξ that we need.

$$(I) \quad \Xi(\Delta(\lambda^a)) \cong \check{\Delta}^F(\lambda^a).$$

We would like to briefly explain the logic behind this condition. First, we explain why it is natural to expect it to hold for $\lambda = 0$. This is thanks to a (prospective) action of the affine Hecke category for G on $\mathcal{O}, \mathcal{F}$ that we want to the derived functor induced by Ξ to intertwine. Then we observe that $\Delta(n\delta) \in \mathcal{O}$ and $\check{\Delta}^F(n\delta)$ are easy examples of objects that transform according to the trivial module over these affine Hecke categories. Then to justify (I) we observe that Λ also acts on both categories $\mathcal{O}$ and $\check{\mathcal{F}}$, the equivalence corresponding to λ sends $\Delta(n\delta)$ to $\Delta(\lambda + n\delta)$ and $\check{\Delta}^F(n\delta)$ to $\check{\Delta}^F(\lambda + n\delta)$. In a sentence, (I) is motivated by the prospective equivariance for the “double affine Hecke action”, the notion that we do not make precise. Another motivation for (I) will be given in Section 1.7.3.

1.7.2. Equivalence of exact categories. We consider the full subcategories $\mathcal{O}^\Delta \subset \mathcal{O}$ (resp., $\check{\mathcal{F}}^\Delta \subset \check{\mathcal{F}}$) consisting of all objects admitting finite filtrations by $\Delta(\lambda^a)\langle n \rangle$ (resp., $\check{\Delta}^F(\lambda^a)\langle n \rangle$)

for various pairs (λ^a, n) ; here $\langle n \rangle$ denote the twist of the T_d -action by n . These are exact categories in the sense of Quillen.

Here's the 2nd property we want to achieve:

(II) Ξ restricts to an equivalence of exact categories:

$$\mathcal{O}^\Delta \xrightarrow{\sim} \check{\mathcal{F}}^\Delta.$$

1.7.3. *Compatibility with functors to modules over W -algebras.* We consider the corresponding category $\mathcal{C}^\heartsuit$ as in Section 1.2.1. Thanks to the Feigin-Frenkel duality, Section 1.6, the category $\mathcal{C}^\heartsuit$ is identified with its analog $\check{\mathcal{C}}^\heartsuit$ for $\check{\mathfrak{g}}$. Consider the following functors:

- $\text{Red} : \mathcal{O} \rightarrow \mathcal{C}_\check{\mathfrak{g}}$
- the derived $\check{I}$ -invariants

$$R\text{Hom}_{\check{I}}(\text{triv}, ?) : \check{\mathcal{F}} \rightarrow \mathcal{C}$$

that we are going to denote by $R\Gamma$.

We will see that

(1) We have

$$(1.22) \quad \text{Red}(\Delta(\lambda^a)) \cong R\Gamma(\check{\Delta}^F(\lambda^a)) = \underline{\Delta}(\lambda^a), \forall \lambda^a \in \Lambda^a.$$

(2) The restrictions of Red to $\mathcal{O}^\Delta$ and $R\Gamma$ to $\check{\mathcal{F}}^\Delta$ are fully faithful functors,

(3) and their essential images coincide.

The claims for the $\mathcal{O}$ -side are generally quite standard and will be proved in Section 3, while the $\check{\mathcal{F}}$ -side requires quite extensive work that will be done in Section 4. A crucial ingredient in our approach to proving (2) and (3) is a suitable modification of the Rouquier-Soergel theory introduced in [18] that we will explain in Section 3.

Once (2) and (3) are established, extending the resulting equivalence Ξ to $\mathcal{O}^\heartsuit \rightarrow \check{\mathcal{F}}^\heartsuit$ is based on the observation that $\mathcal{O}$ is a highest weight category in a suitable sense and $\mathcal{O}^\Delta$ is its category of standardly filtered objects. These constructions are discussed in Section 2.

2. HIGHEST WEIGHT CATEGORIES

2.1. **Categories.** In this section we will recall some generalities about highest weight categories in the setting that we need. The exposition closely follows [18, Section 2] although unlike in that paper we work with big abelian categories (having arbitrary direct sums) rather than with categories of Noetherian objects.

In what follows $\mathbf{R}$ denotes a finitely generated positive graded $\mathbb{K}$ -algebra. Everything explain below in this section carries over to the case when $\mathbf{R}$ is a Noetherian local ring (without a grading).

2.1.1. *Case of finite posets: definition.* By a graded $\mathbf{R}$ -linear abelian category we mean a $\mathbb{K}$ -linear abelian category $\mathcal{C}$ with a grading shift functor $\langle 1 \rangle$ such that the spaces grHom for $\mathcal{C}$ (cf. (1.2)) are naturally $\mathbf{R}$ -modules. We assume that $\mathcal{C}$ has arbitrary direct sums. A basic example is the category of graded modules over a graded $\mathbf{R}$ -algebra.

Let $\mathcal{T}$ be a finite poset with order $\leq$. By a *highest weight structure* with poset $\mathcal{T}$ we mean a collection of objects $\Delta(\tau)$ (called *standard*) satisfying the following properties:

(HW0) For any $\tau \in \mathcal{T}$, and any cofiltered diagram (M_i) of objects in $\mathcal{C}$, where all arrows are monomorphisms, the directed system $\text{grHom}_{\mathcal{C}}(\Delta(\tau), M_i)$ stabilizes meaning that any homomorphism $\Delta(\tau)\langle n \rangle \rightarrow \varinjlim_i M_i$ factors through some M_i . In particular,

$$\varinjlim_i \text{grHom}_{\mathcal{C}}(\Delta(\tau), M_i) \xrightarrow{\sim} \text{grHom}_{\mathcal{C}}(\Delta(\tau), \varinjlim_i M_i).$$

(HW1) For any $M \in \mathcal{C}$, there is $\tau \in \mathcal{T}$ such that $\text{grHom}_{\mathcal{C}}(\Delta(\tau), M) \neq 0$.

(HW2) If $\text{grHom}_{\mathcal{C}}(\Delta(\tau), \Delta(\tau')) \neq \{0\}$, then $\tau \leq \tau'$.

(HW3) $\mathbf{R} \xrightarrow{\sim} \text{grHom}_{\mathcal{C}}(\Delta(\tau), \Delta(\tau))$.

(HW4) For any $\tau, \tau' \in \mathcal{T}$, the graded $\mathbf{R}$ -module $\text{grHom}_{\mathcal{C}}(\Delta(\tau), \Delta(\tau'))$ is finitely generated.

(HW5) For any $\tau \in \mathcal{T}$, there is a projective object P_{τ} in $\mathcal{C}$ together with an epimorphism $P_{\tau} \twoheadrightarrow \Delta(\tau)$ whose kernel admits a finite filtration by objects of the form $\Delta(\tau')\langle i \rangle$ with $\tau' > \tau$.

It makes sense to speak about standardly filtered objects in $\mathcal{C}$, denote their full subcategory by $\mathcal{C}^{\Delta}$. This is an exact category. In particular, $\text{grExt}_{\mathcal{C}}^1(M, N)$ is read off of $\mathcal{C}^{\Delta}$ viewed as an exact category with a grading shift functor.

We note that the isomorphism in (HW0) holds if we replace $\Delta(\tau)$ with any object in $\mathcal{C}^{\Delta}$.

Thanks to (HW5), we have

$$(2.1) \quad \text{grExt}_{\mathcal{C}}^1(\Delta(\tau), \Delta(\tau')) \neq 0 \Rightarrow \tau < \tau'.$$

Let $M \in \mathcal{C}^{\Delta}$ and $\mathcal{T}' \subset \mathcal{T}$ be a poset coideal. Thanks to (2.1) we have a unique subobject $M_{\mathcal{T}'}$ satisfying the following conditions:

- $M_{\mathcal{T}'}$ admits a filtration by $\Delta(\tau')$ with $\tau' \in \mathcal{T}'$,
- and $M/M_{\mathcal{T}'}$ admits a filtration by $\Delta(\tau'')$ with $\tau'' \in \mathcal{T} \setminus \mathcal{T}'$.

2.1.2. Case of finite poset: projective objects. Now we proceed to a discussion of projective objects in $\mathcal{C}$. The proofs of [21, Proposition 4.13, Lemma 4.22] go through and we have the following characterization of projective objects.

Lemma 2.1.1. *The following claims hold:*

- (1) *For a projective object P in $\mathcal{C}$ the following two conditions are equivalent:*
 - (a) $\text{grHom}_{\mathcal{C}}(P, \Delta(\tau))$ is a finitely generated $\mathbf{R}$ -module for any $\tau \in \mathcal{T}$,
 - (b) $P \in \mathcal{C}^{\Delta}$.
- (2) *Moreover, an object P satisfying (a) is projective in $\mathcal{C}$ if and only if it is projective in $\mathcal{C}^{\Delta}$, equivalently, $\text{grExt}_{\mathcal{C}}^1(P, \Delta(\tau)) = 0$ for all $\tau \in \mathcal{T}$.*

Remark 2.1.2. Next note that we have the following construction of a projective object satisfying (HW5), cf. [18, Remark 2.1.9]. Fix $\tau \in \mathcal{T}$ and write $\mathcal{T}(> \tau)$ for $\{\tau' \in \mathcal{T} \mid \tau > \tau'\}$. Suppose we presented $\mathcal{T}(> \tau)$ as the union of poset coideals

$$\emptyset = \mathcal{T}_0 \subsetneq \mathcal{T}_1 \subsetneq \mathcal{T}_2 \subsetneq \dots \mathcal{T}_k = \mathcal{T}(> \tau)$$

so that

(*) the elements in $\mathcal{T}_{i+1} \setminus \mathcal{T}_i$ are pairwise non-comparable.

Then we construct object $P_i(\tau)$ for $i = k, k-1, \dots, 0$, as follows. We set $P_k(\tau) = \Delta(\tau)$. Once $P_i(\tau)$ for $i = 1, \dots, k$, is constructed, we then construct $P_{i-1}(\tau)$ as follows. Let $\tau_1, \dots, \tau_{\ell}$ be all the different elements in $\mathcal{T}_i \setminus \mathcal{T}_{i-1}$. Choose a grade projective cover N_i of

$\text{grExt}_{\mathcal{C}}^1(P_i(\tau), \Delta(\tau_i))$ and let $N_i^* := \text{Hom}_{\mathbf{R}}(N_i, \mathbf{R})$. Then we get $P_{i-1}(\tau)$ from a canonical short exact sequence:

$$0 \rightarrow \bigoplus_{i=1}^{\ell} N_i^* \otimes_{\mathbf{R}} \Delta(\tau_i) \rightarrow P_{i-1}(\tau) \rightarrow P_i(\tau) \rightarrow 0.$$

The object $P_0(\tau)$ is projective.

We finish this section by explaining how to recover $\mathcal{C}$ from $\mathcal{C}^{\Delta}$. Suppose that we presented $\mathcal{T}$ as a union of poset coideals $\emptyset = \mathcal{T}_0 \subset \mathcal{T}_1 \subset \dots \mathcal{T}_k$ satisfying (*) from Remark 2.1.2. Then we get objects $P(\tau)$ for all τ by taking the intersections $\mathcal{T}_i \cap \mathcal{T}(> \tau)$. Form the graded $\mathbf{R}$ -algebra

$$(2.2) \quad A := \bigoplus_{\tau, \tau' \in \mathcal{T}} \text{grHom}(P(\tau), P(\tau')).$$

Lemma 2.1.3. *The functor $\bigoplus_{\tau \in \mathcal{T}} \text{grHom}(P(\tau), ?)$ is an equivalence of abelian categories between $\mathcal{C}$ and the category of right A -modules. A quasi-inverse functor is given by $M \mapsto M \otimes_A \bigoplus_{\tau \in \mathcal{T}} P(\tau)$.*

This claim is standard and the proof is left to the reader.

Remark 2.1.4. Assume that $\text{grHom}_{\mathcal{C}}(P(\tau), P(\tau'))$ is free over $\mathbf{R}$ for all τ, τ' . Then we can consider base changes of $\mathcal{C}$ as follows. For a graded $\mathbf{R}$ -algebra $\mathbf{R}'$ define the category $\mathcal{C}'$ of right modules over $\mathbf{R}' \otimes_{\mathbf{R}} A$. If $\mathbf{R}'$ is positively graded, then the objects $\Delta'(\tau) := \mathbf{R}' \otimes_{\mathbf{R}} \Delta(\tau)$ are standard for a highest weight structure on $\mathcal{C}'$ with poset $\mathcal{T}$.

2.1.3. *Case of coideal finite posets: definition.* Let $\mathcal{T}$ be a poset. We say that $\mathcal{T}$ is *coideal finite* if for all $\tau \in \mathcal{T}$, the coideal $\{\tau' \in \mathcal{T} \mid \tau' > \tau\}$ is finite. We also assume that

- ($\diamond$) there a separated $\mathbb{Z}_{\leq 0}$ -filtration of $\mathcal{T}$ by poset coideals $\dots \subsetneq \mathcal{T}_{-1} \subsetneq \mathcal{T}_0 = \mathcal{T}$ such that the elements of $\mathcal{T}_{i+1} \setminus \mathcal{T}_i$ are pairwise incomparable.

Now let $\mathcal{C}$ be a graded $\mathbf{R}$ -linear abelian category with arbitrary direct sums. By a *highest weight structure* on $\mathcal{C}$ we mean a collection of standard objects $\Delta(\tau), \tau \in \mathcal{T}$, satisfying conditions (HW0)-(HW5).

The claims of Section 2.1.1 hold with the following modifications:

- (a) Objects in $\mathcal{C}^{\Delta}$ still have filtrations indexed by poset coideals of $\mathcal{T}$ as in Section 2.1.1.
- (b) In Lemma 2.1.1, we modify (a) to require that $\text{grHom}_{\mathcal{C}}(P, \Delta(\tau)) = 0$ for all τ but finitely many. Then the claims of the lemma follow.
- (c) The construction of projectives in Remark 2.1.2 goes through.
- (d) We can form the algebra A by (2.2) but it is not unital if $\mathcal{T}$ is infinite, it is only locally unital: it comes with a collection of orthogonal idempotents 1_{τ} such that for every $a \in A$, there is a finite subset $\mathcal{T}' \subset \mathcal{T}$ such that for all finite subsets $\mathcal{T}''$ containing $\mathcal{T}'$

$$\left(\sum_{\tau \in \mathcal{T}''} 1_{\tau} \right) a = a \left(\sum_{\tau \in \mathcal{T}''} 1_{\tau} \right) = a$$

Similarly, in Lemma 2.1.3, we need to consider the category of locally unital A -modules (where for every element $m \in M$ there is a finite subset $\mathcal{T}'' \subset \mathcal{T}$ so that $m(\sum_{\tau \in \mathcal{T}''} 1_{\tau}) = m$), then we still get a category equivalence.

(e) A complete analog of Remark 2.1.4 holds.

2.1.4. *Case of coideal finite posets: highest weight subquotients.* Let $\mathcal{T}$ be a coideal finite poset and $\mathcal{T}_1 \subset \mathcal{T}$ be a poset ideal. Consider the subcategory $\mathcal{C}_{\mathcal{T}_1}$ consisting of all objects $M \in \mathcal{C}$ such that $\text{grHom}_{\mathcal{C}}(\Delta(\tau), M) = 0$ for all $\tau \notin \mathcal{T}_1$. This is a highest weight category with poset $\mathcal{T}_1$ and standard objects $\Delta(\tau), \tau \in \mathcal{T}_1$. Namely, to construct a projective object $P_{1,\tau}$ as in (HW5), we take $P_{\tau}/P_{\tau, \mathcal{T} \setminus \mathcal{T}_1}$, where the meaning of $P_{\tau, \mathcal{T}_1}$ is as in the end of Section 2.1.1.

Since for a projective object $P \in \mathcal{C}^{\Delta}$, the quotient $P/P_{\mathcal{T} \setminus \mathcal{T}_1}$ is projective in $\mathcal{C}_{\mathcal{T}_1}$. It follows that for $M \in \mathcal{C}_{\mathcal{T}_1}^{\Delta}$ and $N \in \mathcal{C}_{\mathcal{T}_1}$ we have

$$(2.3) \quad \text{grExt}_{\mathcal{C}_{\mathcal{T}_1}}^i(M, N) \xrightarrow{\sim} \text{grExt}_{\mathcal{C}}^i(M, N), \forall i.$$

Now suppose that $\mathcal{T}^1 := \mathcal{T} \setminus \mathcal{T}_1$ is finite. Define the category $\mathcal{C}_{\mathcal{T}^1}$ as the Serre quotient of $\mathcal{C}$ by $\mathcal{C}_{\mathcal{T}_1}$. Equivalently, $\mathcal{C}_{\mathcal{T}^1}$ is the category of graded right modules over $\bigoplus_{\tau, \tau' \in \mathcal{T}^1} \text{grHom}_{\mathcal{C}}(P(\tau), P(\tau'))$ and the quotient functor $\pi : \mathcal{C} \rightarrow \mathcal{C}_{\mathcal{T}^1}$ is given by

$$\bigoplus_{\tau \in \mathcal{T}^1} \text{grHom}_{\mathcal{C}}(P(\tau), ?).$$

The category $\mathcal{C}_{\mathcal{T}^1}$ is highest weight with poset $\mathcal{T}^1$ and standard objects $\pi(\Delta(\tau))$ for $\tau \in \mathcal{T}^1$. Note that π restricts to an equivalence between $\mathcal{C}_{\mathcal{T}^1}^{\Delta}$ and the full subcategory of $\mathcal{C}^{\Delta}$ consisting of all objects filtered by $\Delta(\tau)$ with $\tau \in \mathcal{T}^1$.

2.1.5. *Case of coideal finite posets: costandards and relative injectives.* In this section we assume, in addition, that

(HW6) $\Delta(\tau)$ is flat over R .

Thanks to the analog of Lemma 2.1.3 for coideal finite posets, this is equivalent to $\text{grHom}_{\mathcal{C}}(P(\tau'), \Delta(\tau))$ being free over R . In particular, the R -modules $1_{\tau}A, A1_{\tau}$ are free over R .

Consider the locally unital graded right A -module $I(\tau) := \bigoplus_{\tau' \in \mathcal{T}} \text{grHom}_R(1_{\tau}A1_{\tau'}, R)$. Note that it is graded free over R . It is relatively injective in the category of graded locally unital free right A -modules in the sense that $\text{grHom}_A(?, I(\tau))$ is exact on the exact subcategory of modules free over R .

Now pick $\tau \in \mathcal{T}$ and let $\mathcal{T}_1 := \{\tau' \in \mathcal{T} | \tau' \leq \tau\}$. Set $\nabla(\tau) := I_{\mathcal{T}_1}(\tau)$. Similarly to [21, Propositions 4.19, 4.20], we see that we have an inclusion $\nabla(\tau) \hookrightarrow I(\tau)$ whose cokernel is filtered by $\nabla(\tau')$ with $\tau' > \tau$. These are *costandard objects*. We note that by the construction we have

$$(2.4) \quad \text{grExt}_{\mathcal{C}}^i(\Delta(\tau), \nabla(\tau')) = R^{\oplus \delta_{i,0} \delta_{\tau, \tau'}}.$$

2.1.6. *Interval finite posets.* Now we introduce the most general class of highest weight categories considered in this paper. Let $\mathcal{T}$ be a poset that we assume to be *interval finite*: for all $\tau_1, \tau_2 \in \mathcal{T}$, the interval $\{\tau \in \mathcal{T} | \tau_1 \leq \tau \leq \tau_2\}$ is finite. We also assume the following condition:

- ($\diamond$) there an exhaustive separated $\mathbb{Z}$ -filtration of $\mathcal{T}$ by poset coideals $\subsetneq \mathcal{T}_i \subsetneq \mathcal{T}_{i+1} \subsetneq \dots$ such that the elements of $\mathcal{T}_{i+1} \setminus \mathcal{T}_i$ are pairwise incomparable, and each $\mathcal{T}_i$ is coideal finite as a poset.

We will give an example in Section 2.1.7 below. Suppose that $\mathcal{C}$ comes equipped with objects $\Delta(\tau), \tau \in \mathcal{T}$ satisfying conditions (HW0)-(HW4) from Section 2.1.1 (but not necessarily (HW5)). Consider the full subcategory

$$\mathcal{C}_i = \{M \in \mathcal{C} \mid \text{Hom}_{\mathcal{C}}(\Delta(\tau), M) \neq 0 \Rightarrow \tau \in \mathcal{T}_i\}.$$

We impose two conditions:

(HW1') Every object of $\mathcal{C}$ admits an exhaustive filtration, where every filtration subobjects lies in $\mathcal{C}_i$ for some i .

(HW5) Every $\mathcal{C}_i$ is highest weight with poset $\mathcal{T}_i$ and standard objects $\Delta(\tau), \tau \in \mathcal{C}_i$.

We would like to explain analogs of Remark 2.1.2 and Lemma 2.1.3 (whose combination allows to recover $\mathcal{C}$ from $\mathcal{C}^\Delta$) as well as of Remark 2.1.4. Take $\tau \in \mathcal{T}_i$.

The construction of Remark 2.1.2 gives projective objects $P^j(\tau)$ for each $j > i$ with canonical epimorphisms $P^j(\tau) \twoheadrightarrow P^\ell(\tau)$ for each $j > \ell$. Moreover, from (2.1), for all $\tau, \tau' \in \mathcal{T}_i$ and $j > \ell > i$, we get the natural map

$$\text{grHom}_{\mathcal{C}}(P^j(\tau), P^j(\tau')) \twoheadrightarrow \text{grHom}_{\mathcal{C}}(P^\ell(\tau), P^\ell(\tau')).$$

Now form the locally unital algebra A_i for the category $\mathcal{C}_i$ using (2.2). We get canonical epimorphisms $A_j \twoheadrightarrow A_\ell$ for all $j > \ell$. So we can form the complete topological algebra $A := \varprojlim_{\ell \rightarrow \infty} A_\ell$. It comes with a distinguished collection of idempotents $1_\tau, \tau \in \mathcal{T}$.

Definition 2.1.5. By a *smooth locally unital* graded right A -module we mean a graded right A -module M such that for every $m \in M$ the following hold:

- there is ℓ such that $\ker[A \twoheadrightarrow A_\ell]m = 0$,
- and there is a finite subset $\mathcal{T}' \subset \mathcal{T}$ such that for all finite subsets $\mathcal{T}'' \supset \mathcal{T}'$ we have $m(\sum_{\tau \in \mathcal{T}''} 1_\tau) = 0$.

Denote the category of such modules simply by $\text{Mod-}A$.

Here is an analog of Lemma 2.1.3.

Lemma 2.1.6. *The category $\mathcal{C}$ is equivalent to the category of smooth locally unital right A -modules via*

$$\mathcal{F} : M \mapsto \varinjlim_{\ell \rightarrow \infty} \left(\bigoplus_{\tau \in \mathcal{T}_\ell} \text{grHom}_{\mathcal{C}}(P^i(\tau), M) \right).$$

Proof. First, recall that by (HW1'), we can write $M = \bigcup_{i \in \mathbb{Z}} M^i$, where for M^i we can take the maximal subobject of M lying in $\mathcal{C}_i$ (that exists because $\mathcal{C}$ is closed under arbitrary direct sums and the generalization of (HW0) to all standardly filtered objects). From the version of (HW0) for $P^i(\tau) \in \mathcal{C}^\Delta$, we see that

$$\text{grHom}_{\mathcal{C}}(P^i(\tau), M) = \varinjlim_{j \rightarrow \infty} \text{grHom}_{\mathcal{C}_j}(P^i(\tau), M^j).$$

By the construction of M_i , we have

$$\text{grHom}_{\mathcal{C}}(M, N) = \varinjlim_{i \rightarrow \infty} \text{grHom}_{\mathcal{C}_i}(M^i, N^i).$$

The functor $\mathcal{F}$ maps $\mathcal{C}_i$ to the full subcategory $\text{Mod-}A_i \subset \text{Mod-}A$ consisting of all objects where the action of A factors through A_i . Now the claim that $\mathcal{F}$ is fully faithful follows from the straightforward analog of Lemma 2.1.3 for $\mathcal{C}_i$.

To show that $\mathcal{F}$ is essentially surjective we note the following. First, it intertwines filtered colimits under monomorphisms thanks to (HW0). Now note that every object $L \in \text{Mod-}A$ is canonically the union of its maximal submodules L^i in $\text{Mod-}A_i$. Let M^i be the preimage of L^i in $\mathcal{C}_i$. Then the objects M^i form a directed system under inclusions $M^i \hookrightarrow M^{i+1}$ for all $i \in \mathbb{Z}$. Then $\mathcal{F}$ sends $\varinjlim_{i \rightarrow -\infty} M^i$ to $L = \varinjlim_{i \rightarrow -\infty} L^i$. $\square$

Finally, let us discuss a version of Remark 2.1.4. In the setting of that remark assuming all $\text{grHom}_{\mathcal{C}_i}(P^i(\tau), P^i(\tau'))$ are free over R , we can define $\mathcal{C}'$ as the category of smooth locally unital graded right modules over $\varinjlim_{i \rightarrow -\infty} R' \otimes_R A_i$.

2.1.7. Example: category $\mathcal{O}$. Recall the category $\mathcal{O}^\vee$ from Section 1.2.2 with objects $\Delta(\lambda^a)$, where $\lambda \in \Lambda^a$. We equip Λ^a with the usual order: $\lambda_1^a \leq \lambda_2^a$ if $\lambda_2^a - \lambda_1^a$ is the sum of positive roots. We also consider the homomorphism $\text{ht} : \Lambda^a \rightarrow \mathbb{Q}$ sending every simple affine root to 1. Set $\Lambda_i^a := \{\lambda^a \mid \text{ht}(\lambda^a) \leq i\}$.

We note that Λ^a is an interval finite poset, and each Λ_i^a is coideal finite. Let us show that $\mathcal{O}^\vee$ is a graded highest weight category over R . (HW0) follows because each $\Delta(\lambda^a)$ is generated by a single element and we can pick M_i containing the image of this element. To show (HW1), take $m \in M$ and include it into a finite dimensional rational $\tilde{F}$ -subrepresentation $M_0 \subset M$. Suppose λ^a is a maximal $T_\ell \times H$ -weight in that representation and m' is a vector of this weight. Then $\text{grHom}_{\mathcal{O}}(\Delta(\lambda^a), M)$ is nonzero: we can find a nonzero homomorphism mapping the generator of $\Delta(\lambda^a)$ to m' . (HW1') follows similarly: if $M^i \subset M$ denotes the maximal subobject where the heights of all weights are at most i , then $m \in M^i$, where i the maximum of the heights of weights of M_0 .

(HW2) easily follows from the observation that all weights μ^a of $\Delta(\lambda^a)$ satisfy $\mu^a \leq \lambda^a$. (HW3) follows because the weight λ^a component of $\Delta(\lambda^a)$ is identified with R and (HW4) follows because every weight component of $\Delta(\lambda^a)$ is a finitely generated R -module.

(HW5) holds for $\mathcal{O}_n^\vee$: one considers projective objects $P_{\lambda^a, \leq n}$ from Section 1.2.2 in that category.

2.2. Functors. We get back to the general situation of Section 2.1.6. Let $\mathcal{C}$ be a highest weight category as in Section 2.1.6 and $\mathcal{C}'$ be a graded R -linear abelian category with arbitrary direct sums (and hence with arbitrary colimits). Let $\mathcal{G} : \mathcal{C}^\Delta \rightarrow \mathcal{C}'$ be an exact functor compatible with the grading shifts. The main result of this section is the following proposition.

Proposition 2.2.1. *The functor $\mathcal{G}$ uniquely extends (up to a unique isomorphism) to a right exact functor $\mathcal{C} \rightarrow \mathcal{C}'$ that commutes with filtered colimits.*

Proof. Step 1. Our first observation is that $\mathcal{G}$ extends from $\mathcal{C}_i^\Delta$ to a right exact functor $\mathcal{G}_i : \mathcal{C}_i \rightarrow \mathcal{C}'$ commuting with colimits. For this we note that every object in $\mathcal{C}_i$ admits a presentation by direct sum of objects $P^i(\tau)$, $\tau \in \mathcal{T}_i$ and every morphism lifts to this presentation. Since the extension is required to intertwine direct sums and be right exact, the claim follows. By the construction, $\mathcal{G}_i$ is unique up to a unique functor isomorphism.

Step 2. Thanks to the uniqueness part in Step 1, we get an isomorphism $\mathcal{G}_{i+1}|_{\mathcal{C}_i} \cong \mathcal{G}_i$.

Pick an object $M \in \mathcal{C}$ and note that $\mathcal{C}$ is identified with $\text{Mod-}A$ thanks to Lemma 2.1.6. Let $M = \bigcup M^i$ be the canonical filtration of $M \in \mathcal{C}$ by objects in $\mathcal{C}_i$, where M^i is the maximal subobject in $\mathcal{C}$ lying in $\mathcal{C}_i$. Note that the inclusion $M^i \hookrightarrow M^{i+1}$ induces a

morphism

$$(2.5) \quad \mathcal{G}_i(M^i) \xrightarrow{\sim} \mathcal{G}_{i+1}(M^i) \rightarrow \mathcal{G}_{i+1}(M^{i+1})$$

We consider the following functor $\tilde{\mathcal{G}} : \mathcal{C} \rightarrow \mathcal{C}'$:

$$(2.6) \quad M \mapsto \varinjlim_{i \rightarrow \infty} \mathcal{G}_i(M^i).$$

This defines a functor $\tilde{\mathcal{G}}$ on objects, and the definition on morphisms is similar. This functor clearly extends $\mathcal{G}$ from $\mathcal{C}^\Delta$ and commutes with colimits.

Step 3. We claim that $\tilde{\mathcal{G}}$ is also right exact. To see this we identify $\mathcal{C}$ with $\text{Mod-}A$, so that $\mathcal{G}_i$ gets identified with $? \otimes_{A_i} B_i$, where $B_i = \bigoplus_{\tau \in \mathcal{T}_i} \mathcal{G}(P^i(\tau))$. Assume $M \twoheadrightarrow N$ is an epimorphism in $\text{Mod-}A$ and let $n \in N^i$. Pick a preimage m of n in M . It may not lie in M^i but it lies in M^j for some j . From here we easily see that

$$\varinjlim_{i \rightarrow \infty} M^i \otimes_{A_i} B_i \twoheadrightarrow \varinjlim_{i \rightarrow \infty} N^i \otimes_{A_i} B_i.$$

A similar argument shows the exactness in the middle term.

Step 4. For any extension $\tilde{\mathcal{G}}'$ of $\mathcal{G}$ from $\mathcal{C}^\Delta$, using the unique isomorphism $\tilde{\mathcal{G}}'|_{\mathcal{C}_i} \cong \mathcal{G}_i$ combined with the assumption that $\mathcal{G}'$ commutes with filtered colimits, we get a unique isomorphism $\varinjlim_{i \rightarrow \infty} \mathcal{G}_i(M^i) \xrightarrow{\sim} \tilde{\mathcal{G}}'(M)$ yielding $\tilde{\mathcal{G}} \xrightarrow{\sim} \tilde{\mathcal{G}}'$. $\square$

3. ROUQUIER-SOERGER THEORY

3.1. Admissible collections of objects.

3.1.1. *Definition.* Here we work in the setting of Section 1.2.1. For $\mathcal{C}$ we take a suitable derived version of $\mathcal{C}^\heartsuit = \mathcal{C}^\heartsuit(\mathcal{A}, \tilde{F}, \Phi)$ (a derived category of equivariant smooth modules) so that $\mathcal{C}$ comes with a t-structure whose heart is $\mathcal{C}^\heartsuit$. By $\mathcal{C}^{\heartsuit, u}$ and $\mathcal{C}^u$ we denote the ungraded version of the categories $\mathcal{C}^\heartsuit, \mathcal{C}$ where instead of the equivariance for $\tilde{F} = T_c \times (T_\ell \ltimes F)$ we consider that for $T_\ell \ltimes F$. More generally, for a Noetherian $\mathbb{R}$ -algebra $\mathbb{R}'$, we can consider the base change $\mathcal{C}_{\mathbb{R}'}^{\heartsuit, u} = \mathcal{C}^{\heartsuit, u}(\mathbb{R}' \hat{\otimes}_{\mathbb{R}} \mathcal{A}, \tilde{F}, \Phi)$ and if $\mathbb{R}'$ is graded, we can consider the category $\mathcal{C}_{\mathbb{R}'}^\heartsuit$.

Consider an abelian group Γ equipped with a linear action on $\mathfrak{h}^{da}$ by linear transformations fixing $c, \hbar$. Suppose that this action lifts to a weak action of Γ on $\mathcal{C}$ by graded $\mathbb{R}$ -semilinear equivalences. Fix an object $\Delta(0) \in \mathcal{C}^\heartsuit$ and set $\Delta(\gamma) := \gamma \cdot \Delta(0)$. We make the following assumptions on the objects $\Delta(\gamma)$. Let $\mathbb{F}$ denote the full graded localization ring of $\text{Frac}(\mathbb{R})$, i.e., the localization of $\mathbb{R}$ by all nonzero homogeneous elements.

- (a1) The object $\Delta(0)$ is finitely generated over $\mathcal{A}$.
- (a2) The grading on $\Delta(0)$ (coming from the T_c -action) is bounded from below (note that this does not follow from (a1) because $\mathcal{A}$ is not required to be non-negatively graded).
- (a3) The objects $\Delta(\gamma)$ lie in $\mathcal{C}^\heartsuit$ and are flat over $\mathbb{R}$ for all $\gamma \in \Gamma$.
- (a4) For all $\gamma_1, \gamma_2 \in \Gamma$, the $\mathbb{K}$ -vector spaces

$$\text{grExt}_{\mathcal{C}_{\mathbb{K}}}^i(\Delta_0(\gamma_1), \Delta_0(\gamma_2))$$

are finite dimensional for $i = 0, 1$, where we write $\Delta_0(\gamma) = \mathbb{K} \otimes_{\mathbb{R}} \Delta(\gamma)$ with $\mathbb{R}$ acting on $\mathbb{K}$ by 0.

(a5) $\text{grExt}_{\mathcal{C}_{\mathbb{F}}}^1(\Delta_{\mathbb{F}}(\gamma_1), \Delta_{\mathbb{F}}(\gamma_2)) = 0$ for $\gamma_1 \neq \gamma_2$ (where $\Delta_{\mathbb{F}}(\gamma_i) := \mathbb{F} \otimes_{\mathbb{R}} \Delta_{\mathbb{R}}(\gamma_i)$).

Definition 3.1.1. In the presence of the aforementioned structures and assumptions (a1)-(a5), we say that the collection of objects $\Delta(\gamma)$ is *admissible*.

We write $\mathcal{C}^{\Delta}$ for the full subcategory of $\mathcal{C}$ that admit finite filtrations with successive quotients $\Delta(\gamma)$.

Example 3.1.2. We take $\mathcal{C} = \mathcal{O}$, cf. Sections 1.2.2, 2.1.7. We take $\Gamma = \Lambda^a$ and for $\lambda^a \in \Lambda^a$ consider the Verma module define $\Delta(\lambda^a)$ as in Section 1.2.2. It is easy to see that the objects $\Delta(\lambda^a)$ satisfy (a1)-(a3). To prove (a4) note that for $i = 0, 1$ by the Yoneda description of Ext^1 , both $\text{grExt}^0 = \text{grHom}$ and grExt^1 can be computed in the category $\mathcal{O}_i$ (i.e., the full subcategory of objects whose λ^a weight components are zero unless $\langle \check{\rho}^a, \lambda^a \rangle \leq i$) for i satisfying $\lambda_1^a, \lambda_2^a \in \lambda_i^a$. The category $\mathcal{O}_i$ is highest weight with coideal finite poset, and (a4) follows. (a5) is just a rephrasing of a classical observation that there are no extensions between Verma modules with generic highest weights.

In Section 4 we will see that the objects $\Delta^{\mathbb{F}}(\lambda^a) \in \mathbb{F}$ (with $\lambda^a \in \Lambda^a$) advertised in Section 1.7.1 are also admissible. It is not clear whether in our main case of interest, when $\mathcal{C} = \mathbb{F}$, all objects $\Delta(\gamma)$ are finitely generated. So we do not require this, which leads to some technical complications later in this section.

3.1.2. Commutative algebra constructions. Before we proceed to our discussion of admissible modules we need to make some remarks on various commutative algebra constructions to be used below. We also introduce some notation. Let $\mathbb{R}$ denote a $\mathbb{Z}$ -graded Noetherian ring.

For a graded $\mathbb{R}$ -module M and a graded $\mathbb{R}$ -algebra $\tilde{\mathbb{R}}$ we will write $M_{\tilde{\mathbb{R}}} := \tilde{\mathbb{R}} \otimes_{\mathbb{R}} M$.

For a graded quotient $\mathbb{R}_0$ of $\mathbb{R}$, we will write $\mathbb{F}_0$ for the full graded fraction ring of $\mathbb{R}_0$, where we invert all homogeneous nonzero divisors, so that we have an inclusion of graded rings $\mathbb{R}_0 \hookrightarrow \mathbb{F}_0$. If $\mathbb{R}_0 = \mathbb{R}$, we write $\mathbb{F}$ instead of $\mathbb{F}_0$.

We will also need graded localizations of $\mathbb{R}_0$. An example of a graded localization is as follows: for a graded prime ideal $\mathfrak{p} \subset \mathbb{R}$, we set $\mathbb{R}_{\mathfrak{p}}$ to be the fraction ring of $\mathbb{R}$, where we invert all homogeneous elements not in $\mathfrak{p}$.

For example, suppose that $\mathfrak{p}$ has height 1, equivalently, is generated by a single irreducible homogeneous elements. Then every homogeneous ideal in $\mathbb{R}_{\mathfrak{p}}$ is principal and every graded $\mathbb{R}_{\mathfrak{p}}$ -module is the direct sum of cyclic modules. This is proved by a direct analog of the usual arguments.

Now we define the notion of *graded depth* for a graded finitely generated $\mathbb{R}'_0$ -module M , where $\mathbb{R}'_0$ is a graded localization of a graded quotient of $\mathbb{R}$. By definition, it is the maximal length of a regular sequence in $\mathbb{R}'_0$ consisting of homogeneous elements that is a regular sequence in M .

A standard argument, see e.g. [22, Lemma 15.24.18] shows the following:

Lemma 3.1.3. *Let $\mathbb{R}'$ be a graded localization of $\mathbb{R}$ and M be a finitely generated graded $\mathbb{R}'$ -module. Suppose M is of graded depth ≥ 2 . Then the following hold:*

- (1) $M_{\mathfrak{p}}$ is graded free for any graded height 1 prime ideal $\mathfrak{p} \subset \mathbb{R}'$.
- (2) $M = \bigcap_{\mathfrak{p}} M_{\mathfrak{p}}$, where the intersection is in $M_{\mathbb{F}}$ and it is taken over all graded height 1 prime ideals in $\mathbb{R}$.

3.1.3. Finite generation of Hom's.

Proposition 3.1.4. *Assume the objects $\Delta(\gamma)$ are admissible in the sense of Definition 3.1.1. Let M, N be two objects of $\mathcal{C}^\heartsuit$ filtered by base changes of the objects $\Delta(\gamma)$ to graded quotients of R . Then $\mathrm{grHom}_{\mathcal{C}}(M, N)$ is finitely generated over R .*

Proof. It is enough to consider the case when $M = \Delta(\gamma_1), N = \Delta_{R_0}(\gamma_2) := R_0 \otimes_R \Delta(\gamma_2)$. Note that

$$\gamma_2^{-1}(R_0 \otimes_R^L \Delta(\gamma_2)) = R_0 \otimes_R^L \Delta(0).$$

By (a1), $\Delta(\gamma)$'s are flat over R , so the derived tensor product is the same as the usual one.

So, we can assume that $N = \Delta_{R_0}(0)$. From (a2), it follows that N coincides with the limit $\varprojlim_{n \rightarrow \infty} N/\mathfrak{m}^n N$ taken in the abelian category of rational representations of T_d and hence also in $\mathcal{C}^\heartsuit$. Since by (a3), $\Delta(\gamma_1)$ also lies in $\mathcal{C}^\heartsuit$, so $\mathrm{grHom}_{\mathcal{C}^\heartsuit}(M, N)$ is the inverse limit $\varprojlim_n \mathrm{grHom}_{\mathcal{C}}(M, N/\mathfrak{m}^n N)$ (in the category of graded R -modules).

Note that

$$\mathrm{grHom}_{\mathcal{C}^\heartsuit}(M, N_{\mathbb{K}}) = \mathrm{grHom}_{\mathcal{C}_{\mathbb{K}}^\heartsuit}(M_{\mathbb{K}}, N_{\mathbb{K}}).$$

(a4) implies that the right hand side is finite dimensional. In particular, the grading coming from the T_d -action is bounded from below. It follows that the grading on

$$\mathrm{grHom}_{\mathcal{C}}(M, N) = \varprojlim_{n \rightarrow \infty} \mathrm{grHom}_{\mathcal{C}}(M, N/\mathfrak{m}^n N)$$

is bounded from below. Again using that $\mathrm{grHom}_{\mathcal{C}}(M, N_{\mathbb{K}})$ is finite dimensional and applying the graded Nakayama lemma several times, we arrive at the claim of the proposition. $\square$

3.1.4. Homs and Exts vs localization.

Proposition 3.1.5. *We continue to assume that the objects $\Delta(\gamma)$ are admissible in the sense of Definition 3.1.1. Suppose that R'_0 is a localization of a quotient R_0 of R by homogeneous non-zero divisors, and $M, N \in \mathcal{C}^\Delta$. Then the natural homomorphism*

$$(3.1) \quad R'_0 \otimes_{R_0} \mathrm{grHom}_{\mathcal{C}_{R_0}}(M_{R_0}, N_{R_0}) \rightarrow \mathrm{grHom}_{\mathcal{C}_{R'_0}}(M_{R'_0}, N_{R'_0}),$$

is an isomorphism, while the natural homomorphism

$$(3.2) \quad R'_0 \otimes_{R_0} \mathrm{grExt}_{\mathcal{C}_{R_0}}^1(M_{R_0}, N_{R_0}) \rightarrow \mathrm{grExt}_{\mathcal{C}_{R'_0}}^1(M_{R'_0}, N_{R'_0})$$

is injective.

Proof. We note first of all, that a module in $\mathcal{C}_{R'_0}^\heartsuit$ is the same thing as a module in $\mathcal{C}_{R_0}^\heartsuit$ where all homogeneous $b \in R_0$ such that $b^{-1} \in R'_0$ act by invertible operators. So the targets in (3.1) and (3.2) do not change if we replace the category $\mathcal{C}_{R'_0}$ with $\mathcal{C}_{R_0}$.

We start by proving that (3.2) is injective. Since N_{R_0} is flat over R_0 and in particular nonzero divisors in R_0 are nonzero divisors in N_{R_0} , (3.1) is injective. We use a version of a 5-lemma and induction on the number of terms in the filtration of M_{R_0} to reduce to the case when $M = \Delta(\gamma)$ for some γ and then to the case when $\gamma = 0$ by applying the equivalence γ^{-1} – notice that $\Delta_{R'_0}(\gamma) \cong \gamma(\Delta_{R'_0}(0))$ since γ commutes with taking colimits.

Let $0 \rightarrow N_{R_0} \rightarrow \tilde{N} \rightarrow \Delta_{R_0}(0) \rightarrow 0$ be a short exact sequence in $\mathcal{C}_{R_0}^\heartsuit$. Let κ be the class of this extension. We need to show that if the short exact sequence

$$0 \rightarrow N_{R'_0} \rightarrow \tilde{N}_{R'_0} \rightarrow \Delta_{R'_0}(0) \rightarrow 0$$

induced by κ splits in $\mathcal{C}_{R'_0}^\heartsuit$ then $b\kappa$ is 0 in $\text{grExt}^1(\Delta_{R_0}(0), N_{R_0})$ for some homogeneous nonzero divisor $b \in R_0$ such that $b^{-1} \in R'_0$. So let π denote the projection $\tilde{N} \rightarrow \Delta_{R_0}(0)$ and let ι denote a splitting $\Delta_{R'_0}(0) \rightarrow \tilde{N}_{R'_0}$. Note that nonzero divisors in R_0 are also nonzero divisors in $\tilde{N}$, so $\tilde{N} \hookrightarrow \tilde{N}_{R'_0}$. And $\Delta_{R_0}(0)$ is generated by finitely many homogeneous elements. So we can find a homogeneous nonzero divisor $b \in R_0$ such that $b^{-1} \in R'_0$ and the image of $b\iota$ is in $\tilde{N}_0$. This precisely means that the extension class $b\kappa$ is trivial. So we have proved that (3.2) is injective.

To prove that (3.1) is an isomorphism, we again argue by induction on the length of the filtration of M and use the 5-lemma itself (possible since we already proved that (3.2) is injective) to reduce to the case when $M = \Delta(\gamma)$. Then we again can assume that $\gamma = 0$. The claim that (3.1) is an isomorphism follows similarly to the previous paragraph thanks to the fact that $\Delta(0)$ is finitely generated. $\square$

3.1.5. Properties of Hom spaces. We want to describe some properties of Hom's in $\mathcal{C}_{R_0}^\Delta$, where R_0 is a graded quotient of a graded localization of R . The following is elementary.

Lemma 3.1.6. *We continue to assume that the collection $\Delta(\gamma)$ is weakly admissible. Let R_0 be a quotient of R . The following claims are true:*

- (1) *If $a \in R_0$ is a homogeneous nonzero divisor, then a is not a zero divisor in $\text{grHom}_{\mathcal{C}}(M_{R_0}, N_{R_0})$ for all $M, N \in \mathcal{C}^\Delta$, where the meaning of M_{R_0}, N_{R_0} are base changes of M, N to R_0 .*
- (2) *Let R' be a graded localization of R . Let $M, N \in \mathcal{C}_{R'}^\Delta$. Then $\text{grHom}_{\mathcal{C}_{R'}}(M, N)$ is finitely generated and has graded depth at least 2.*

Proof. (1) is straightforward.

Let us prove part (2). Let $a, b \in R'$ be a regular sequence of homogeneous elements. Set $R_0 = R'/(b)$. It follows from Proposition 3.1.4 that the R' -modules $\text{grHom}_{\mathcal{C}_{R'}}(M_0, N_0)$ and $\text{grHom}_{\mathcal{C}_{R'}}(M, N)$ are finitely generated. We have

$$\text{grHom}_{\mathcal{C}_{R'}}(M, N)/b \text{grHom}_{\mathcal{C}_{R'}}(M, N) \hookrightarrow \text{grHom}_{\mathcal{C}_{R_0}}(M_0, N_0).$$

From here and part (1) we see that a, b form a regular sequence in M , which finishes the proof of (2). $\square$

3.2. Rouquier-Soergel functors in general. In this section we will present modifications of results from [18, Section 3].

3.2.1. Definition. Let $\mathcal{C}$ and $\Delta(\gamma), \gamma \in \Gamma$, be as in Section 3.1. We assume that the objects $\Delta(\gamma)$ are admissible, i.e., satisfy conditions (a1)-(a5) from Section 3.1.

Now let $\underline{\mathcal{C}}$ be another graded R -linear category as in Section 1.2.1. In fact, in our applications it will be the category associated to the affine W-algebra, see Section 1.5. We choose a collection of objects $\underline{\Delta}(\gamma) \in \underline{\mathcal{C}}$. Recall that $\mathbb{F}$ stands for the full graded localization of R . We assume, in addition that:

- (b) $\text{grExt}_{\mathcal{C}_{\mathbb{F}}}^1(\Delta_{\mathbb{F}}(0), \Delta_{\mathbb{F}}(0)) = 0$.

- (c1) All $\underline{\Delta}(\gamma)$ are finitely generated and flat over R with PBW grading bounded from below.
- (c2) $\text{grExt}_{\underline{\mathcal{C}}}^i(\underline{\Delta}(\gamma_1), \underline{\Delta}(\gamma_2))$ is a finitely generated R -module for all $\gamma_1, \gamma_2 \in \Gamma, i = 0, 1$.
- (c3) The objects $\underline{\Delta}_{\mathbb{F}}(\gamma)$ are simple and pair-wise non-isomorphic. Moreover,

$$(3.3) \quad \text{grExt}_{\underline{\mathcal{C}}_{\mathbb{F}}}^1(\underline{\Delta}_{\mathbb{F}}(\gamma_1), \underline{\Delta}_{\mathbb{F}}(\gamma_2)) \neq 0$$

for all $\gamma_1 \neq \gamma_2$.

- (c4) For any graded localization R' of R and any objects $\underline{M}, \underline{N} \in \underline{\mathcal{C}}^{\Delta}$, we have

$$R' \otimes_R \text{grExt}_{\underline{\mathcal{C}}}^i(\underline{M}, \underline{N}) \xrightarrow{\sim} \text{grExt}_{\underline{\mathcal{C}}_{R'}}^i(\underline{M}_{R'}, \underline{N}_{R'}), i = 0, 1.$$

- (c5) For any graded quotient R_0 of R and any $\gamma \in \Gamma$, we have

$$(3.4) \quad R_0 \xrightarrow{\sim} \text{grHom}_{\underline{\mathcal{C}}_{R_0}}(\underline{\Delta}_{R_0}(\gamma), \underline{\Delta}_{R_0}(\gamma))$$

We are going to present a modified version of the definition from [18, Section 3.1].

Definition 3.2.1. Let $\pi : \mathcal{C} \rightarrow \underline{\mathcal{C}}$ be an exact graded R -linear functor that is left t-exact, so that the 0-th cohomology functor commutes with filtered colimits of monomorphisms. We say that is a *Rouquier-Soergel functor* (abbreviated as an *RS functor*) if the following hold:

- (RS1) $\pi(\Delta(\gamma)) \cong \underline{\Delta}(\gamma)$ for all $\gamma \in \Gamma$.
- (RS2) (RS1) combined with (a3) implies that $\pi(\mathbb{K} \otimes_R \Delta(\gamma)) \cong \mathbb{K} \otimes_R \underline{\Delta}(\gamma)$. We require that the map

$$\text{grHom}_{\mathcal{C}^{\heartsuit}}(\mathbb{K} \otimes_R \Delta(\gamma), \mathbb{K} \otimes_R \Delta(\gamma')) \rightarrow \text{grHom}_{\underline{\mathcal{C}}^{\heartsuit}}(\mathbb{K} \otimes_R \underline{\Delta}(\gamma), \mathbb{K} \otimes_R \underline{\Delta}(\gamma'))$$

is injective for all γ, γ' .

Remark 3.2.2. In [18, Section 3.1] a morally weaker version of (RS2) was used: instead of requiring the faithfulness at the point 0, it was required to hold after specializing to height ≤ 2 prime ideals. We use the stronger version of (RS2) here because it is easy to check it in the cases of interest for us and is less technical.

Lemma 3.2.3. *For any graded localization R'_0 of a graded quotient of R by nonzero divisors, we have*

$$\pi(R'_0 \otimes_R \Delta(\gamma)) \cong R'_0 \otimes_R \underline{\Delta}(\gamma).$$

Proof. We use a spectral sequence argument in the case when R'_0 is a graded quotient and the condition that the zero cohomology of π commutes with filtered colimits of monomorphisms for the general case. $\square$

3.2.2. Example: functor Red. We take $\Gamma = \Lambda^a, \mathcal{C} = \mathcal{O}$ and $\underline{\mathcal{C}} = \mathcal{C}$. Recall the functor $\text{Red} : \mathcal{O}^{\heartsuit} \rightarrow \mathcal{C}^{\heartsuit}$, see Section 1.5.2 The main result of this section is the following proposition.

Proposition 3.2.4. *The following claims hold:*

- (1) *The objects $\Delta(\lambda^a)$ satisfy condition (b).*
- (2) *The objects $\underline{\Delta}(\lambda^a)$ satisfy conditions (c1)-(c5).*
- (3) *The functor $\text{Red} : \mathcal{O} \rightarrow \mathcal{C}$ is an RS functor.*

Proof. (1) is a classical fact that a Verma module in the category $\mathcal{O}$ does not have higher self-extensions.

(2): (c1) follows directly from the construction in Section 1.5.1 and the PBW theorem for W-algebras, Theorem 1.4.4.

To prove (c2) and (c4) we argue as follows.

For $n \in \mathbb{Z}$, let $\mathcal{C}_{R', \leq n}^\heartsuit$ denote the full subcategory of $\mathcal{C}_{R'}^\heartsuit$ of all modules with energy degrees bounded from above by n . It is closed under extensions, so

$$\mathrm{grExt}_{\mathcal{C}_{R', \leq n}^\heartsuit}^1(\underline{\Delta}_{R'}(\lambda_1^a), \underline{\Delta}_{R'}(\lambda_2^a)) \xrightarrow{\sim} \mathrm{grExt}_{\mathcal{C}_{R'}}^1(\underline{\Delta}_{R'}(\lambda_1^a), \underline{\Delta}_{R'}(\lambda_2^a)).$$

We can produce a projective object P in $\mathcal{C}_{R', \leq n}^\heartsuit$ surjecting onto $\underline{\Delta}_{R'}(\lambda_1^a)$ as follows. We can assume without loss of generality that $\lambda_1^a = (\lambda_1, 0)$. Let $J'_{>n}$ denote the closed left ideal in $\mathcal{W} \widehat{\otimes} R'$ topologically generated by elements x_i^j with $i > n$. Then $P := [\mathcal{W} \widehat{\otimes} R'] / J'_{>n}$ is a projective object, it represents the functor $\mathcal{C}_{R', \leq n}^\heartsuit \rightarrow R'$ -grmod sending M to its energy graded component of degree 0. Let $\underline{\Delta}_i$ denote the quotient $[\mathcal{W} \widehat{\otimes} R'] / J'_{>0}$, where the generator is in energy degree i (and PBW degree 0). Then P surjects onto $\underline{\Delta}_0$ and the kernel admits a finite filtration by $\underline{\Delta}_i$ with $0 < i \leq n$. So we have epimorphisms $P \twoheadrightarrow \underline{\Delta}_0 \twoheadrightarrow \underline{\Delta}(\lambda_1^a)$. Let K_0 denote the kernel of $\underline{\Delta}_0 \twoheadrightarrow \underline{\Delta}(\lambda_1^a)$ and K denote the kernel of $P \twoheadrightarrow \underline{\Delta}(\lambda_1^a)$.

(c2) reduces to showing that $\mathrm{grHom}_{\mathcal{C}_{R'}}(?, \underline{\Delta}(\lambda_2^a))$ is finitely generated over R' , where $?$ is K_0 or $\underline{\Delta}_i$. We will prove the finite generation for K_0 , the case of $\underline{\Delta}_i$ is similar and easier. Note that $K_0 = [(\mathcal{W} \widehat{\otimes} R') / J_{>0}] \otimes_{R_h \mathcal{Z}(\mathfrak{g}) \otimes R'} M$, where M is the kernel of the homomorphism $R_h \mathcal{Z}(\mathfrak{g}) \otimes R' \rightarrow R'$ base-changed from γ_{λ_1} . So

$$(3.5) \quad \mathrm{grHom}_{\mathcal{C}}(K_0, \underline{\Delta}_{R'}(\lambda_2)) \xrightarrow{\sim} \mathrm{grHom}_{R_h \mathcal{Z}(\mathfrak{g}) \otimes R'}(M, \underline{\Delta}_{R'}(\lambda_2)_0^{J'_{>0}}),$$

where the superscript $J'_{>0}$ stands for the annihilator of this left ideal and the subscript 0 is for the corresponding energy graded component. The right hand side of (3.5) embeds into $\mathrm{grHom}_{R_h \mathcal{Z}(\mathfrak{g}) \otimes R'}(M, \underline{\Delta}_{R'}(\lambda_2)_0)$. The latter is a finitely generated graded R' -module because, as can be seen from Corollary 1.4.4, the energy graded components of $\underline{\Delta}_{R'}(\lambda_2)$ are finitely generated over R' . This finishes the proof of (c2).

Now we proceed to proving (c4). We will handle the case $i = 1$ because $i = 0$ is easier. We form the modules P, M over R_0 . It is enough to show that

$$R' \otimes \mathrm{grHom}_{\mathcal{C}_{R_0}}(?, \underline{\Delta}_{R_0}(\lambda_2^a)) \xrightarrow{\sim} \mathrm{grHom}_{\mathcal{C}_{R'}}(?_{R'}, \underline{\Delta}_{R'}(\lambda_2^a)), \quad ? = P, M.$$

The case of P follows because both $P, P_{R'}$ represent the functor of taking the 0-th energy graded component.

Now we handle the more difficult case of M . Let P_i denote the projective object representing the functor on $\mathcal{C}_{R_0}$ of taking i -th energy graded component, so that $P_0 = P$. Notice that M admits a surjection from a finite direct sum of the objects P_i : we filter M by M_0 and $\underline{\Delta}_i$ and cover M_0 by the direct sum of several copies of P_0 . Let P^1 denote the corresponding direct sum of projectives surjecting onto M . The kernel of $P^1 \twoheadrightarrow M$ admits a finite filtration by $\underline{\Delta}_j$'s and a module of the form

$$[(\mathcal{W} \widehat{\otimes} R') / J_{>0}] \otimes_{R_h \mathcal{Z}(\mathfrak{g}) \otimes R_0} N,$$

where N is a finitely generated graded $R_h \mathcal{Z}(\mathfrak{g}) \otimes R_0$ -module. In particular, the kernel admits an epimorphism from P^2 , a finite direct sum of P_j 's. Using the presentation $P^2 \rightarrow P^1 \rightarrow M$ we reduce the case of M to the case of P considered above.

(c5) follows similarly to (3.5) from (1.11).

Now we show (c3). That the objects $\underline{\Delta}_{\mathbb{F}}(\lambda^a)$ are pairwise nonisomorphic simple – one can see this, for example, from Proposition 1.5.5 combined with Proposition 1.5.4. To prove (3.3) suppose that $\gamma_1 = (\lambda_1, n_1), \gamma_2 = (\lambda_2, n_2)$. If $n_1 > n_2$ or $n_1 = n_2$ and $\lambda_1 \neq \lambda_2$, then the Ext vanishing follows from the highest weight theory. And if $n_1 < n_2$, then we apply the functor D from Section 1.5.5 to reduce to the previous case.

Finally, we need to check conditions (RS1) and (RS2) for the functor Red . (RS1) follows from Proposition 1.5.4. To prove (RS2) notice that (1.7) implies that for $M \in \mathbf{O}_{\mathbb{K}}$ we have $\text{Red}(M) \cong M|_{S^+}$, where $S^+ = (et^{-1} + \mathfrak{z}_{\mathfrak{g}}(f)[[t]])dt$. Notice that IS^+ is open in $\mathfrak{i}^{\perp} = (\mathfrak{n}t^{-1} \oplus \mathfrak{g}[[t]])dt$. So restricting to S^+ cannot kill a homomorphism to an object of $\mathbf{O}_{\mathbb{K}}$ that is a free module over $\mathbb{K}[\mathfrak{i}^{\perp}]$ such as an object of $\mathbf{O}_{\mathbb{K}}^{\Delta}$. This finishes checking (RS2) and the proof of the proposition. $\square$

3.2.3. Basic properties. Let R_0 be a graded quotient of R . We write $\Delta_{R_0}(\gamma)$ for $R_0 \otimes_R \Delta(\gamma)$. By Lemma 3.2.3, $\pi(\Delta_{R_0}(\gamma)) \cong \underline{\Delta}_{R_0}(\gamma)$. For $M \in \mathcal{C}^{\Delta}$, we write $\underline{M}$ for its image under π .

Proposition 3.2.5. *The following claims hold:*

- (1) *For any graded quotient R_0 of R , the functor π is faithful on $\mathcal{C}_{R_0}^{\Delta}$ (the full subcategory of $\mathcal{C}_{R_0}^{\heartsuit}$ of all objects admitting finite filtrations by $\Delta_{R_0}(\gamma)$'s).*
- (2) *π is fully faithful on $\mathcal{C}^{\Delta}$.*

Proof. First of all, note that by Proposition 3.1.4 all Hom spaces involved are finitely generated over R .

(1): one uses three reduction steps, first (I), and then interchangingly (II) and (III) to reduce to the case $R_0 = \mathbb{K}$, where our claim is (RS2):

I) Reduction to $M = \Delta(\gamma_1), N = \Delta(\gamma_2)$ using exact (on the left) sequences of Hom's.

II) to the case when R_0 is a domain using a filtration $R_0 = I_0 \supset I_1 \supset \dots \supset I_k = \{0\}$ by graded ideals on R_0 whose successive quotients are graded domain quotients $R/\mathfrak{p}_i$ of R_0 . Thanks to the flatness of M, N over R , we get filtrations on M_{R_0}, N_{R_0} with successive quotients $M_{R/\mathfrak{p}_i}, N_{R/\mathfrak{p}_i}$. We notice that $\text{grHom}_{\mathcal{C}^{\heartsuit}}(M_{R/\mathfrak{p}_i}, N_{R/\mathfrak{p}_j})$ is 0 if $\mathfrak{p}_j \not\subset \mathfrak{p}_i$ and is equal to $\text{grHom}_{\mathcal{C}_{R/\mathfrak{p}_j}^{\heartsuit}}(M_{R/\mathfrak{p}_j}, N_{R/\mathfrak{p}_j})$ else. Then we again use exact (on the left) sequences of Hom's to reduce to the case when $R_0 = R/\mathfrak{p}$ for a graded prime ideal $\mathfrak{p}$.

III) Reducing the dimension of R_0 . Namely, suppose R_0 is a domain and $a \in R_0$ is a nonzero homogeneous element. Let $R_1 := R_0/(a)$. Then thanks (1) of Lemma 3.1.6, a is a nonzero divisor in $\text{grHom}_{\mathcal{C}^{\heartsuit}}(M_{R_0}, N_{R_0})$. So we have an inclusion

$$(3.6) \quad \text{grHom}_{\mathcal{C}^{\heartsuit}}(M_{R_0}, N_{R_0})/a \text{grHom}_{\mathcal{C}^{\heartsuit}}(M_{R_0}, N_{R_0}) \hookrightarrow \text{grHom}_{\mathcal{C}^{\heartsuit}}(M_{R_1}, N_{R_1}).$$

Similarly, we have

$$(3.7) \quad \text{grHom}_{\mathcal{C}^{\heartsuit}}(\underline{M}_{R_0}, \underline{N}_{R_0})/a \text{grHom}_{\mathcal{C}^{\heartsuit}}(\underline{M}_{R_0}, \underline{N}_{R_0}) \hookrightarrow \text{grHom}_{\mathcal{C}^{\heartsuit}}(\underline{M}_{R_1}, \underline{N}_{R_1}).$$

We write π_{R_i} for the homomorphisms

$$\text{grHom}_{\mathcal{C}^{\heartsuit}}(M_{R_i}, N_{R_i}) \rightarrow \text{grHom}_{\mathcal{C}^{\heartsuit}}(\underline{M}_{R_i}, \underline{N}_{R_i}), i = 0, 1.$$

They intertwine (3.6) and (3.7). So once we know that π_{R_1} is injective, we see that π_{R_0} induces an embedding

$$\mathrm{grHom}_{\mathcal{C}^\heartsuit}(M_{R_0}, N_{R_0})/a \mathrm{grHom}_{\mathcal{C}^\heartsuit}(M_{R_0}, N_{R_0}) \hookrightarrow \mathrm{grHom}_{\underline{\mathcal{C}}^\heartsuit}(\underline{M}_{R_0}, \underline{N}_{R_0})/a \mathrm{grHom}_{\underline{\mathcal{C}}^\heartsuit}(\underline{M}_{R_0}, \underline{N}_{R_0}).$$

Since this is true for any nonzero homogeneous $a \in R_0$ and the domain of π_{R_0} is a finitely generated graded R_0 -module, we see that the injectivity of π_{R_1} implies that of π_{R_0} .

(2): By (2) of Lemma 3.1.6, both $P := \mathrm{grHom}_{\mathcal{C}}(M, N)$ and $\underline{P} := \mathrm{grHom}_{\underline{\mathcal{C}}}(\underline{M}, \underline{N})$ are of graded depth ≥ 2 . By III) above, we already know that for any nonzero homogeneous element $a \in R$, $\pi_R : P \rightarrow \underline{P}$ induces an embedding $P/aP \hookrightarrow \underline{P}/a\underline{P}$. Thanks to Lemma 3.1.3, to show that π_R is an isomorphism it is sufficient to prove that the map $\mathbb{F} \otimes_R P \rightarrow \mathbb{F} \otimes_R \underline{P}$ is an isomorphism. By Proposition 3.1.5,

$$P_{\mathbb{F}} \xrightarrow{\sim} \mathrm{grHom}_{\mathcal{C}_{\mathbb{F}}}(M_{\mathbb{F}}, N_{\mathbb{F}}), \underline{P}_{\mathbb{F}} \xrightarrow{\sim} \mathrm{grHom}_{\underline{\mathcal{C}}_{\mathbb{F}}}(\underline{M}_{\mathbb{F}}, \underline{N}_{\mathbb{F}}).$$

So we need to show that

$$\pi_{\mathbb{F}} : \mathrm{grHom}_{\mathcal{C}_{\mathbb{F}}}(M_{\mathbb{F}}, N_{\mathbb{F}}) \xrightarrow{\sim} \mathrm{grHom}_{\underline{\mathcal{C}}_{\mathbb{F}}}(\underline{M}_{\mathbb{F}}, \underline{N}_{\mathbb{F}}).$$

Thanks to (a5) and (b), we reduce this to the case when $M = N = \Delta(\gamma)$ for some γ . By (c3), $\mathbb{F} \xrightarrow{\sim} \mathrm{grHom}_{\mathcal{C}_{\mathbb{F}}}(\underline{M}_{\mathbb{F}}, \underline{N}_{\mathbb{F}})$ and by part (1) of this lemma, $\pi_{\mathbb{F}}$ is injective. It follows that $\pi_{\mathbb{F}}$ is an isomorphism finishing the proof. $\square$

Corollary 3.2.6. *For all $M, N \in \mathcal{C}^\Delta$, the following claims hold:*

- (1) π induces an inclusion $\mathrm{grExt}_{\mathcal{C}}^1(M, N) \hookrightarrow \mathrm{grExt}_{\underline{\mathcal{C}}}^1(\underline{M}, \underline{N})$.
- (2) There is a homogeneous nonzero element $a \in R$ depending on M, N such that a annihilates the R -torsion submodule of $\mathrm{grExt}_{\underline{\mathcal{C}}}^1(\underline{M}, \underline{N})$. For such a we have $a \mathrm{grExt}_{\mathcal{C}}^1(M, N) = 0$.

Proof. (1) follows directly from (2) of Proposition 3.2.5. From (1) and (c2), it follows that $\mathrm{grExt}_{\mathcal{C}}^1(M, N), \mathrm{grExt}_{\underline{\mathcal{C}}}^1(\underline{M}, \underline{N})$ are finitely generated R -modules. From (b) and Proposition 3.1.5 it follows that $\mathrm{grExt}_{\mathcal{C}}^1(\Delta(\gamma), \Delta(\gamma))$ is torsion. Similarly, (a4) implies that $\mathrm{grExt}_{\underline{\mathcal{C}}}^1(\Delta(\gamma_1), \Delta(\gamma_2))$ is torsion if $\gamma_1 \neq \gamma_2$. Now (2) follows from (1). $\square$

Lemma 3.2.7. *Let R' be a graded localization of R . The functor $\pi : \mathcal{C}^\Delta \rightarrow \underline{\mathcal{C}}^\Delta$ induces a fully faithful embedding $\mathcal{C}_{R'}^\Delta \rightarrow \underline{\mathcal{C}}_{R'}^\Delta$. In particular, for $M, N \in \mathcal{C}_{R'}^\Delta$, we have*

$$\mathrm{grExt}_{\mathcal{C}_{R'}}^1(M_{R'}, N_{R'}) \hookrightarrow \mathrm{grExt}_{\underline{\mathcal{C}}_{R'}}^1(\underline{M}_{R'}, \underline{N}_{R'}).$$

Proof. We have an induced functor thanks to $\pi(\Delta_{R'}(\gamma)) \cong \underline{\Delta}_{R'}(\gamma)$, which follows from Lemma 3.2.3. Denote it by $\pi_{R'}$. By the Hom part of Proposition 3.1.5, the analog of (1) of Proposition 3.2.5 holds for $\pi_{R'}$. The analog of (2) follows for the same reason as in the proof of that proposition. The claim about the embedding of grExt^1 follows. $\square$

3.2.4. Further properties. Our goal in this section is to obtain a version of results in [18, Section 3.2.2]: we want to compute the graded R -module $\mathrm{grExt}_{\mathcal{C}^\heartsuit}^1(M, N)$. First, we need to describe an abelian (but not Serre) subcategory of $\underline{\mathcal{C}}^\heartsuit$. Namely, we define the subcategory $\underline{\mathcal{C}}_{gss}^\heartsuit$ (with “gss” abbreviating “generically semisimple”) as the subcategory of all modules M such that $M_{\mathbb{F}}$ is a finite direct sum of $\underline{\Delta}_{\mathbb{F}}(\gamma_i)$ ’s. To show that it is closed

under finite direct sums, kernels and cokernels is an exercise (based on the exactness of localization). Also

$$(3.8) \quad \text{grExt}_{\underline{\mathcal{C}}_{gss}}^1(\bullet, ?) \text{ is the } R\text{-torsion submodule of } \text{grExt}_{\underline{\mathcal{C}}}^1(\bullet, ?).$$

Lemma 3.2.8. *We have $\pi(\mathcal{C}^\Delta) \subset \underline{\mathcal{C}}_{gss}^\heartsuit$.*

Proof. Let $M \in \mathcal{C}^\Delta$. From (2) of Corollary 3.2.6 it follows that there is a homogeneous element $a \in R$ (killing all extensions between terms of a Δ -filtration in M) and an inclusion from a direct sum of $\Delta(\gamma)$'s to M whose cokernel is killed by a . Using $\pi(\Delta(\gamma)) \cong \underline{\Delta}(\gamma)$ we finish the proof. $\square$

So our task is to describe $\text{grExt}_{\mathcal{C}}^1(M, N)$ as a submodule in $\text{grExt}_{\underline{\mathcal{C}}_{gss}}^1(\underline{M}, \underline{N})$, where $\underline{M} = \pi(M)$, $\underline{N} = \pi(N)$. By the definition of $\underline{\mathcal{C}}_{gss}^\heartsuit$, the graded R -module $\text{grExt}_{\underline{\mathcal{C}}_{gss}}^1(\underline{M}, \underline{N})$ is torsion. By (c2), this module is finitely generated. So we can find a nonzero homogeneous element a annihilating $\text{grExt}_{\underline{\mathcal{C}}_{gss}}^1(\underline{M}, \underline{N})$, and hence $\text{grExt}_{\mathcal{C}}^1(M, N)$. Set $R_0 := R/(a)$.

Therefore we have the following commutative diagram with exact rows and vertical arrows induced by π and (a) stands for the image of the multiplication by a in a R -module:

$$(3.9) \quad \begin{array}{ccccccc} 0 & \longrightarrow & \text{grHom}_{\mathcal{C}}(M, N)/(a) & \longrightarrow & \text{grHom}_{\mathcal{C}_{R_0}}(M_{R_0}, N_{R_0}) & \longrightarrow & \text{grExt}_{\mathcal{C}}^1(M, N) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \text{grHom}_{\underline{\mathcal{C}}}(\underline{M}, \underline{N})/(a) & \longrightarrow & \text{grHom}_{\underline{\mathcal{C}}_{R_0}}(\underline{M}_{R_0}, \underline{N}_{R_0}) & \longrightarrow & \text{grExt}_{\underline{\mathcal{C}}_{gss}}^1(\underline{M}, \underline{N}) \longrightarrow 0 \end{array}$$

Here the left vertical arrow is an isomorphism by (2) of Proposition 3.2.5.

Remark 3.2.9. We have a direct analog of (3.9) for the category $\mathcal{C}_{R'}$, where R' is a graded localization of R , cf. Lemma 3.2.7.

We have the following analog of [18, Proposition 3.2.2].

Lemma 3.2.10. *Recall that $R_0 := R/(a)$. The map*

$$(3.10) \quad \text{grHom}_{\mathcal{C}_{R_0}}(M_{R_0}, N_{R_0}) \rightarrow \text{grHom}_{\underline{\mathcal{C}}_{R_0}}(\underline{M}_{R_0}, \underline{N}_{R_0})$$

is injective and the cokernel has positive graded depth.

Proof. The injectivity follows by applying (1) of Proposition 3.2.5 to R_0 . To prove the positive graded depth claim, take a homogeneous nonzero divisor $b \in R_0$ and set $R_1 := R_0/(b)$. Then we apply (1) of Proposition 3.2.5 to R_1 and use the following commutative diagram with injective (by (1) of Lemma 3.1.6) horizontal arrows:

$$\begin{array}{ccc} \text{grHom}_{\mathcal{C}_{R_0}}(M_{R_0}, N_{R_0})/b \text{grHom}_{\mathcal{C}_{R_0}}(M_{R_0}, N_{R_0}) & \longrightarrow & \text{grHom}_{\mathcal{C}_{R_1}}(M_{R_1}, N_{R_1}) \\ \downarrow & & \downarrow \\ \text{grHom}_{\underline{\mathcal{C}}_{R_0}}(\underline{M}_{R_0}, \underline{N}_{R_0})/b \text{grHom}_{\underline{\mathcal{C}}_{R_0}}(\underline{M}_{R_0}, \underline{N}_{R_0}) & \longrightarrow & \text{grHom}_{\underline{\mathcal{C}}_{R_1}}(\underline{M}_{R_1}, \underline{N}_{R_1}) \end{array}$$

$\square$

Let $\mathfrak{p}_i, i = 1, \dots, k$ be the minimal prime ideals of a , all of them are graded. We set $\mathbb{L} := \bigoplus_{i=1}^k \mathbb{R}_{\mathfrak{p}_i}$, so that $\mathbb{F}_0 := \mathbb{L}/a\mathbb{L}$ is the full graded fraction ring of $\mathbb{R}_0$. The following proposition gives a desired answer for $\text{grExt}_{\mathcal{C}}^1(M, N)$.

Proposition 3.2.11. *The following diagram is Cartesian*

$$(3.11) \quad \begin{array}{ccc} \text{grExt}_{\mathcal{C}}^1(M, N) & \longrightarrow & \text{grExt}_{\mathcal{C}_{\mathbb{L}}}^1(M_{\mathbb{L}}, N_{\mathbb{L}}) \\ \downarrow & & \downarrow \\ \text{grExt}_{\mathcal{C}_{gss}^{\heartsuit}}^1(\underline{M}, \underline{N}) & \longrightarrow & \text{grExt}_{\mathcal{C}_{gss, \mathbb{L}}^{\heartsuit}}^1(\underline{M}_{\mathbb{L}}, \underline{N}_{\mathbb{L}}). \end{array}$$

Proof. Since the cokernel of (3.10) is of positive graded depth, we see that the following diagram is Cartesian

$$(3.12) \quad \begin{array}{ccc} \text{grHom}_{\mathcal{C}_{\mathbb{R}_0}}(M_{\mathbb{R}_0}, N_{\mathbb{R}_0}) & \longrightarrow & \text{grHom}_{\mathcal{C}_{\mathbb{F}_0}}(M_{\mathbb{F}_0}, N_{\mathbb{F}_0}) \\ \downarrow & & \downarrow \\ \text{grHom}_{\mathcal{C}_{\mathbb{R}_0}}(\underline{M}_{\mathbb{R}_0}, \underline{N}_{\mathbb{R}_0}) & \longrightarrow & \text{grHom}_{\mathcal{C}_{\mathbb{F}_0}}(\underline{M}_{\mathbb{F}_0}, \underline{N}_{\mathbb{F}_0}). \end{array}$$

Recall, Proposition 3.1.5 and (c4), that $\text{grHom}_{\mathcal{C}_?}(M_?, N_?), \text{grHom}_{\mathcal{C}_?}(\underline{M}_?, \underline{N}_?)$ commute with graded localization. Combining this with (3.9) and (3.12), we arrive at the following Cartesian diagram:

$$(3.13) \quad \begin{array}{ccc} \text{grExt}_{\mathcal{C}}^1(M, N) & \longrightarrow & \mathbb{L} \otimes_{\mathbb{R}} \text{grExt}_{\mathcal{C}}^1(M, N) \\ \downarrow & & \downarrow \\ \text{grExt}_{\mathcal{C}_{gss}^{\heartsuit}}^1(\underline{M}, \underline{N}) & \longrightarrow & \mathbb{L} \otimes_{\mathbb{R}} \text{grExt}_{\mathcal{C}_{gss}^{\heartsuit}}^1(\underline{M}, \underline{N}). \end{array}$$

It remains to show that the right column of (3.13) coincides with that of (3.11).

According to (c4), $\mathbb{L} \otimes_{\mathbb{R}} \text{grExt}_{\mathcal{C}}^1(\underline{M}, \underline{N}) \xrightarrow{\sim} \text{grExt}_{\mathcal{C}_{\mathbb{L}}}^1(\underline{M}_{\mathbb{L}}, \underline{N}_{\mathbb{L}})$ Thanks to (3.8) and its direct analog over $\mathbb{L}$, we get

$$\mathbb{L} \otimes_{\mathbb{R}} \text{grExt}_{\mathcal{C}_{gss}^{\heartsuit}}^1(\underline{M}, \underline{N}) \xrightarrow{\sim} \text{grExt}_{\mathcal{C}_{gss, \mathbb{L}}^{\heartsuit}}^1(\underline{M}_{\mathbb{L}}, \underline{N}_{\mathbb{L}}).$$

By Lemma 3.2.7, π induces a fully faithful functor $\pi_{\mathbb{L}} : \mathcal{C}_{\mathbb{L}}^{\Delta} \rightarrow \mathcal{C}^{\Delta} \cap \mathcal{C}_{gss}^{\heartsuit}$ and

$$(3.14) \quad \text{grExt}_{\mathcal{C}_{\mathbb{L}}}^1(M_{\mathbb{L}}, N_{\mathbb{L}}) \hookrightarrow \text{grExt}_{\mathcal{C}_{gss, \mathbb{L}}^{\heartsuit}}^1(\underline{M}_{\mathbb{L}}, \underline{N}_{\mathbb{L}}).$$

In particular, the domain is annihilated by a . It follows that it coincides with the cokernel of

$$\text{grHom}_{\mathcal{C}_{\mathbb{L}}}(M_{\mathbb{L}}, N_{\mathbb{L}}) \rightarrow \text{grHom}_{\mathcal{C}_{\mathbb{F}_0}}(M_{\mathbb{F}_0}, N_{\mathbb{F}_0}),$$

cf. Remark 3.2.9. Hence $\mathbb{L} \otimes_{\mathbb{R}} \text{grExt}_{\mathcal{C}}^1(M, N) \xrightarrow{\sim} \text{grExt}_{\mathcal{C}_{\mathbb{L}}}^1(M_{\mathbb{L}}, N_{\mathbb{L}})$. Under the corresponding identifications (3.14) becomes the right vertical map in (3.11). This finishes the proof. $\square$

We will also need the following corollary of (3.9).

Lemma 3.2.12. *For all $\gamma \in \Gamma$ and any graded localization R' of R , we have*

$$(3.15) \quad \mathrm{grExt}_{\mathcal{C}_{R'}}^1(\Delta_{R'}(\gamma), \Delta_{R'}(\gamma)) = 0,$$

$$(3.16) \quad \mathrm{grExt}_{\mathcal{C}_{R', gss}^1}^1(\underline{\Delta}_{R'}(\gamma), \underline{\Delta}_{R'}(\gamma)) = 0.$$

Proof. It is enough to prove (3.16): thanks to Lemma 3.2.7, $\pi_{R'}$ is a full embedding $\mathcal{C}_{R'}^\Delta \hookrightarrow \mathcal{C}_{R', gss}^\Delta$, (3.15) follows from (3.16).

And (3.16) follows from the exactness of the bottom row of (3.9) (cf. Remark 3.2.9), (c4), and (c5) applied to $R_0 = R$ and $R/(a)$. $\square$

3.3. Rouquier-Soergel functors of $\mathfrak{sl}_2$ -type. In this section we define a special class of Rouquier-Soergel functors and check that $\mathrm{Red} : \mathcal{O} \rightarrow \mathcal{C}$ belongs to this class. By $\mathfrak{p} \subset R$ we denote a height one graded prime ideal in R . Let f be an automatically homogeneous generator. We write R' for the graded localization $R_{\mathfrak{p}}$, R_0 for $R/\mathfrak{p}$, and $\mathbb{F}_0$ for the full graded fraction ring of R_0 . Let $R'_k := R'/(f^{k+1})$ so that $R'_0 = \mathbb{F}_0$. Further let $(\mathcal{C}, \Delta(\gamma))$ and $(\underline{\mathcal{C}}, \underline{\Delta}(\gamma))$ and π have the same meaning as in Section 3.2.1.

3.3.1. Prime ideals of fully faithful type.

Definition 3.3.1. We say that $\mathfrak{p}$ is of *fully faithful type* if $\pi_{\mathbb{F}_0} : \mathcal{C}_{\mathbb{F}_0}^\Delta \rightarrow \underline{\mathcal{C}}_{\mathbb{F}_0}^\Delta$ is fully faithful.

The following result should be compared to [?, Proposition 4.42]. We note that our setting is more difficult as the categories of interest are not those of modules over finitely generated R' -algebras.

Lemma 3.3.2. *If $\mathfrak{p}$ is of fully faithful type, then $\pi_{R'}$ is an equivalence $\mathcal{C}_{R'}^\Delta \xrightarrow{\sim} \underline{\mathcal{C}}_{R'}^\Delta$.*

Proof. For $M, N \in \mathcal{C}_{R'}^\Delta$, we can find $k > 0$ such that $a := f^k$ annihilates $\mathrm{grExt}_{\mathcal{C}_{gss, R'}^1}^1(M, N)$.

Thanks to Proposition 3.2.11, it is enough to show that

$$\pi_{R'_k} : \mathcal{C}_{R'_k}^\Delta \rightarrow \underline{\mathcal{C}}_{R'_k}^\Delta$$

is fully faithful for all $k > 0$.

The proof is by induction on k . The base, $k = 0$, is the assumption of the proposition. The step is done considering the long exact sequence of Ext 's for the short exact sequence

$$0 \rightarrow N_{\mathbb{F}_0} \xrightarrow{f^{k-1}} N_{R'_k} \rightarrow N_{R'_{k-1}} \rightarrow 0.$$

Using the 5-lemma, we reduce the proof to showing that

$$(3.17) \quad \mathrm{grExt}_{\mathcal{C}_{R'_k}^1}^1(M_{R'_k}, N_{\mathbb{F}_0}) \hookrightarrow \mathrm{grExt}_{\underline{\mathcal{C}}_{R'_k}^1}^1(\underline{M}_{R'_k}, \underline{N}_{\mathbb{F}_0}).$$

First, notice that we have the following commuting square:

$$(3.18) \quad \begin{array}{ccc} \mathrm{grExt}_{\mathcal{C}_{R'_k}^1}^1(M_{R'_k}, N_{\mathbb{F}_0}) & \longrightarrow & \mathrm{grExt}_{\mathcal{C}_{R'_1}^1}^1(M_{\mathbb{F}_0}, N_{\mathbb{F}_0}) \\ \downarrow & & \downarrow \\ \mathrm{grExt}_{\underline{\mathcal{C}}_{R'_k}^1}^1(\underline{M}_{R'_k}, \underline{N}_{\mathbb{F}_0}) & \longrightarrow & \mathrm{grExt}_{\underline{\mathcal{C}}_{R'_1}^1}^1(\underline{M}_{\mathbb{F}_0}, \underline{N}_{\mathbb{F}_0}). \end{array}$$

The horizontal maps are constructed in the following way. Take a short exact sequence

$$(3.19) \quad 0 \rightarrow N_{\mathbb{F}_0} \rightarrow \tilde{M}_{R'_k} \rightarrow M_{R'_k} \rightarrow 0$$

in $\mathcal{C}_{R'_k}$. Its Ext class is annihilated by f because $N_{\mathbb{F}_0}$ is. So this short exact sequence splits on $fM_{R'_k} \cong M_{R'_{k-1}}$ (an isomorphism is up to a grading shift). Fix an embedding $\iota : fM_{R'_k} \hookrightarrow \tilde{M}_{R'_k}$ splitting $\tilde{M}_{R'_k} \twoheadrightarrow M_{R'_k}$. We send the class of (3.19) to the class of

$$(3.20) \quad 0 \rightarrow N_{\mathbb{F}_0} \rightarrow \tilde{M}_{R'_k} / \text{im } \iota \rightarrow M_{\mathbb{F}_0} \rightarrow 0,$$

note that up to an isomorphism of short exact sequences (3.20) is independent of the choice of ι . Indeed, the difference between two splittings ι, ι_1 is homomorphism $M_{R'_{k-1}} \rightarrow N_{\mathbb{F}_0}$. If ϖ denotes the projection $\tilde{M}_{R'_k} \twoheadrightarrow M_{R'_k}$, then ι, ι_1 are intertwined by the automorphism $\text{id} + \varphi \circ (f\varpi)$ of $\tilde{M}_{R'_k}$, where we view f as an epimorphism $M_{R'_k} \twoheadrightarrow M_{R'_{k-1}}$.

The bottom horizontal map in (3.18) is constructed similarly.

We claim that if (3.20) splits, then so does (3.19), so that the horizontal maps in (3.18) are injective. Indeed, take a splitting $\hat{\iota}$ of (3.20) and take the preimage of $\hat{\iota}(M_{\mathbb{F}_0})$ in $\tilde{M}_{R'_k}$. The projection $\tilde{M}_{R'_k} \twoheadrightarrow M_{R'_k}$ identifies the preimage with $M_{R'_k}$ therefore giving a splitting of (3.19). So, it remains to show that the map $\text{grExt}_{\mathcal{C}_{R'_1}}^1(M_{\mathbb{F}_0}, N_{\mathbb{F}_0}) \rightarrow \text{grExt}_{\mathcal{C}_{R'_1}}^1(\underline{M}_{\mathbb{F}_0}, \underline{N}_{\mathbb{F}_0})$ induced by $\pi_{R'_1}$ is injective.

In order to do this we notice that there is a natural exact sequence

$$(3.21) \quad 0 \rightarrow \text{grExt}_{\mathcal{C}_{\mathbb{F}_0}}^1(M_{\mathbb{F}_0}, N_{\mathbb{F}_0}) \rightarrow \text{grExt}_{\mathcal{C}_{R'_1}}^1(M_{\mathbb{F}_0}, N_{\mathbb{F}_0}) \rightarrow \text{grHom}_{\mathcal{C}_{\mathbb{F}_0}}(M_{\mathbb{F}_0}, N_{\mathbb{F}_0}),$$

where the first map is the inclusion and the second map is induced by the multiplication by f on a short exact sequence $0 \rightarrow N_{\mathbb{F}_0} \rightarrow \tilde{M}_{R'_1} \rightarrow M_{\mathbb{F}_0} \rightarrow 0$. Of course, $\pi_{R'_1}$ intertwines (3.21) with its analog for $\mathcal{C}$. Since $\pi_{\mathbb{F}_0}$ is fully faithful on $\mathcal{C}_{\mathbb{F}_0}^\Delta$, we see that $\text{grExt}_{\mathcal{C}_{\mathbb{F}_0}}^1(M_{\mathbb{F}_0}, N_{\mathbb{F}_0}) \hookrightarrow \text{grExt}_{\mathcal{C}_{\mathbb{F}_0}}^1(\underline{M}_{\mathbb{F}_0}, \underline{N}_{\mathbb{F}_0})$. The injectivity of (3.17) follows finishing the proof. $\square$

3.3.2. Prime ideals of $\mathfrak{sl}_2$ type. Our notation is the same as in the beginning of Section

3.3.1. We consider the following relation on Γ : $\gamma_1 \uparrow_{\mathfrak{p}} \gamma_2$ if $\text{grHom}_{\mathcal{C}_{\mathbb{F}_0}}(\underline{\Delta}_{\mathbb{F}_0}(\gamma_1), \underline{\Delta}_{\mathbb{F}_0}(\gamma_2)) \neq 0$. Let $\sim_{\mathfrak{p}}$ be the equivalence relation on Γ generated by $\uparrow_{\mathfrak{p}}$. We note that thanks to (3.9) if $\gamma_1 \not\sim_{\mathfrak{p}} \gamma_2$, then there no graded Hom's or extensions between $\Delta_{R'}(\gamma_1), \Delta_{R'}(\gamma_2)$ in $\mathcal{C}_{R'}^\heartsuit$ and between $\underline{\Delta}_{R'}(\gamma_1), \underline{\Delta}_{R'}(\gamma_2)$ in $\mathcal{C}_{R', gss}^\heartsuit$. Therefore both $\mathcal{C}_{R'}^\Delta$ and $\mathcal{C}_{R', gss}^\Delta$ split into the direct sums of exact categories indexed by the equivalence classes, Θ , for $\sim_{\mathfrak{p}}$, let $\mathcal{C}_{R', \Theta}^\Delta, \mathcal{C}_{R', \Theta}^\Delta$ denote the summands corresponding to Θ . Note that $\pi_{R'}$ restricts to $\mathcal{C}_{R', \Theta}^\Delta \rightarrow \mathcal{C}_{R', \Theta}^\Delta$. Let $\pi_{R', \Theta}$ denote the restriction.

Definition 3.3.3. We say that $\mathfrak{p}$ is of $\mathfrak{sl}_2$ -type if $\mathcal{C}_{\mathbb{F}_0}^\heartsuit$ is not semisimple, and for every equivalence class Θ such that $\pi_{R', \Theta}$ is not an equivalence of exact categories the following conditions hold:

- (1) Θ consists of exactly of two elements, denote them by γ_1, γ_2 .
- (2) $\text{grHom}_{\mathcal{C}_{R'_k}}(\underline{\Delta}_{R'_k}(\gamma_i), \underline{\Delta}_{R'_k}(\gamma_{3-i})) \cong \mathbb{F}_0$ or 0 for all $k > 0$ and $i = 1, 2$.
- (3) $\text{grHom}_{\mathcal{C}_{\mathbb{F}_0}}(\Delta_{\mathbb{F}_0}(\gamma_i), \Delta_{\mathbb{F}_0}(\gamma_{3-i})) = 0$ for exactly one choice of $i = 1, 2$ (thanks to (3) the other grHom -module is of rank 1). For the choice, where this grHom vanishes we will write $\gamma_{3-i} \uparrow_{\mathfrak{p}} \gamma_i$.

The name $\mathfrak{sl}_2$ -type is chosen because conditions (1)-(4) are modelled after those of the Soergel functor or the identity functor for the deformed principal block of the category $\mathcal{O}$ for $\mathfrak{sl}_2$. We also note that a prime ideal can be simultaneously of fully faithful and of $\mathfrak{sl}_2$ -type.

Lemma 3.3.4. *Suppose that Θ is such that $\pi_{R',\Theta}$ is not an equivalence of exact categories. Then the essential image of $\pi_{R',\Theta}$ consists of all objects M that fit into a short exact sequence of the form*

$$0 \rightarrow \Delta_{R'}(\gamma_i)^{\oplus n} \rightarrow M \rightarrow \Delta_{R'}(\gamma_{3-i})^{\oplus m} \rightarrow 0,$$

where $\gamma_{3-i} \uparrow_{\mathfrak{p}} \gamma_i$.

Proof. This follows in a straightforward way from the following isomorphisms:

$$(3.22) \quad \mathrm{grExt}_{\mathcal{C}_{R'}}^1(\Delta_{R'}(\gamma_j), \Delta_{R'}(\gamma_j)) = 0, \forall j \in \{1, 2\},$$

$$(3.23) \quad \mathrm{grExt}_{\mathcal{C}_{R'}}^1(\Delta_{R'}(\gamma_i), \Delta_{R'}(\gamma_{3-i})) = 0, \text{ if } \gamma_{3-i} \uparrow \gamma_i.$$

$$(3.24) \quad \mathrm{grExt}_{\mathcal{C}_{R'}}^1(\Delta_{R'}(\gamma_{3-i}), \Delta_{R'}(\gamma_i)) \xrightarrow{\sim} \mathrm{grExt}_{\mathcal{C}_{R', gss}}^1(\underline{\Delta}_{R'}(\gamma_{3-i}), \underline{\Delta}_{R'}(\gamma_i)), \text{ if } \gamma_i \uparrow \gamma_{3-i}.$$

(3.22) follows from Lemma 3.2.12. (3.23) follows from Remark 3.2.9 combined with the condition (3) in Definition 3.3.3. Finally, we deduce (3.24) from the commutativity of (3.9) in the setting of Remark 3.2.9. Namely, it is enough to prove that the middle vertical arrow in the localized version of (3.9), that we know by (1) of Proposition 3.2.5 is injective (combined with Proposition 3.1.5 and the $i = 0$ case of (c4)), is actually an isomorphism. This is because the target R' -module has length 1 thanks to (2) of Definition 3.3.3, while the source R' -module is nonzero by (3) of Definition 3.3.3. $\square$

3.3.3. Prime ideals of semisimple type.

Definition 3.3.5. We say that $\mathfrak{p}$ is of *semisimple type*, if $\mathcal{C}_{\mathbb{F}_0}^{\heartsuit}$ is semisimple with pairwise non-isomorphic simple objects $\Delta_{\mathbb{F}_0}(\gamma)$.

Here is an easier analog of Lemma 3.3.4 in this setting. The proof is similar (and easier), and so is omitted.

Lemma 3.3.6. *Suppose $\mathfrak{p}$ is of semisimple type. Then the essential image of $\pi_{R'}$ consists of the direct sums of the objects $\underline{\Delta}_{R'}(\gamma)$ for $\gamma \in \Gamma$.*

We would like to emphasize that an ideal can be simultaneously of semisimple and fully faithful type, but cannot be simultaneously of semisimple and of $\mathfrak{sl}_2$ -type.

3.3.4. Definition and example.

Definition 3.3.7. We say that an RS functor π is of $\mathfrak{sl}_2$ -type if every graded height 1 prime ideal in R is of fully faithful type or of $\mathfrak{sl}_2$ -type or of semisimple type.

Here is the main result of this section.

Proposition 3.3.8. *The functor $\mathrm{Red} : \mathcal{O} \rightarrow \mathcal{C}$ (that is RS by Proposition 3.2.4) is of $\mathfrak{sl}_2$ -type. Moreover,*

- (1) *the ideal $(\hbar)$ is of semisimple type and not of fully faithful type,*
- (2) *the ideals $(\check{\alpha} - n\hbar)$ for $n \in \mathbb{Z}$ are of $\mathfrak{sl}_2$ -type and not of fully faithful type,*

- (3) The ideal (c) is fully faithful type, but not of $\mathfrak{sl}_2$ -type or of semisimple type,
- (4) The ideals $(\tilde{\alpha} - n\hbar - mc)$ for $n, m \in \mathbb{Z}$ with $m \neq 0$ are of $\mathfrak{sl}_2$ - and fully faithful type,
- (5) and the remaining ideals are of fully faithful and semisimple type.

And the relation $\uparrow_{\mathfrak{p}}$ in (2) and (4) is such that $\lambda_1^a \uparrow_{\mathfrak{p}} \lambda_2^a \Rightarrow \lambda_1^a < \lambda_2^a$.

Proof. The proof is in several steps. We will write π for Red .

Step 1. We prove (1). Let us start by showing $(\hbar)$ of semisimple type. Set $R_0 := R/(\hbar)$. Indeed, $\mathcal{O}_{R_0}$ is nothing else but the category of $T_{\ell,d} \rtimes I$ -equivariant quasi-coherent sheaves on $(\hat{\mathfrak{g}}/\mathfrak{i}^0)^*$. The functor of restricting to $\text{Spec}(\mathbb{F}_0) \subset \mathfrak{h}^{a,*}$ defines an equivalence between the category of $T_{\ell,d} \rtimes I$ -equivariant quasi-coherent sheaves on $(\hat{\mathfrak{g}}/\mathfrak{i}^0)^* \times_{\mathfrak{h}^{a,*}} \text{Spec}(\mathbb{F}_0)$ and the category of $T_{\ell,d} \times H$ -equivariant $\mathbb{F}_0$ -modules because the I -action gives a scheme isomorphism

$$I \times^H \text{Spec}(\mathbb{F}_0) \xrightarrow{\sim} (\hat{\mathfrak{g}}/\mathfrak{i}^0)^* \times_{\mathfrak{h}^{a,*}} \text{Spec}(\mathbb{F}_0).$$

It follows that $(\hbar)$ is of semisimple type.

Since $\mathcal{C}_{R_0}$ is the category of bi-graded smooth modules over a completion of a polynomial algebra in infinitely many variables, it is not semisimple. It follows that $(\hbar)$ is not of fully faithful type.

Step 2. So we can assume that $\hbar$ is invertible in $\mathbb{F}_0$. We can pass to the representations of $\hat{\mathfrak{g}}$ over the characteristic 0 field $\mathbf{f} := \mathbb{F}_0^{T_d}$, cf. Section 1.2.3. We are interested in blocks of the usual category $\mathcal{O}$ with weights in $\mu^\circ + \Lambda_r^a$, where μ° lies on at most one root hyperplane and Λ_r^a is the affine root lattice. Denote the sum of these blocks by $\mathcal{O}(\mu^\circ)$.

One can use [17, Theorem 2] to understand the decomposition of $\mathcal{O}(\mu^\circ)$ into blocks. We get the following:

- (i) If μ° is of critical level, then two Verma modules lie in the same block if and only if their highest weights differ by a multiple of δ .
- (ii) If there is a real affine coroot whose value on μ° is an integer, then each block has either one or two simple objects.

Step 3. Recall the homomorphisms $\gamma_\lambda : R_\hbar \mathcal{Z}(\mathfrak{g})[c] \rightarrow R$ from Definition 1.5.2. We claim that

- (*) if all γ_λ are pairwise distinct in $\mathbb{F}_0$, then $\mathfrak{p}$ is of fully faithful type.

We will prove this claim in this and the next two steps.

Namely, take indecomposable $M, N \in \mathcal{O}(\mu^\circ)^\Delta$. We need to show that π induces an isomorphism

$$\text{Hom}_{\mathcal{O}(\mu^\circ)}(M, N) \xrightarrow{\sim} \text{Hom}_{\mathcal{O}_{fin, \mathbf{f}}^{\mathcal{W}}}(\underline{M}, \underline{N}),$$

where the category $\mathcal{O}_{fin, \mathbf{f}}^{\mathcal{W}}$ is the equivalent ungraded variant of the category $\mathcal{O}_{fin, \mathbb{F}_0}^{\mathcal{W}}$ from Section 1.5.5, $\underline{M} := \text{Red}(M)$, $\underline{N} := \text{Red}(N)$.

Suppose that M is filtered by $\Delta_{\mathbf{f}}(\lambda_i + m_i\delta)$ (=the Verma module with highest weight $\lambda_i + m_i\delta$) for $\lambda_i \in \Lambda, m_i \in \mathbb{Z}$. Let m be the minimum of m_i 's. In particular, M has no Ext^1 's in $\mathcal{O}(\mu^\circ)$ to objects where all energy degrees are less than m . An analogous result holds for $\underline{M}$.

We claim that we can find costandardly filtered objects N_1, N_2 and morphisms $\varphi_1 : N \rightarrow N_1, \varphi_2 : N_1 \rightarrow N_2$ such that

- (a) $\varphi_2 \circ \varphi_1 = 0$

(b) the energy degrees in $\ker \varphi_1, \ker \varphi_2 / \operatorname{im} \varphi_1$ are strictly less than m .

If $\mathfrak{p}$ is as in (ii) of Step 2, we just take the first two terms of an injective resolution of N : it exists because $\mathcal{O}(\mu^\circ)$ is the direct sum of highest weight categories with finite (in fact, two-element) posets.

In case (i) we argue as follows. Take a block of $\mathcal{O}(\mu^\circ)$, the labels of Vermas in this block are of the form $\mu^1 + n\delta$ for $n \in \mathbb{Z}$. We can assume that M, N are indecomposable objects in a single block. Take an interval $[m, m']$ such that the labels of standards occurring in M, N are of the form $\mu^1 + i\delta$ for $i \in [m, m']$. Consider the highest weight subquotient category $\mathcal{O}(\mu^\circ)_{[m, m']}$ of the corresponding block in $\mathcal{O}(\mu^\circ)$ and let ϖ denote the quotient functor $\mathcal{O}(\mu^\circ)_{(-\infty, m']} \rightarrow \mathcal{O}(\mu^\circ)_{[m, m']}$, cf. Section 2.1.4. By the construction ϖ hits a module with an idempotent – it has both left and right adjoints, to be denoted by $\varpi^!$ and ϖ^* . Then for N'_1, N'_2 we take the first two terms of the injective resolution of $\varpi(N)$. We set $N_j := \varpi^*(N'_j)$ for $j = 1, 2$. The resulting complex $N \rightarrow N_1 \rightarrow N_2$ satisfies (a) and (b).

Step 4. We write $\underline{N}_j := \pi(N_j)$. We claim that

$$(3.25) \quad \operatorname{Hom}_{\mathcal{O}(\mu^\circ)}(M, N) = \ker[\operatorname{Hom}_{\mathcal{O}(\mu^\circ)}(M, N_1) \rightarrow \operatorname{Hom}_{\mathcal{O}(\mu^\circ)}(M, N_2)]$$

$$(3.26) \quad \operatorname{Hom}_{\mathcal{O}_{fin, \mathbf{f}}^{\mathcal{W}}}(\underline{M}, \underline{N}) = \ker[\operatorname{Hom}_{\mathcal{O}_{fin, \mathbf{f}}^{\mathcal{W}}}(\underline{M}, \underline{N}_1) \rightarrow \operatorname{Hom}_{\mathcal{O}_{fin, \mathbf{f}}^{\mathcal{W}}}(\underline{M}, \underline{N}_2)].$$

This is because $\ker \varphi_1, \ker \varphi_2 / \operatorname{im} \varphi_1$ are in energy degrees $< m$, and so have no Hom's or Ext¹'s from M – and the analogous claim holds in $\mathcal{O}_{fin, \mathbf{f}}^{\mathcal{W}}$.

Step 5. Now we finish the proof of (*) in Step 3. We need to show that the map induced by $\pi_{\mathbf{f}}$

$$(3.27) \quad \operatorname{Hom}_{\mathcal{O}(\mu^\circ)}(M, N_j) \xrightarrow{\sim} \operatorname{Hom}_{\mathcal{O}_{fin, \mathbf{f}}^{\mathcal{W}}}(\underline{M}, \underline{N}_j).$$

is an isomorphism. This boils down to showing that the dimensions (over $\mathbf{f}$) of the domain and codomain of (3.27) coincide and that (3.27) is injective.

To see the injectivity, we will show that $\pi_{\mathbf{f}} : \mathcal{O}(\mu^\circ) \rightarrow \mathcal{O}_{\mathbf{f}}^{\mathcal{W}}$ is faithful. Recall that π is t-exact. So it is faithful if and only if it does not annihilate any simple object. By Proposition 1.5.5, this boils down to the condition that $\langle \mu^\circ, \check{\alpha} \rangle \notin \mathbb{Z}$ for all finite type roots α . This follows from the condition that all γ_λ are pairwise distinct in $\mathbb{F}_0$.

To prove that the dimensions of the domain and codomain of (3.27) are the same, we argue by the induction on the lengths of the filtrations of M and N_j 's by standards and costandards respectively. The induction step is based on the 5-lemma combined with the observation that Ext^1 from a standard to a costandard object in $\mathcal{O}(\mu^\circ)$ vanishes. The base case is when $M = \Delta_{\mathbf{f}}(\mu^\circ + \lambda_1), N_j = \nabla_{\mathbf{f}}(\mu^\circ + \lambda_2)$, so that $\dim \operatorname{Hom}_{\mathcal{O}(\mu^\circ)}(M, N_j) = \delta_{\lambda_1, \lambda_2}$. The equality $\dim \operatorname{Hom}_{\mathcal{O}_{\mathbf{f}}^{\mathcal{W}}}(\underline{M}, \underline{N}_j) = \delta_{\lambda_1, \lambda_2}$ follows from [2, Theorem 5.5.4, (iii)] combined with the condition that all γ_λ are pairwise distinct in $\mathbb{F}_0$.

Step 6. The assumption of (*) holds in cases (3), (4), (5), so they are all of fully faithful type. In case (3), $\mathfrak{W}_{0, h}$ is the center of $V_{0, h}(\mathfrak{g})$, and so is a polynomial algebra in infinitely many variables. So (3) is not of $\mathfrak{sl}_2$ or semisimple type. In case (4), the blocks of category $\mathcal{O}_{\mathbf{f}}$ are either semisimple with one simple object or equivalent to principal blocks of category $\mathcal{O}$ for $\mathfrak{sl}_2$. So we get $\mathfrak{sl}_2$ -type. And in case (5), $\mathcal{O}_{\mathbf{f}}$ is semisimple.

Step 7. Now we proceed to examining the case when (*) does not hold, equivalently, there is a finite type root α such that $\langle \mu^\circ + \rho, \check{\alpha} \rangle \in \mathbb{Z}$, which is case (2). By changing μ° we can assume that $\langle \mu^\circ, \check{\alpha} \rangle = 0$. Then the following hold:

- (A) Verma modules $\Delta_{\mathbf{f}}(\mu^\circ + \lambda_1^a), \Delta_{\mathbf{f}}(\mu^\circ + \lambda_2^a)$ lie in the same block if and only if $s_\alpha(\lambda_1^a) = \lambda_2^a$. We have two cases.
- (I) If $s_\alpha(\lambda_1^a) = \lambda_1^a$, then the block is semisimple with one simple object, $\Delta_{\mathbf{f}}(\lambda_1^a)$.
 - (II) If $\lambda_1^a \neq \lambda_2^a = s_\alpha \lambda_1^a$, then we can assume that $\lambda_1^a < \lambda_2^a$. In this case, we have an inclusion $\Delta_{\mathbf{f}}(\mu^\circ + \lambda_1^a) \hookrightarrow \Delta_{\mathbf{f}}(\mu^\circ + \lambda_2^a)$, the module $\Delta_{\mathbf{f}}(\mu^\circ + \lambda_1^a)$ is irreducible, while $L_{\mathbf{f}}(\mu^\circ + \lambda_2^a)$ (=the irreducible quotient of $\Delta_{\mathbf{f}}(\mu^\circ + \lambda_2^a)$) is the cokernel of $\Delta_{\mathbf{f}}(\mu^\circ + \lambda_1^a) \hookrightarrow \Delta_{\mathbf{f}}(\mu^\circ + \lambda_2^a)$.
- (B) By Proposition 1.5.5, $\underline{\Delta}(\mu^\circ + \lambda_1^a) = \underline{\Delta}(\mu^\circ + \lambda_2^a)$ is simple.

For Θ as in Case (I), $\pi_{\mathbb{F}_0, \Theta}$ is fully faithful. Now consider the $\Theta = \{\lambda_1^a, \lambda_2^a\}$ as in Case (II). Then $\lambda_1^a \uparrow_{\mathfrak{p}} \lambda_2^a$. We need to check conditions (1)-(3) in Definition 3.3.3. (1) is straightforward, and (3) follows from (A.II). It remains to establish (2). The highest energy degrees in $\underline{\Delta}_{R'_k}(\lambda_1^a)$ and $\underline{\Delta}_{R'_k}(\lambda_2^a)$ are the same, say equal n , so, similarly to (1.11), we get

$$\mathrm{grHom}_{\underline{\mathcal{C}}_{R'_k}}(\underline{\Delta}_{R'_k}(\lambda_i^a), \underline{\Delta}_{R'_k}(\lambda_{3-i}^a)) = \mathrm{grHom}_{R_h \mathcal{Z}(\mathfrak{g}) \otimes_{\mathbb{K}[h]} R'_k}(\underline{\Delta}_{R'_k}(\lambda_i^a)_n, \underline{\Delta}_{R'_k}(\lambda_{3-i}^a)_n),$$

where the subscript n in the right hand side indicate the n th energy graded component. The module $\underline{\Delta}_{R'_k}(\lambda_j^a)_n$ is the algebra of functions on the graph of $\gamma_{\lambda_j} : \mathrm{Spec}(R'_k) \rightarrow \mathrm{Spec}(R_h \mathcal{Z}(\mathfrak{g}))$. The two graphs in question intersect transversally at $\mathrm{Spec}(\mathbb{F}_0)$. (2) follows finishing the proof. $\square$

3.3.5. Equivalences of exact categories. We use Proposition 3.2.11 to establish an equivalence between certain exact categories. Namely, we suppose that for $i = 1, 2$ we have categories $\mathcal{C}^i$ with admissible collections of objects $\Delta^i(\gamma)$ satisfying (b) from Section 3.2.1, where γ runs over a group Γ . Further, let $\pi^i : \mathcal{C}^i \rightarrow \underline{\mathcal{C}}$ be RS functors (for admissible collection of objects $\underline{\Delta}(\gamma), \gamma \in \Gamma$, satisfying (c1)-(c5)).

The following is a version of [18, Theorem 3.2.1] adapted to RS functors of $\mathfrak{sl}_2$ -type.

Theorem 3.3.9. *Suppose we partitioned the set $\mathfrak{I}$ of height 1 graded prime ideals in R into three disjoint sets $\mathfrak{I}_{ff} \sqcup \mathfrak{I}_{ss} \sqcup \mathfrak{I}_{\mathfrak{sl}_2}$ such that*

- (1) *the ideals from $\mathfrak{I}_{ff}$ are of fully faithful type for both π^1 and π^2 ,*
- (2) *the ideals from $\mathfrak{I}_{ss}$ are of semisimple type for both π^1 and π^2 ,*
- (3) *the ideals from $\mathfrak{I}_{\mathfrak{sl}_2}$ are of $\mathfrak{sl}_2$ type for both π^1 and π^2 . Moreover, the relations $\uparrow_{\mathfrak{p}}$ on Γ coincide for π^1 and π^2 .*

Then there is a unique equivalence of exact categories $\iota : \mathcal{C}^{1,\Delta} \xrightarrow{\sim} \mathcal{C}^{2,\Delta}$ intertwining the functors π^i to $\underline{\mathcal{C}}^\Delta \cap \underline{\mathcal{C}}_{gss}^\Delta$.

Proof. Theorem boils down to showing that the essential images of $\mathcal{C}^{i,\Delta}, i = 1, 2$, in $\underline{\mathcal{C}}^\Delta \cap \underline{\mathcal{C}}_{gss}^\Delta$ coincide. As in the proof of Theorem [18, Theorem 3.2.1], we reduce the claim to

- (*) For $M^i, N^i \in \mathcal{C}^{i,\Delta}$ with identifications $\pi^i(M^i) \xrightarrow{\sim} \underline{M}, \pi^i(N^i) \xrightarrow{\sim} \underline{N}$, the images of $\mathrm{grExt}_{\mathcal{C}}^1(M^i, N^i)$ in $\mathrm{grExt}_{\underline{\mathcal{C}}_{gss}}^1(\underline{M}, \underline{N})$ coincide.

Note that $\pi^i(M_{R_{\mathfrak{p}}}^i) \xrightarrow{\sim} \underline{M}_{R_{\mathfrak{p}}}, \pi^i(N_{R_{\mathfrak{p}}}^i) \xrightarrow{\sim} \underline{N}_{R_{\mathfrak{p}}}$ for all height 1 graded prime ideals $\mathfrak{p}$. So we need to show that the images of $\mathrm{grExt}_{\mathcal{C}_{R_{\mathfrak{p}}}}^1(M_{R_{\mathfrak{p}}}^i, N_{R_{\mathfrak{p}}}^i)$ in $\mathrm{grExt}_{\underline{\mathcal{C}}_{gss, R_{\mathfrak{p}}}}^1(\underline{M}_{R_{\mathfrak{p}}}, \underline{N}_{R_{\mathfrak{p}}})$ coincide, equivalently the essential images of $\mathcal{C}_{R_{\mathfrak{p}}}^{i,\Delta}$ in $\underline{\mathcal{C}}_{R_{\mathfrak{p}}, gss}^\Delta$ coincide. This splits into three cases:

- (1) if $\mathfrak{p} \in \mathfrak{I}_{ff}$, then the claim follows from Lemma 3.3.2;
- (2) if $\mathfrak{p} \in \mathfrak{I}_{ss}$, then the claim follows from Lemma 3.3.6;
- (3) and if $\mathfrak{p} \in \mathfrak{I}_{sl_2}$, the claim follows from Lemma 3.3.4.

□

4. FLE CATEGORY

Let G be a simple group.

Recall that we have the equivariant affine W-algebra $\mathcal{EW}_{c,\hbar}$ for $\mathfrak{g}$, the universal enveloping algebra of the vertex algebra $\mathfrak{EW}_{c,\hbar}$ introduced in Definition 1.4.2. It is acted on by $T_{\ell,d} \ltimes I$, so we can consider the category associated to $(\mathcal{EW}_{c,\hbar}, T_{\ell,d} \ltimes I)$ as in Section 1.2.1. Denote the derived version by F and the heart by $F^\heartsuit$. In this section we will study this category in detail culminating in the construction of the main functor of this paper.

4.1. Spaces X and $X_{\mathfrak{h}^a}$.

4.1.1. Affine flag varieties and cotangent bundles. We will write $\mathcal{F}l$ for the affine flag variety $\mathcal{F}l_G$. For some results below we need a construction, cf. [23, Section 1.2]. Fix an embedding $G \hookrightarrow \mathrm{GL}_n$. Then $\mathcal{F}l$ embeds as a closed sub ind-scheme into $\mathcal{F}l_{\mathrm{GL}_n}$. We can present $\mathcal{F}l_{\mathrm{GL}_n}$ as the union of closed projective subschemes as follows. Let $0 \subset V_1 \subset \dots \subset V_{n-1} \subset \mathbb{K}^n$ be the standard flag in $\mathbb{K}^n$. We assume that the Borel subgroup $B \subset G$ is the stabilizer of this flag. Fix the corresponding standard affine flag $(U_i^{st})_{i \in \mathbb{Z}}$, so that I is the stabilizer of this affine flag in $\mathcal{L}G$. We assume that this standard flag lies in $\mathcal{F}l$.

For each $k > 0$, we can consider the subscheme $\mathcal{F}l_{\mathrm{GL}_n,k}$ of all affine flags $(U_i)_{i \in \mathbb{Z}}$ such that $t^k U_i \subset U_i^{st} \subset t^{-k} U_i$ for all i . Then we can define $\mathcal{F}l_k$ to be the scheme-theoretic intersection of $\mathcal{F}l_{\mathrm{GL}_n,k}$ and $\mathcal{F}l$.

We note that $\mathcal{F}l$ is a formally smooth reduced ind-scheme whose irreducible components are parameterized by $\pi_1(G)$. This easily follows from the affine Grassmannian case, which is addressed in [7, Thm 4.5.1, Prop 4.5.4].

We can consider the cotangent bundle $T^*\mathcal{F}l$, an ind-scheme of infinite type given whose points are pairs $\{(I_1, \xi)\}$, where I_1 is an Iwahori subgroup and $\xi \in [\mathcal{L}\mathfrak{g}/\mathfrak{i}]^*$. The projection to the first coordinate establishes $T^*\mathcal{F}l$ as a pro-vector bundle over $\mathcal{F}l$. In particular, it is a formally smooth and classical (as opposed to derived) ind-scheme.

We will also need a deformation of $T^*\mathcal{F}l$ over the space $\mathfrak{h}^{a,*}$ to be denoted by $T_{\mathfrak{h}^a}^*\mathcal{F}l$. It parameterizes pairs (I_1, ξ) , where now $\xi \in (\hat{\mathfrak{g}}/\mathrm{Lie}(I_1^0))^*$, here and below we write I_1^0 for the pro-unipotent radical of I_1 . Let $\hat{\mathfrak{i}}_1$ denote the preimage of $\mathrm{Lie}(I_1) \subset \mathcal{L}\mathfrak{g}$ in $\hat{\mathfrak{g}}$. The spaces $\hat{\mathfrak{i}}_1/\mathrm{Lie}(I_1^0)$ are canonically identified for different choices of I_1 and, in particular, are identified with $\mathfrak{h}^a$. In particular, we have a natural morphism $T_{\mathfrak{h}^a}^*\mathcal{F}l \rightarrow \mathfrak{h}^{a,*}$, which is flat and whose fiber at 0 is $T^*\mathcal{F}l$. Note that $T_{\mathfrak{h}^a}^*\mathcal{F}l \rightarrow \mathcal{F}l$ is a pro-vector bundle. Also we have the morphism $T_{\mathfrak{h}^a}^*\mathcal{F}l \rightarrow \mathcal{L}\mathfrak{g}^*$ the composition of the moment map $T_{\mathfrak{h}^a}^*\mathcal{F}l \rightarrow \hat{\mathfrak{g}}^*$ for the action of $\hat{G}$ and the $\mathcal{L}G$ -equivariant projection $\hat{\mathfrak{g}}^* \rightarrow \mathcal{L}\mathfrak{g}^*$.

4.1.2. Main definitions. As before, cf. Section 1.3, we fix the loop rotation invariant nondegenerate character $\psi : \mathcal{L}\mathfrak{n}^- \rightarrow \mathbb{K}$. The space X (resp., $X_{\mathfrak{h}^a}$) we care about is the Hamiltonian reduction of $T^*\mathcal{F}l$ (resp., $T_{\mathfrak{h}^a}^*\mathcal{F}l$) by the action of $\mathcal{L}N^-$ with character ψ . Equivalently, we can write it as follows. Let S be the Kostant slice in $\mathfrak{g}^*(\cong \mathfrak{g})$, i.e., in the notation of Step 1 of the proof of Proposition 1.3.2, $S = e + \mathfrak{z}_{\mathfrak{g}}(f)$. Let ψ and $\mathfrak{z}$ denote

the images of $e, \mathfrak{z}_{\mathfrak{g}}(f)$ in $\mathfrak{g}^*$ (this ψ extends the homomorphism $\mathfrak{n}^- \rightarrow \mathbb{C}$ also denoted by ψ). Set $\mathcal{L}S := [\psi + \mathfrak{z}((t))] \frac{dt}{t} \subset (\mathcal{L}\mathfrak{g})^*$. Then

$$X := \mathcal{L}S \times_{\mathcal{L}\mathfrak{g}^*}^R T^*\mathcal{F}l, \quad X_{\mathfrak{h}^a} := \mathcal{L}S \times_{\mathcal{L}\mathfrak{g}^*}^R T_{\mathfrak{h}^a}^*\mathcal{F}l,$$

so that X is the derived fiber of 0 under the projection $X_{\mathfrak{h}^a} \rightarrow \mathfrak{h}^{a,*}$. In other words, X is the derived ind-stack parameterizing pairs

$$\{(gI, \xi) | gI \in \mathcal{F}l, \xi \in \mathcal{L}S \cap (\mathcal{L}\mathfrak{g}/\text{Ad}(g)\mathfrak{i})^*\}.$$

Note that by the construction $T_{\ell,d}$ acts on $X, X_{\mathfrak{h}^a}$ (the loop rotation and the modified dilation actions, as in Section 1.3.1).

Here is a basic result about the structure of $X, X_{\mathfrak{h}^a}$.

Lemma 4.1.1. *The derived ind-scheme X is the filtered colimit $\text{colim } Z_i$, where Z_i is the total space of a trivial $T_{\ell,d}$ -equivariant pro-vector bundle on a finite type derived scheme $\underline{Z}_i$, under closed inclusions of derived schemes. The same is true for $X_{\mathfrak{h}^a}$.*

Proof. We will handle the case of X , the case of $X_{\mathfrak{h}^a}$ is similar. Let $\varpi : X_{\mathfrak{h}^a} \rightarrow \mathcal{F}l$ denote the projection. Recall the closed subscheme $\mathcal{F}l_k \subset \mathcal{F}l_G$. We take $Z_k := \varpi^{-1}(\mathcal{F}l_k)$ with its natural structure of a derived scheme. Clearly, $Z_k \subset Z_{k+1}$ is closed, and $X = \text{colim}_k Z_k$. Notice that Z_k admits a closed embedding into $\mathcal{F}l_k \times (\psi + t^{-2k}\mathfrak{z}[[t]]) \frac{dt}{t}$ thanks to the construction of $\mathcal{F}l_k$. For $\underline{Z}_k$ we take the intersection of the image with $\mathcal{F}l_k \times (\psi + \mathfrak{z} \text{Span}_{\mathbb{K}}(t^i) - 2k \leq i \leq 2k) \frac{dt}{t}$. Then $Z_k \rightarrow \underline{Z}_k$ is the trivial pro-vector bundle with fiber $t^{2k+1}\mathfrak{z}[[t]] \frac{dt}{t}$. $\square$

4.1.3. Categories of sheaves. Here we discuss categories of coherent and ind-coherent sheaves on X and $X_{\mathfrak{h}^a}$ and their connection to suitable specializations of the category $\mathbf{F}$.

We can talk about the (derived) categories of $T_{\ell,d}$ -equivariant ind-coherent sheaves on the derived ind-schemes on $X, X_{\mathfrak{h}^a}$, denote them by $\text{IndCoh}^{T_{\ell,d}}(X), \text{IndCoh}^{T_{\ell,d}}(X_{\mathfrak{h}^a})$. These categories come with natural t-structures. On the other hand, we can consider the graded quotients $\mathbf{R}_0 = \mathbf{R}/(\hbar) (= S(\mathfrak{h}^a))$ and $\mathbb{K}$. We form the derived categories $\mathbf{F}_{\mathbf{R}_0}, \mathbf{F}_{\mathbb{K}}$, see Section ??.

Lemma 4.1.2. *We have t-exact equivalences $\mathbf{F}_{\mathbf{R}_0} \cong \text{IndCoh}^{T_{\ell,d}}(X_{\mathfrak{h}^a})$ (graded linear over $S(\mathfrak{h}^a)$) and $\mathbf{F}_{\mathbb{K}} \cong \text{IndCoh}^{T_{\ell,d}}(X)$.*

Proof. We will prove the former equivalence, the latter is similar. Consider the 1-parameter deformation $\mathcal{L}_c(G \times S)$ of $\mathcal{L}(G \times S)$, cf. the discussion before Lemma 1.4.5. The Kac-Moody group $\widehat{G}$ acts on $\mathcal{L}_c(G \times S)$ in a Hamiltonian way. Restrict this action to the Iwahori subgroup $I^0 \subset \widehat{G}$ and let $\mu : \mathcal{L}_c(G \times S) \rightarrow \mathfrak{i}^{0,*}$ be the moment map. Note that the action of I on $\mu^{-1}(0)$ is free and $X_{\mathfrak{h}^a}$ is the quotient $\mu^{-1}(0)/I$. The category $\mathbf{F}_{\mathbf{R}_0}$ is nothing else but the category of I -equivariant ind-coherent sheaves on $\mu^{-1}(0)$ and the equivalence we need is by equivariant descent. $\square$

Finally, let us define coherent complexes of sheaves on X and $X_{\mathfrak{h}^a}$. consider the case of X (the case of $X_{\mathfrak{h}^a}$ is completely analogous). Recall the presentation $X = \text{colim}_n Z_n$, where the derived schemes Z_n were defined in the proof of Lemma 4.1.1. Let $\varrho_n : Z_n \rightarrow \underline{Z}_n$ denote the projection and $\iota_n : Z_n \hookrightarrow X$ denote the inclusion, the trivial vector bundle.

An object of $\mathrm{IndCoh}^{T_{\ell,d}}(X)$ is said to be *coherent* if it is of the form $\iota_{n,*}\varrho_n^*(\mathcal{F})$, where $\mathcal{F} \in D^b(\mathrm{Coh}^{T_{\ell,c}}\underline{Z}_n)$ for some n (and automatically for all larger n).

Remark 4.1.3. Below we will need the notion of coherence for a related derived ind-stack. Namely, consider the derived ind-stack $Y := T^*(I^0 \backslash \hat{G}/I^0)/(H \times H \times \mathbb{G}_m)$, where $\mathbb{G}_m$ is the connected component of the center of $\hat{G}$. It parametrizes pairs (I_1, ξ) of an Iwahori subgroup $I_1 \subset \mathcal{L}G$ and $\xi \in (\hat{\mathfrak{g}}/\mathfrak{i}_1^0 + \mathfrak{i}^0)^*$ modulo the action of I . It admits a presentation as a union of $T_{\ell,d} \ltimes I$ -equivariant pro-vector bundles, Y_j , over finite type derived stacks $\underline{Y}_j$ equipped with rational actions of quotients of $T_{\ell,d} \ltimes I$, denote them by I_j ; here $j \geq 0$. In particular, the definition of coherent sheaves on such derived stacks goes through: we start with a bounded complex of $\tilde{I}_j$ -equivariant coherent sheaves on $\underline{Y}_j$, where $\tilde{I}_j$ is a quotient of $T_{\ell,d} \ltimes I$ surjecting onto I_j , pull it back to Y_j and then push it forward to Y .

4.2. Object $\Delta^F(0)$. Our goal in this section is to define an object $\Delta^F(0)$ that, as we will see below, belongs to an admissible collection of objects (see Section 3.1.1) that we will need to construct the main functor of this paper.

4.2.1. Category of D -modules on $\mathcal{L}G$ and reduction functor. We need to introduce one more category of the form considered in Section 1.2.1.

We consider the vertex algebra $\mathrm{CDO}_{c,h}(G)$, see Example 1.1.2. Its universal enveloping algebra can be thought of the algebra of homogenized c -twisted differential operators on $\mathcal{L}G$. We consider the group $F = I \times I$ with $F^0 = I^0 \times I^0$ and maximal $H \times H$. We note that that we have a strong action of F on $\mathrm{CDO}_{c,h}$. Then as usual we take $\tilde{F} = T_{\ell,d} \ltimes F$. We consider the category $\mathbf{H}$ (the “double affine Hecke category”) associated to the data of $\mathrm{CDO}_{c,h}(G)$ and the action of $\tilde{F}$. We note that $\mathbf{H}$ is graded linear over $S(\mathfrak{h})^{\otimes 2}[c, \hbar]$.

Now we are going to construct a t-exact functor $\mathrm{Red} : \mathbf{H} \rightarrow \mathbf{F}$, cf. Section 1.7.1. We consider the $\mathbb{K}[c, \hbar]$ -linear quantum comoment map $\Phi_\ell : R_\hbar U(\hat{\mathfrak{g}}) \rightarrow \tilde{U}(\mathrm{CDO}_{c,h}(G))$. It gives rise to a forgetful functor $\mathbf{H} \rightarrow \mathbf{O}$. So for an object $M \in \mathbf{H}$, we can consider $\mathrm{Red}(\Phi_\ell^*(M))$. Completely analogously to Section 1.5.2, this is naturally an object in $\mathbf{F}$, giving us the desired functor $\mathrm{Red} : \mathbf{H} \rightarrow \mathbf{F}$.

Remark 4.2.1. The following remark will play a very important role in what follows. Any object, $\mathcal{F}$, of $\mathbf{H}$ comes with two commuting actions of $\mathbf{R}$ to be referred to as left and right actions (that agree on $\hbar$, while the two actions of c differ by -1). Therefore $\mathrm{Red}(\mathcal{F})$ comes with two $\mathbf{R}$ -actions. On the other hand, any object of $\mathbf{F}$ comes with a single action of $\mathbf{R}$, and on $\mathrm{Red}(\mathcal{F})$ this action is the right action.

4.2.2. δ -function object. Note that $R_\hbar U(\mathfrak{i}^0)$ acts on $\mathbb{K}[I, \hbar]$ (via differentiating the left translation action) commuting with the action by right translations.

Define the $R_\hbar U(\hat{\mathfrak{g}})$ -module $\delta_{c,h}^I$ by

$$\delta_{c,h}^I := R_\hbar U(\hat{\mathfrak{g}}) \otimes_{R_\hbar U(\mathfrak{i}^0)} \mathbb{K}[I, \hbar].$$

Let us explain how to equip $\delta_{c,h}^I$ with the structure of a module over $\mathrm{CDO}_{c,h}(G)$. Being a $\mathrm{CDO}_{c,h}$ -module is the same things as being

- (i) a smooth module over $\tilde{U}(V_{c,h}(\mathfrak{g}))$ (this structure comes from the construction of $\delta_{c,h}^I$ as an induced module),
- (ii) a smooth module over $\mathbb{K}[\mathcal{L}G]$,

- (iii) so that the two structures are compatible via the commutation relation $\xi f - f\xi = \hbar L_\xi f$.

The action in (ii) is defined using the commutation relation in (iii) and the natural action of $\mathbb{K}[\mathcal{L}G]$ on $\mathbb{K}[I]$.

The module $\delta_{c,\hbar}^I$ carries a natural action of $\tilde{F}$ making it a weakly $\tilde{F}$ -equivariant $\tilde{U}(\text{CDO}_{c,\hbar}(G))$ -module. It is strongly equivariant for the action of I^0 by left translations. Let us show that it is also strongly equivariant for the action of I^0 by right translations. It is enough to show that $1 \otimes 1 \in \delta_{c,\hbar}^I$ is annihilated by the elements of the form $\Phi_r(x)$ for $x \in \mathfrak{i}^0$. This is true thanks to the Λ^a -grading on $\delta_{c,\hbar}^I$: the degree component in $\mathbb{K}[I]$ is zero unless the degree is a nonnegative combination of negative roots, so the same is true for $\delta_{c,\hbar}^I$. Since 1 is in degree 0, and our claim follows.

As any object in $\mathbf{H}$, $\delta_{c,\hbar}^I$ carries two commuting actions of $\mathfrak{h}$ by graded endomorphisms. Our next time is to compare them. For $x \in \mathfrak{h}$, let x_ℓ, x_r denote the graded endomorphisms coming from the left and right H -actions.

Lemma 4.2.2. *We have $x_\ell + x_r = -\hbar \langle 2\rho, x \rangle$ for all $x \in \mathfrak{h}$.*

Proof. Step 1. We have the inclusion of $\mathbf{R}$ into the algebra of graded endomorphisms of $\delta_{c,\hbar}^I$, i.e., $\text{grHom}_{\mathbf{H}}(\delta_{c,\hbar}^I, \delta_{c,\hbar}^I)$ with $x \in \mathfrak{h} \subset \mathbf{R}$ mapping to x_ℓ .

We claim that the algebra of graded endomorphism of $\delta_{c,\hbar}^I$ is the image of this inclusion. Indeed, $\delta_{c,\hbar}^I$ is generated by the element $1 \otimes 1$ and so a graded endomorphism is determined by the image of this element. The element $1 \otimes 1$ is $T_\ell \ltimes (I \times I)$ -invariant, hence its image is so. The submodule of invariants for the action of I from the right is $R_\hbar U(\hat{\mathfrak{g}}) \otimes_{R_\hbar U(\mathfrak{i}^0)} \mathbb{K}[\hbar] = \Delta(0)$. And the submodule of $T_\ell \ltimes I$ -invariants in $\Delta(0)$ is the highest weight subspace, i.e., $\mathbf{R}$.

Since the PBW degree of the graded endomorphism x_r is 1, we see that

$$(4.1) \quad x_r \in \mathfrak{h}^{da} \subset \mathbf{R}.$$

Step 2. Let us consider the specialization to $\hbar = 0$. The quotient $\tilde{U}(\text{CDO}_{c,\hbar}(G))/(\hbar)$ is $\mathbb{K}[\mathcal{L}_c T^*G]$. The inclusions of $\mathbb{K}[\hat{\mathfrak{g}}^*]$ into $\mathbb{K}[\mathcal{L}_c T^*G]$ is via the pullbacks μ_ℓ^*, μ_r^* , where μ_ℓ, μ_r are the moment maps $\mathcal{L}_c T^*G \rightarrow \hat{\mathfrak{g}}^*$ associated to the actions of $\mathcal{L}G$ from the left and from the right. We identify $\mathcal{L}_c T^*G$ with $\mathcal{L}G \times \hat{\mathfrak{g}}^*$, so that the action of $\mathcal{L}G$ from the right is by the right translations on the first factor. Then the moment maps in question are given by

$$\mu_\ell(g, x) = x, \mu_r(g, x) = -\text{Ad}^*(g)x.$$

The module $\delta_{c,\hbar}^I$ is the functions on the (deformed) conormal bundle to $I \subset \mathcal{L}G$, i.e., the locus (g, x) with $g \in I, x \in (\hat{\mathfrak{g}}/\mathfrak{i}^0)^*$. It is easy to see that for $x \in \mathfrak{h}$ we have $\mu_\ell^*(x) + \mu_r^*(x) = 0$. Combining this with (4.1), we see that $x_\ell + x_r = \langle \chi, x \rangle \hbar$ for some $\chi \in \mathfrak{h}^*$. Our task therefore is to show that $\chi = 0$. For this it is enough to specialize c and $\mathfrak{h}^*$ to 0.

Step 3. The 0-mode algebra of $\text{CDO}_{0,\hbar}(G)$ is $R_\hbar D(G)$. Namely, we have associative algebra homomorphisms $A^0(V_{0,\hbar}), A^0(\mathbb{K}[\mathcal{J}G]) \rightarrow A^0(\text{CDO}_{0,\hbar}(G))$, with the former induced by Φ_ℓ . It is easy to see that they combine to $R_\hbar D(G) \rightarrow A^0(\text{CDO}_{0,\hbar}(G))$. Modulo $\hbar$, this is an isomorphism $\mathbb{K}[T^*G] \rightarrow A^0(\mathbb{K}[\mathcal{J}T^*G])$. Since the PBW grading on $\tilde{U}(\text{CDO}_{0,\hbar}(G))$ is bounded from below, the homomorphism $R_\hbar D(G) \rightarrow A^0(\text{CDO}_{0,\hbar}(G))$ is surjective. Since the algebra $D(G)$ is simple, it must also be injective.

Now we claim that the map $A^0(\Phi_r) : R_{\hbar}U(\mathfrak{g}) \rightarrow R_{\hbar}D(G)$ is the quantum comoment map induced by the right action of G on $D_{\hbar}(G)$. Indeed, the two maps coincide modulo $\hbar$, so the image of the difference is in $\hbar\mathbb{K}[G]$. Also both maps are invariant under the left action of G . So the difference of images lies in $\mathbb{K}[\hbar]$. Since $\mathfrak{g}$ is semisimple, the difference is actually 0.

Step 4. The energy degree 0 component of $\mathbb{K} \otimes_{S(\mathfrak{h})} \delta_{c,\hbar}^I$ is $R_{\hbar}U(\mathfrak{g}) \otimes_{R_{\hbar}U(\mathfrak{b})} \mathbb{K}[B, \hbar]$, in other words, the Rees module of $U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}[B]$. The latter is the δ -function D-module of $B \subset G$ with the right B -action twisted by -2ρ . The δ -function D-module is strongly $B \times B$ -equivariant, which implies that on this D-module x_{ℓ} and x_r both act by 0 finishing the proof. $\square$

Finally, we will need a certain filtration on $\delta_{c,\hbar}^I$ viewed as an object of $\mathcal{O}$ via Φ_{ℓ}^* .

Lemma 4.2.3. *The object $\delta_{c,\hbar}^I \in \mathcal{O}$ has an exhaustive ascending filtration $\delta_{c,\hbar}^I = \bigcup_{\alpha} \delta_{c,\hbar,\leq \nu}^I$ by subobjects indexed by the elements ν in the $\mathbb{Z}_{\geq 0}$ -span on the positive affine roots, such that $\delta_{c,\hbar,\leq \nu}^I / \delta_{c,\hbar,< \nu}^I$ is the direct sum of objects $\Delta(\lambda^a)$ (note that there are no shifts in the PBW degrees, so that the highest weights of these modules are in PBW degree 0).*

Proof. We note that the grading on $\mathbb{K}[I]$ by the adjoint action of $T_{\ell} \times T$ is by the $\mathbb{Z}_{\leq 0}$ -span of positive roots. For an element μ in this span consider the span $\mathbb{K}[I, \hbar]_{\geq \mu}$ of the corresponding graded components. This is an $R_{\hbar}U(\mathfrak{i}^0)$ -submodule. We arrive at the exhaustive ascending filtration on the induced module $\delta_{c,\hbar}^I$ which satisfies the conditions in the lemma because $\mathbb{K}[I]$ is in PBW degree 0. $\square$

4.2.3. *Construction and properties of $\Delta^F(0)$.* We set

$$(4.2) \quad \Delta^F(0) := \text{Red}(\delta_{c,\hbar}^I).$$

We will need the following properties of $\Delta^F(0)$, which are exactly (a1) and (a2) from Section 3.1.

Lemma 4.2.4. *The following claims hold:*

- (1) *The PBW grading on $\Delta^F(0)$ is bounded from below by 0.*
- (2) *As a $\mathcal{EW}_{c,\hbar} \otimes S(\mathfrak{h})$ -module, $\Delta^F(0)$ is generated by a single I -invariant element.*
- (3) *As a $\mathcal{EW}_{c,\hbar}$ -module, $\Delta^F(0)$ is generated by finitely many I -invariant elements.*

Proof. (1): Recall that the functor F is t-exact and commutes with filtered colimits. Thanks to Lemma 4.2.3, $\Delta^F(0)$ admits an exhaustive ascending filtration whose associated graded is the direct sum of the objects $\text{Red}(\Delta(\lambda^a))$. By Proposition 1.5.4, the latter is isomorphic to $\underline{\Delta}(\lambda^a)$. The PBW degrees in each $\underline{\Delta}(\lambda^a)$ are bounded below by 0 from their construction in Section 1.5.1.

(2): First, we show the claim for $\Delta^F(0)/\hbar\Delta^F(0)$. Recall from Step 2 of the proof of Lemma 4.2.2 that $\delta_{c,\hbar}^I/\hbar\delta_{c,\hbar}^I$ is the algebra of functions on the locus $I \times (\hat{\mathfrak{g}}/\mathfrak{i}^0)^* \subset \mathcal{LG} \times \hat{\mathfrak{g}}^*$. Combining (1.7) and Lemma 1.5.3, we see that $\Delta^F(0)/\hbar\Delta^F(0)$ is the algebra of functions on $I \times S_{\mathfrak{h}^a}^+$ as a module over $\mathbb{K}[(\mathcal{LG} \times \mathcal{L}_c S) \times \mathfrak{h}^*]$. Since $I \times S_{\mathfrak{h}^a}^+$ is a closed I -stable subscheme $(\mathcal{LG} \times \mathcal{L}_c S) \times \mathfrak{h}^*$, for its generator we can take the unit function. Note that it is in PBW degree 0. Thanks to (1), we can uniquely lift this element to a PBW degree 0 element in $\Delta^F(0)$, denote it by v . This element is I -invariant and, again, thanks to (1), generates $\Delta^F(0)$ as a module over $\mathcal{EW}_{c,\hbar} \otimes S(\mathfrak{h})$.

(3): Recall from (1.12) that $S_{\mathfrak{h}^a}^+$ is of the form $S_c^+ \times_{\mathfrak{h}^*/W} \mathfrak{h}^*$ for the T_ℓ -contracting locus S_c^+ (and, in particular, a closed subscheme) in $\mathcal{L}_c S$. The proof of (2) shows that for generators we can take elements of the form $f_i v$, where f_i is a homogeneous basis of the free $S(\mathfrak{h})^W$ -module $S(\mathfrak{h})$. $\square$

Remark 4.2.5. Let $L_{\mathfrak{h}^a}$ denote the classical (as opposed to the derived) fiber of the point $1I$ under the projection $X_{\mathfrak{h}^a} \rightarrow \mathcal{F}l$. This fiber should be thought of as a family of lagrangian subvarieties over $\mathfrak{h}^{a,*}$, hence the notation. The computation of $\Delta^F(0)/\hbar\Delta^F(0)$ in (2) shows that under the identification of F with $\text{IndCoh}^{T_{\ell,d}}(X_{\mathfrak{h}^a})$, the object $\Delta^F(0)/\hbar\Delta^F(0)$ gets identified with $\mathcal{O}_{L_{\mathfrak{h}^a}}$.

We define $\Delta^F(n\delta) := \Delta^F(0) \otimes \mathbb{C}_n$, where $\mathbb{C}_n$ is the representation of T_ℓ corresponding to $t \mapsto t^n$.

4.3. Extensions to $\Delta^F(0)/\hbar\Delta^F(0)$. In proving various properties of $\Delta^F(0)$ (and more general modules $\Delta^F(\tilde{\lambda}^a)$ to be introduced later) we need to understand the properties of $\text{grExt}_{F_{R_0}}^*(?, \Delta_{R_0}^F(0))$, where $R_0 = R/(\hbar)$ and so $\Delta_{R_0}^F(0) = \Delta^F(0)/\hbar\Delta^F(0)$. Thanks to Lemma 4.1.2 we can view objects of F_{R_0} as $T_{\ell,d}$ -equivariant ind-coherent sheaves on $X_{\mathfrak{h}^a}$. Our main results are as follows.

Proposition 4.3.1. *Suppose that $\mathcal{F}$ is a $T_{\ell,d}$ -equivariant coherent sheaf on $X_{\mathfrak{h}^a}$ and $\mathcal{F}_0$ is a $T_{\ell,d}$ -equivariant coherent sheaf on X (where the coherence is in the sense of Section 4.1.3). Then the $\mathbb{K}$ -vector space*

$$\bigoplus_i \text{grExt}_{\text{IndCoh}^{T_{\ell,d}}(X)}^i(\mathcal{F}_0, \Delta_{\mathbb{K}}^F(0))$$

is finite dimensional and the graded $S(\mathfrak{h}^a)$ -module

$$\bigoplus_i \text{Ext}_{\text{IndCoh}^{T_{\ell,d}}(X_{\mathfrak{h}^a})}^i(\mathcal{F}, \Delta_{R_0}^F(0))$$

is finitely generated.

Lemma 4.3.2. *We have*

$$\dim \text{grExt}_{\text{IndCoh}^{T_{\ell,d}}(X)}^i(\Delta_{\mathbb{K}}(0), \Delta_{\mathbb{K}}(0)) = \delta_{i,0},$$

while

$$\text{grExt}_{\text{IndCoh}^{T_{\ell,d}}(X_{\mathfrak{h}^a})}^i(\Delta_{R_0}(0), \Delta_{R_0}(0))$$

is zero for $i \neq 0$, and $S(\mathfrak{h}^a)$ for $i = 0$.

4.3.1. Structure of formal neighborhoods. Here we prove some results about the structure of formal neighborhoods that go into the proofs of the results above. Our goal is to describe the formal neighborhood of $\Lambda_{1,\mathfrak{h}^a}$ in $X_{\mathfrak{h}^a}$.

Let $T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l$ denote the fiber of $1I$ under the projection $T_{\mathfrak{h}^a}^* \mathcal{F}l \rightarrow \mathcal{F}l$. It is identified with the pro-affine space $(\hat{\mathfrak{g}}/\mathfrak{i}^0)^*$. We start by describing the formal neighborhood of $T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l$ in $T_{\mathfrak{h}^a}^* \mathcal{F}l$ and identify it with the formal spectrum of a suitable completion of an algebra of polynomials in infinitely many variables. First, consider the algebra of regular functions $\mathbb{K}[T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l]$, which is the algebra of polynomials in infinitely many variables. The group $T_{\ell,d}$ acts rationally on $\mathbb{K}[T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l]$, the weights for the T_ℓ -action are non-positive and each weight space is finite dimensional.

Now consider the ind-unipotent group ind-scheme $I^{0,-}$, the ind-unipotent radical of the opposite Iwahori subgroup.

Lemma 4.3.3. *For any connected reductive group G , the formal neighborhood of 1 in $I^{-,0}$ is identified with the formal neighborhood $Z_-^\wedge$ of 0 in $\mathfrak{i}^{-,0}$ via the exponential map.*

Proof. First, we consider the case when $G = \mathrm{GL}_n$. For a $\mathbb{K}$ -algebra R , the points $I^{-,0}(R)$ is the matrices $(g_{ij})_{i,j=1}^n$ with invertible determinant, $g_{ij} \in R[t^{-1}]$ for $i > j$, $g_{ii} \in 1 + t^{-1}R[t^{-1}]$ and $g_{ij} \in t^{-1}R[t^{-1}]$ for $i < j$. In particular, if the coefficients of $g_{ij} - \delta_{i,j}$ are nilpotent in R , then such a matrix is automatically invertible. Moreover, it makes sense to speak about its logarithm, which is a point $\mathfrak{i}^{-,0}(R)$ with nilpotent coefficients. Conversely, for a point in $\mathfrak{i}^{-,0}(R)$ with nilpotent coefficients its exponential is a matrix in $I^{-,0}(R)$ with nilpotent $g_{ij} - \delta_{i,j}$. Also if the degrees in t^{-1} of matrices in $I^{-,0}(R)$ and $\mathfrak{i}^{-,0}(R)$ (with nilpotent coefficients) are bounded from above, then the same is true of their exp and log. This shows that exp is an ind-scheme isomorphism from $Z_-^\wedge$ to the formal neighborhood of 1 in $I^{-,0}$ with inverse log.

Now consider the case of general G . Embed G into GL_n , and let $\tilde{I}^-$ denote the opposite Iwahori for GL_n . For usual reasons, the maps exp and log between formal neighborhoods of 0 and 1 in $\mathfrak{i}^{-,0}$ and $\tilde{I}^{-,0}$ to those in $\mathfrak{i}^{-,0}$ and $I^{-,0}$. Now we are done by the previous paragraph. $\square$

The algebra of functions $\mathbb{C}[Z_-^\wedge]$ on $Z_-^\wedge$ can be described as follows: pick a weight topological basis for the loop rotation action in $\mathfrak{i}^{0,-,*}$, denote it by $y_j, j \in J$. Note that these elements have non-negative energy gradings. The elements of $\mathbb{C}[Z_-^\wedge]$ are all infinite sums of monomials in the variables y_j such that for every $N \in \mathbb{Z}_{\geq 0}$, there are only finitely many monomials of both energy degree and the degree of the monomial $\leq N$. For example, $\sum_{j \in J} y_j$ is a legitimate element of this algebra.

The action of $I^{0,-}$ on $T_{\mathfrak{h}^a}^* \mathcal{F}l$ restricts to a morphism

$$T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l \times Z_-^\wedge \rightarrow T_{\mathfrak{h}^a}^* \mathcal{F}l, (\alpha, \xi) \mapsto \exp(\xi) \cdot \alpha,$$

which, in turn yields an isomorphism between $T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l \times Z_-^\wedge$ (the product is taken in the category of ind-schemes, hence on the level of topological algebras corresponds to the completed tensor product) and the formal neighborhood of $T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l$ in $T_{\mathfrak{h}^a}^* \mathcal{F}l$.

Our next step is to understand the derived fiber product

$$(4.3) \quad (T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l \times Z_-^\wedge) \times_{\mathcal{L}_{\mathfrak{g}}^*}^R \mathcal{L}S.$$

Proposition 4.3.4. *(4.3) is $T_{\ell,d}$ -equivariantly and $\mathbb{K}[\mathfrak{h}^{a,*}]$ -linearly identified with the ind-spectrum of the following topological algebra: we complete $\mathbb{K}[\mathfrak{h}^{a,*}, \bigoplus_{k>0} \mathfrak{z}t^k, \bigoplus_{n<0} \mathfrak{z}_{\mathfrak{g}}(e)t^n]$ with respect to the sequence of ideals $\mathcal{I}_N$ generated by all elements of energy degree $\geq N$. Here $\mathfrak{z} = \mathfrak{z}_{\mathfrak{g}}(f)$.*

Proof. We introduce coordinates on $T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l \times Z_-^\wedge$ as follows. The coordinates are going to be $T_{\ell,d}$ -eigenvectors, where the action of T_d is the Kazhdan action. The coordinates on $T_{\mathfrak{h}^a, 1I}^*$ will be affine functions centered at e and will split into two groups. We fix an invariant form on $\hat{\mathfrak{g}} \oplus \mathbb{K}\mathrm{eu}$ (where eu is the Euler vector field) that allows us to identify $T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l = (\hat{\mathfrak{g}}/\mathfrak{i}^0)^*$ with $\mathfrak{i} \oplus \mathbb{K}\mathrm{eu}$. We fix a $T_{\ell,d}$ -stable vector space complement V_+ of $\mathfrak{h} \oplus \mathbb{K}\mathrm{eu} \oplus \bigoplus_{k>0} \mathfrak{z}t^k$ in $\mathfrak{i} \oplus \mathbb{K}\mathrm{eu}$.

We choose $T_{\ell,d}$ -weight bases

- x_j^1 (with j in an indexing set J_1) in $(\mathfrak{h} \oplus \mathbb{K}e)^* \oplus \bigoplus_{k>0} (\mathfrak{z}^{t^k})^*$ compatible with the direct sum decomposition, so that there is a subset J'_1 labeling a basis in the 2nd summand with $J_1 \setminus J'_1$ labeling a basis in the 1st summand,
- and x_j^2 (with $j \in J_2$) of $\{\xi - \langle \xi, e \rangle | \xi \in V_+^*\}$.

We choose coordinates on $\mathfrak{i}^{0,-}$ in a similar way: we choose a $T_{\ell,d}$ -stable complement V_- to $\bigoplus_{n<0} \mathfrak{z}_{\mathfrak{g}}(e)t^n$ in $\mathfrak{i}^{0,-}$, and then topological weight bases $y_j^1, j \in J'_1$, and $y_j^2, j \in J_2$, in the duals of these spaces. We assume that the energy degrees of x_j^1, y_j^1 are opposite for all $j \in J'_1$, and similarly for x_j^2, y_j^2 . Our goal is to modify the topological generators x_j^2, y_j^2 (still getting free topological generators that are weight vectors for $T_{\ell,d}$) so that the equations that cut (4.3) in $T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l \times Z_-^\wedge$ are $x_j^2 = y_j^2 = 0$. This will imply the claim of the proposition.

For this we need to compute the moment map $T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l \times Z_-^\wedge \rightarrow \mathcal{L}\mathfrak{g}$. Under the identifications $T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l \cong \mathfrak{i} \oplus \mathbb{K}d$ and $\mathfrak{g} \cong \mathfrak{g}^*$ from before, the map becomes

$$(x, y) \mapsto \text{pr}(\exp(\text{ad}(y))(e + x)),$$

where we write pr for the projection $\mathcal{L}\mathfrak{g} \oplus \mathbb{K}d \mapsto \mathcal{L}\mathfrak{g}$. The condition that this element lies in $e + \mathcal{L}\mathfrak{z}$ imposes a collection of equations on (x, y) that are affine linear in x .

We introduce a partial order on monomials in x_j^2 and y_j^2 of fixed energy degree, say, m , in the following manner: two monomials are first ordered in the increasing order of energy of the y -part, then by the total degree in y 's, again in the increasing order, and finally in the increasing order in the degrees in x 's. For example, the largest monomials of energy $m > 0$ are always the variables y_j^2 .

We consider separately equations on x, y that have positive energy degree, negative energy degree, and zero energy degree.

First, the positive degree case. After replacing this (infinite) system of equations with suitable non-degenerate linear combinations (in each energy degree separately), we can assume that there is the unique maximal monomial in each of the equations and this monomial is y_j^2 with $j \in J_2$, which comes from the summand $[e, y]$. This gives a bijection between the positive degree equations and the subset of J_2 corresponding to the variables y_j^2 of positive energy degree.

Similarly, for the equations of negative energy degree (after a passing to non-degenerate linear combinations), the unique maximal monomial is some x_j^2 (coming from the summand x) with $j \in J_2$. This gives a bijection between such equations and the subset of J'_1 corresponding to the variables x_j^2 of negative energy degree.

Finally, we consider the energy degree 0 equations. There are $\dim \mathfrak{g} - \text{rk } \mathfrak{g}$ of them. After passing to their non-degenerate linear combinations, we can assume that they are homogeneous with respect to the principal grading on $\mathfrak{g}$. And after a further non-degenerate linear combination, we see that each has a unique minimal monomial: for the non-positive principal degrees, it is of the form y_j^2 , and for the positive principal degree it is of the form x_j^2 (for j such that the corresponding elements are of energy degree 0).

Now we replace the variables x_j^2, y_j^2 with $j \in J_2$ with the left hand sides of the corresponding equations. Thanks to our ordering on the monomials above, the modified elements together with $x_i^1, y_{i'}^1$ (where $i \in J_1, i' \in J'_1$) still form free topological generators of $\mathbb{K}[T_{\mathfrak{h}^a, 1I}^* \mathcal{F}l \times Z_-^\wedge]$ so that every element is uniquely written as an infinite converging sum of monomials in these variables and the convergence is determined in the same way

as before: for the infinite sum of monomials to converge, for each N we must have only finitely many monomials with both the total degree and the total energy of the y -part $\leq N$. This finishes the proof. $\square$

4.3.2. *Roadmap.* Let us explain how we are going to prove Proposition 4.3.1 and Lemma 4.3.2. We will handle the case of $X_{\mathfrak{h}^a}$, the case of X formally follows – or can be proved in a similar way.

Let us explain how the computation goes. We modify the category in which the Ext's are taken as follows. We write $\Lambda^\wedge$ for the formal neighborhood of $\Lambda_{1,\mathfrak{h}^a}$, by Proposition 4.3.4, its algebra of functions, $\mathbb{K}[\Lambda^\wedge]$, is the completion of $\mathbb{K}[\mathfrak{h}^{a,*}, \bigoplus_{k>0} \mathfrak{z}t^k, \bigoplus_{n<0} \mathfrak{z}_{\mathfrak{g}}(e)t^n]$, the inverse limit over $N \in \mathbb{Z}_{>0}$ by the sequence of ideals generated by all elements of energy $\geq N$. To simplify the notation, we write U^- for $\bigoplus_{k>0} (\mathfrak{z}t^k)^*$ and U^+ for $\bigoplus_{n<0} (\mathfrak{z}_{\mathfrak{g}}(e)t^n)^*$. So $\mathbb{K}[\Lambda^\wedge]$ is the completion of $\text{Sym}(\mathfrak{h}^a \oplus U^- \oplus U^+)$. The elements of $\mathfrak{h}^a$ (resp., U^- , resp. U^+) have zero (resp., negative, resp., positive) energy degrees.

We reduce the computation of Ext's to a suitable pro-completion of the category of discrete modules over $\mathbb{K}[\Lambda^\wedge]$. Note that $\Delta_{R_0}(0)$ corresponds to the $\mathbb{K}[\Lambda^\wedge]$ -module $\text{Sym}(\mathfrak{h}^a \oplus U^-)$. We'll see that we need to compute the energy degree 0 Ext over $\mathbb{K}[\Lambda^\wedge]$ to this module from a module $\tilde{M}$ of the following form. We have direct sum decompositions into energy graded subspaces $U^\pm = U_1^\pm \oplus U_2^\pm$ with $\dim U_2^\pm < \infty$. We have

$$(4.4) \quad \tilde{M} = \text{Sym}(U_1^-) \hat{\otimes}_{\mathbb{K}} M,$$

where M is a finitely generated module over $\mathbb{K}[\mathfrak{h}^{a,*}, U_2^{-,*}][[U_2^{+,*}]]$, and $\hat{\otimes}$ stands for the completed tensor product; in particular U_1^+ acts on $\tilde{M}$ by 0.

4.3.3. *Construction for schemes.* Here we consider the following situation. Let $\underline{Z}$ be a finite type scheme over $\mathbb{K}$ and $\underline{\Lambda}$ be a closed subscheme of $\underline{Z}$ isomorphic to an affine space. We will also assume that $\underline{Z}$ comes with a $T_{\ell,d}$ -action that preserves $\underline{\Lambda}$.

Now let Z be the total space of a $T_{\ell,d}$ -equivariantly trivial pro-vector bundle $\mathcal{E}$ on $\underline{Z}$, in particular, Z is a scheme, a priori of infinite type. Let V denote the fiber of the fixed point so that $Z = \underline{Z} \times V$. Let Λ be the preimage of $\underline{\Lambda}$, this is an affine space, possibly infinite dimensional. We still assume that T_ℓ contracts Λ to a point. Let $\pi : Z \rightarrow \underline{Z}$ denote the projection. We will be interested in quasi-coherent sheaves of the form $\mathcal{F} := \pi^*(\underline{\mathcal{F}})$, where $\underline{\mathcal{F}}$ is a $T_{\ell,d}$ -equivariant coherent sheaf on $\underline{Z}$.

Let $\underline{\Lambda}^i$ denote the i -th infinitesimal neighborhood of $\underline{\Lambda}$ in $\underline{Z}$, it is given by the i -th power of the defining ideal of $\underline{\Lambda}$ in $\underline{Z}$. Let $\Lambda^i := \pi^{-1}(\underline{\Lambda}^i)$. Then we can consider the formal neighborhoods $\underline{\Lambda}^\wedge := \text{colim } \underline{\Lambda}^i$ and $\Lambda^\wedge := \text{colim } \Lambda^i$.

We define the pullback of $\mathcal{F}$ to $\Lambda^\wedge$ as the pro-object $\varprojlim \mathcal{F} \otimes_{\mathcal{O}_Z}^L \mathcal{O}_{\Lambda^i}$. Note that we have compatible (for various i) isomorphisms:

$$\mathcal{F} \otimes_{\mathcal{O}_Z}^L \mathcal{O}_{\Lambda^i} \xrightarrow{\sim} \pi^*(\underline{\mathcal{F}} \otimes_{\mathcal{O}_{\underline{Z}}}^L \mathcal{O}_{\underline{\Lambda}^i}) \xrightarrow{\sim} (\underline{\mathcal{F}} \otimes_{\mathcal{O}_{\underline{Z}}}^L \mathcal{O}_{\underline{\Lambda}^i}) \otimes_{\mathbb{K}} \mathbb{K}[V]$$

Note that the inverse limit $\varprojlim \mathcal{F} \otimes_{\mathcal{O}_Z}^L \mathcal{O}_{\Lambda^i}$ is just the restriction of $\underline{\mathcal{F}}$ to $\underline{\Lambda}^\wedge$ (to be denoted by $\underline{\mathcal{F}}^\wedge$), as an object in the derived category of coherent sheaves on $\underline{\Lambda}^\wedge$. So, as a pro-sheaf on $\Lambda^\wedge$, $\varprojlim \mathcal{F} \otimes_{\mathcal{O}_Z}^L \mathcal{O}_{\Lambda^i}$, is the completed tensor product $\mathbb{K}[V] \hat{\otimes}_{\mathbb{K}} \underline{\mathcal{F}}^\wedge$.

Now we are going to explain an application to the computation of certain Ext spaces.

Lemma 4.3.5. *For each k we have a natural isomorphism*

$$\mathrm{Ext}_{\Lambda^\wedge}^k(\mathcal{F}^\wedge, \mathcal{O}_\Lambda) \xrightarrow{\sim} \mathrm{Ext}_Z^k(\mathcal{F}, \mathcal{O}_\Lambda).$$

The analogous claim holds for the $T_{\ell,d}$ -equivariant categories, where we twist $\mathcal{O}_\Lambda$ by a character of $T_{\ell,d}$.

Proof. We will only prove the non-equivariant statement, the equivariant version is analogous.

Note that the left hand side is by definition

$$\mathrm{colim}_i \mathrm{Ext}_{\Lambda^i}^k(\mathcal{F} \otimes_{\mathcal{O}_Z}^L \mathcal{O}_{\Lambda^i}, \mathcal{O}_\Lambda).$$

In particular each of the terms in the colimit admits a homomorphism to the right hand side and these homomorphisms form a directed system. Hence we get a natural homomorphism from the left hand side to the right hand side, and we need to show that it is an isomorphism. Recall that $\mathcal{F} = \pi^* \underline{\mathcal{F}}$, hence we have natural isomorphisms:

$$\begin{aligned} \mathrm{Ext}_Z^k(\mathcal{F}, \mathcal{O}_\Lambda) &\xrightarrow{\sim} \mathrm{Ext}_{\underline{Z}}^k(\underline{\mathcal{F}}, \pi_* \mathcal{O}_\Lambda), \\ \mathrm{Ext}_{\Lambda^i}^k(\mathcal{F}, \mathcal{O}_\Lambda) &\xrightarrow{\sim} \mathrm{Ext}_{\underline{Z}}^k(\underline{\mathcal{F}} \otimes_{\mathcal{O}_Z}^L \mathcal{O}_{\Lambda^i}, \pi_* \mathcal{O}_\Lambda). \end{aligned}$$

Note that $\pi_* \mathcal{O}_\Lambda = \mathcal{O}_{\underline{\Lambda}} \otimes \mathbb{K}[V]$. It is easy to see (for example, by considering locally free/ projective resolutions) that both Ext^k commute with infinite direct sums in the 2nd argument. Then our claim reduces to computing Ext 's between coherent sheaves, where it is classical. $\square$

4.3.4. Construction for ind-schemes. Now suppose that we are in the following situation: we have a sequence of closed embeddings of

$$Z_0 \hookrightarrow Z_1 \hookrightarrow Z_2 \hookrightarrow \dots \hookrightarrow Z_\ell \hookrightarrow \dots$$

of possibly infinite type schemes. Consider the ind-scheme $Z := \mathrm{colim}_\ell Z_\ell$.

We assume that these schemes come equipped with $T_{\ell,d}$ -actions and each Z_i is a $T_{\ell,d}$ -equivariantly trivial pro-vector bundle over a finite type scheme $\underline{Z}_i$. We also assume that $\underline{Z}_0 = \Lambda$ is an infinite dimensional affine space. We then can consider $\Lambda_\ell^i \subset Z_\ell$ defined as in Section 4.3.3, and set

$$\Lambda_\ell^\wedge := \mathrm{colim}_i \Lambda_\ell^i, \quad \Lambda^\wedge := \mathrm{colim}_\ell \Lambda_\ell = \mathrm{colim}_{\ell,i} \Lambda_\ell^i.$$

We refer to $\Lambda^\wedge$ as the formal neighborhood of Λ in Z .

Now consider a sheaf $\mathcal{F}$ on Z_1 of the form explained in Section 4.3.3. We can view it as a sheaf on Z via pushforward. On the other hand, we can form the completion $\mathcal{F}^\wedge$ and view it as a pro-sheaf on $\Lambda_1^\wedge$ and then on $\Lambda^\wedge$ via pushforward. We then have the following generalization of Lemma 4.3.5.

Lemma 4.3.6. *For each k we have a natural isomorphism*

$$\mathrm{Ext}_{\Lambda^\wedge}^k(\mathcal{F}^\wedge, \mathcal{O}_\Lambda) \xrightarrow{\sim} \mathrm{Ext}_Z^k(\mathcal{F}, \mathcal{O}_\Lambda).$$

Proof. The left hand side is, by definition,

$$\mathrm{colim}_{\ell,i} \mathrm{Ext}_{\Lambda_\ell^i}^k(\mathcal{F} \otimes_{\mathcal{O}_{Z_1}}^L \mathcal{O}_{\Lambda_1^i}, \mathcal{O}_\Lambda) \xrightarrow{\sim} \mathrm{colim}_\ell \mathrm{Ext}_{\Lambda_\ell^\wedge}^k(\mathcal{F}^\wedge, \mathcal{O}_\Lambda)$$

while the right hand side is $\mathrm{colim}_\ell \mathrm{Ext}_{Z_\ell}^k(\mathcal{F}, \mathcal{O}_\Lambda)$. Now the present lemma follows from Lemma 4.3.5 by changing the order in taking (filtered) colimits. $\square$

4.3.5. *Completion of coherent sheaves on X .* We can apply Lemma 4.3.6 to $Z := X_{\mathfrak{h}^a}$, $\Lambda := \Lambda_{\mathfrak{h}^a}$. Our goal now is to describe $\mathcal{F}^\wedge$ as announced in Section 4.3.2.

Proposition 4.3.7. *In Proposition 4.3.4, one can modify the embedding of U^+ into $\mathbb{K}[\Lambda^\wedge]$ (see the conventions in the beginning of Section 4.3.2) with the following property:*

- *there is a $T_{\ell,d}$ -stable decomposition $U^+ = U_1^+ \oplus U_2^+$ with finite dimensional U_1^+ such that U_2^+ acts by 0 on $\mathcal{F}^\wedge$.*

Proof. The action of $\mathbb{K}[\Lambda^\wedge]$ on $\mathcal{F}^\wedge$ factors through $\mathbb{K}[\Lambda_\ell^\wedge]$ for suitable ℓ (we can assume it to be equal to 1). We embed $\mathfrak{h}^{a,*}$ into $X_{\mathfrak{h}^a}$ as the T_ℓ -fixed point component corresponding to $x = 1$. We claim that only finitely many weights of the loop rotation $\mathbb{G}_m$ on the fibers of the normal bundle $N_{\mathfrak{h}^a,*}(\Lambda_\ell^\wedge) = N_{\mathfrak{h}^a,*}Z_\ell$ are non-positive. Indeed, recall that Z_ℓ is the total space of an $T_{\ell,d}$ -equivariantly trivial pro-vector bundle over a finite type derived scheme $\underline{Z}_\ell$. The weights of T_ℓ in the fiber of this vector bundle over the points of the T_ℓ -fixed component $\mathfrak{h}^{a,*}$ are all positive, so our claim follows because $\underline{Z}_\ell$ is finite type.

Let s be the maximal energy degree that appears in $N_{\mathfrak{h}^a,*}(\Lambda_\ell^\wedge)$. For U_1^+ we take the direct sum of energy graded components in $U^+(\hookrightarrow \mathbb{K}[\Lambda^\wedge])$ with degree $\leq s$ and for U_2^+ we take the direct sum of the remaining graded components. If $y_j \in U_2^+$, then the image of y_j in the quotient $\mathbb{K}[\Lambda_\ell^\wedge]$ of $\mathbb{K}[\Lambda^\wedge]$ coincides with the image of an infinite sum of monomials in x_i 's and y_i for $i \in J'_1$ (this excludes the coordinates on $\mathfrak{h}^{a,*}$) with coefficients in $\mathbb{K}[\mathfrak{h}^{a,*}]$, where all monomials have degree at least 2 and the energy degree equal to that of y_j . This is because y_j vanishes in the normal bundle.

For each j with $y_j \in U_2^+$, we can replace y_j with the difference of y_j and this infinite sum. The resulting elements y_j together with the x_i 's are still free topological generators of $\mathbb{K}[\Lambda^\wedge]$ as a topological algebra $\mathbb{K}[\mathfrak{h}^{a,*}]$. On the other hand, by construction, the image of U_2^+ in $\mathbb{K}[\Lambda_\ell^\wedge]$ is equal to 0. $\square$

Since $\mathcal{F}$ is obtained as a pullback from a finite type scheme $\underline{Z}_\ell$, we arrive at the following description of $\mathcal{F}^\wedge$.

Corollary 4.3.8. *$\tilde{M} := \mathcal{F}^\wedge$ admits decomposition (4.4) satisfying the conditions after this equation.*

4.3.6. *Ext computation.* Suppose $\mathcal{F}^\wedge$ is described as in Corollary 4.3.8. To simplify the notation, let $A := \mathbb{K}[\mathfrak{h}^{a,*}, U_2^{-,*}][[U_2^{+,*}]]$.

Consider the restricted dual $U_1^{+,(*)}$, the direct sum of the duals of (finite dimensional) energy graded components. So we can talk about the (homologically graded) free algebra $\text{Sym}(U_1^-) \otimes \Lambda(U_1^{+,(*)})$, where the elements of $U_1^{+,(*)}$ are in cohomological degree 1. Note that the energy degrees in this algebra are in $\mathbb{Z}_{\leq 0}$, and the weight spaces are finite dimensional. The degree 0 subspace consists of constants.

Then we have the following result computing the Ext's.

Proposition 4.3.9. *The following claims hold:*

- (1) *We have the following natural (in particular, energy graded) isomorphism*

$$(4.5) \quad \text{grExt}_{\mathbb{F}_{R_0}}^*(\mathcal{F}, \Delta_{R_0}(n\delta)) \xrightarrow{\sim} \left[\mathbb{C}_n \otimes \text{Ext}_A^*(M, \mathbb{K}[\mathfrak{h}^{a,*}, U_2^{-,*}]) \otimes \text{Sym}(U_1^-) \otimes \Lambda(U_1^{+,(*)}) \right]^{T_\ell},$$

where we write $\mathbb{C}_n$ for the 1-dimensional space on which T_ℓ by $t \mapsto t^n$.

- (2) (4.5) is a finitely generated $S(\mathfrak{h}^a)$ -module.
- (3) If $\mathcal{F} = \Delta_{R_0}^F(n\delta)$, then (4.5) is $\mathbb{K}[\mathfrak{h}^{a,*}]^{\oplus \delta_{n,0}}$ in cohomological degree 0.

We note that (2) implies Proposition 4.3.1, while (3) implies Lemma 4.3.2.

Proof. By Lemma 4.3.6, we have an identification

$$\mathrm{Ext}_{X_{\mathfrak{h}^a}}^*(\mathcal{F}, \Delta_{R_0}(0)) \xrightarrow{\sim} \mathrm{Ext}_{\Lambda^\wedge}^*(\mathcal{F}^\wedge, \Delta_{R_0}(0)).$$

Now (1) easily follows from (4.4).

Part (3) follows in a straightforward way. To prove part (2), we need study the functor $\mathrm{Ext}_A^*(M, ?)$. Note that A has finite homological dimension. Also $\mathrm{Ext}_A^*(M, \mathbb{K}[\mathfrak{h}^{a,*}, U_2^{-,*}])$ is a finitely generated A -module, where the A -action factors through its quotient $\mathbb{K}[\mathfrak{h}^{a,*}, U_2^{-,*}]$. The energy grading on $\mathbb{K}[\mathfrak{h}^{a,*}, U_2^{-,*}]$ is non-positive, and each degree component is finitely generated over $\mathbb{K}[\mathfrak{h}^{a,*}]$. (2) follows from the facts about $\mathrm{Sym}(U_1^-) \otimes \Lambda(U_1^{+,*})$ quoted before the proposition. $\square$

4.4. Affine braid group action and objects $\Delta^F(\check{\lambda}^a)$. We consider the extended affine braid group Br^{ea} for G . Let $p : \mathrm{Br}^{ea} \twoheadrightarrow W^{ea} = W \ltimes \check{\Lambda}$ denote the natural projection. For a simple affine root α_i , let T_i be the corresponding generator of Br^{ea} .

The goal of this section is to construct (weak) actions of Br^{ea} on $\mathbf{F}, \mathbf{H}$ by self-equivalences of graded triangulated categories and study its properties.

4.4.1. Construction in the strongly equivariant setting. We will start by constructing an action of Br^{ea} on related but different categories and establish some properties. Then we will use a Koszul duality construction to pass to the categories we care about.

Set $\tilde{G} := T_\ell \ltimes \mathcal{L}G$ and let $\tilde{I} := T_\ell \ltimes I$, so that $\tilde{I} = \tilde{H} \ltimes I^0$, where $\tilde{H} = T_\ell \times H$. Note that Λ^a is exactly the character lattice of $\tilde{H}$. In $\tilde{\mathfrak{h}}^*$ we have the element $-\rho^a$.

Consider the ind-stack $\tilde{I} \backslash \tilde{G} / I^0 = I \backslash \mathcal{L}G / I^0$. It makes sense to speak about twisted D-modules on this ind-stack and also about twisted mixed Hodge modules. We choose the twist to be $-\rho^a$. It also makes sense to talk about weakly $\tilde{H}$ -equivariant structures on such mixed Hodge modules.

The derived category of $(-\rho^a)$ -twisted weakly $\tilde{H}$ -equivariant mixed Hodge modules on $\tilde{I} \backslash \tilde{G} / I^0$ is denoted by $\mathbf{MHM}(\tilde{I} \backslash \tilde{G} / I^0)$. On the other hand, we can consider the (derived) category $\mathbf{H}_{\tilde{G},0}$ of weakly $H \times T_\ell \times T_d$ -equivariant and strongly $I \times I^0$ -equivariant $\mathrm{CDO}_{0,\hbar}(G)$ -modules.

Now suppose $\tilde{J} \subset \tilde{G}$ is a standard parahoric with standard Levi $\tilde{L}$. Inside $\tilde{L}$ we consider the Borel subgroup $B_L := \tilde{I} \cap \tilde{L}$. we can form the category $\mathbf{H}'_{\tilde{J}}$ of strongly $(B_L \times B_L, (-\rho^a \hbar, -\rho^a \hbar))$ -equivariant and weakly T_d -equivariant $D_{\hbar}(\tilde{L})$ -modules. Note that $\tilde{I} \backslash \tilde{J} / \tilde{I}$ is actually a finite type smooth quotient stack.

Note that we have forgetful functors $\varphi'_{\tilde{J}} : \mathbf{MHM}(\tilde{I} \backslash \tilde{J} / \tilde{I}) \rightarrow \mathbf{H}'_{\tilde{J}}$. Namely, from a mixed Hodge module we get a D-module with the Hodge filtration, take the corresponding module over the Rees algebra. Thanks to the twist we impose the functors $\varphi'_{\tilde{J}}$ are monoidal: this is based on the compatibility of the functors $\varphi'_?$ with proper pushforwards and smooth pullbacks. By the construction, the functors $\varphi'_?$ are t-exact (and the objects in the image are flat over $\mathbb{K}[\hbar]$).

Remark 4.4.1. Note that we have the pull-push functor $H'_{\tilde{J}_1} \hookrightarrow H_{\tilde{J}_2}$ for every inclusion of standard parahorics $\tilde{J}_1 \rightarrow \tilde{J}_2$ so that we have the following commutative diagram, where the vertical arrows are $\varphi'_?$:

$$(4.6) \quad \begin{array}{ccc} \mathrm{MHM}(\tilde{I} \backslash \tilde{J}_1 / \tilde{I}) & \longrightarrow & \mathrm{MHM}(\tilde{I} \backslash \tilde{J}_2 / \tilde{I}) \\ \downarrow & & \downarrow \\ H'_{\tilde{J}_1} & \longrightarrow & H'_{\tilde{J}_2}. \end{array}$$

We note that the functors in this diagram are monoidal.

The first named author and Fedorov in [11] generalized the construction of the functors φ to certain ind-stacks including $\tilde{I} \backslash \tilde{G} / I^0$: in this situation we get the forgetful functor

$$(4.7) \quad \varphi_{\tilde{G},0} : \mathrm{MHM}(\tilde{I} \backslash \tilde{G} / I^0) \rightarrow H_{\tilde{G},0}.$$

that has the following properties:

- (a) $\varphi_{\tilde{G},0}$ is t-exact;
- (b) The construction in Section ?? equips $H_{\tilde{G},0}$ with the structure of a module over $H'_{\tilde{J}}$ (via convolution) for every standard parahoric $\tilde{J}$. The category $\mathrm{MHM}(\tilde{I} \backslash \tilde{J} / \tilde{I})$ acts on $H_{\tilde{G},0}$ via the monoidal functor $\varphi'_{\tilde{J}}$. Then $\varphi_{\tilde{G},0}$ is a functor between module categories over $\mathrm{MHM}(\tilde{I} \backslash \tilde{J} / \tilde{I})$.

We will need a generalization. Let $A_{\hbar}$ be a graded complete $\mathbb{C}[\hbar]$ -algebra equipped with a strong $\tilde{G}$ -action. In particular, we have a categorical action of $\tilde{J}$ on the (dg) category of smooth graded $A_{\hbar}$ -modules and consider the strongly $\tilde{I}$ -equivariant $A_{\hbar}$ -modules, cf. Section ?? . In particular, for $A_{\hbar} := \mathcal{E}\mathcal{W}_{0,\hbar}$, we get the strongly equivariant (still with the twist by $-\rho^a \hbar$) analog F' . As in (b), it is a module over $H'_{\tilde{J}}$ for every proper parabolic $\tilde{J}$.

Note that we have the reduction functor $\mathrm{Red} : H_{\tilde{G},0} \rightarrow F'$. It comes from a $\tilde{J}$ -equivariant functor between the category of strongly $\tilde{J}$ -equivariant graded smooth $D_{0,\hbar}(\mathcal{L}G)$ -modules and the category of graded smooth $\mathcal{E}\mathcal{W}_{0,\hbar}$ -modules. In particular by Lemma ?? we have the following:

Lemma 4.4.2. *The functor $\mathrm{Red} : H_{\tilde{G},0} \rightarrow F'$ is $H'_{\tilde{J}}$ -equivariant for every proper parahoric subgroup $\tilde{J}$.*

With this preparation we can proceed to the (weak) action of Br^{ea} . First, we define the action of the affine braid group Br^a . Let $\tilde{J}_i$ be the minimal parahoric in $\tilde{G}$ corresponding to the simple (affine) root α_i . Consider the object $j'_{i,!} \in \mathrm{MHM}(\tilde{I} \backslash \tilde{J}_i / \tilde{I})$, the shriek push-forward from the open orbit. We define the endofunctors T_i of $H_{\tilde{G},0}, F'$ as the action of $\varphi'_{\tilde{J}_i}(j'_{i,!})$.

Lemma 4.4.3. *The endofunctors T_i define weak actions of Br^a on $H_{G,0}, F'$. Moreover, Red intertwines these actions: $\beta * \mathrm{Red}(?) \cong \mathrm{Red}(\beta * ?)$.*

Proof. The inverse T_i^{-1} is given by convolution with $\varphi'_{\tilde{J}_i}(j'_{i,*})$. Moreover, thanks to Remark 4.4.1, T_i is isomorphic to the functor of convolving with $\varphi'_{\tilde{J}}(j'_{i,!})$ for any proper parahoric $\tilde{J}$. It remains to note that all braid relations can be checked inside the monoidal category $H'_{\tilde{J}}$ for some proper parahoric $\tilde{J}$. $\square$

Now we explain how to extend the action to Br^{ea} . Let σ be a length 0 element in W^{ea} . We can lift it to a $\mathbb{K}$ -point of $\tilde{G}$ stabilizing $\tilde{H}$ and $\tilde{I}$ to be denoted by the same letter. It is defined up to multiplication by a point of $\tilde{H}$. Note that in the general setting of A_h above, σ acts on A_h by a graded algebra automorphism. Since σ normalizes $\tilde{I}$, the action extends to the derived category of strongly I -equivariant graded A_h -modules by an equivalence and, up to isomorphism, this self-equivalence is independent of the choice of a lift. We make σ to act by this self-equivalence. It is easy to see that this action extends the Br^a -action in Lemma 4.4.3 to Br^{ea} .

4.4.2. Strongly equivariant setting: properties. Our first task is to produce certain objects in $H_{\tilde{G},0}, F'$.

Note that $\check{\Lambda}$ embeds into Br^{ea} : for dominant $\check{\lambda} \in \check{\Lambda}$ the element $t_{\check{\lambda}}$ is sent to $T_{t_{\check{\lambda}}}$, this uniquely specifies the embedding. Let $X_{\check{\lambda}}$ denote the corresponding element of the braid group. The corresponding object in $\mathrm{MHM}(\tilde{I} \backslash \tilde{G} / \tilde{I})$, i.e., the image of the monoidal unit under the action of $X_{\check{\lambda}}$, is known as the *Wakimoto sheaf*, denote it by $\mathcal{X}_{\check{\lambda}}$. It lies in the heart of the natural t-structure.

Consider the lattice $\check{\Lambda}^a := \check{\Lambda} \oplus \mathbb{Z}\delta$. For $\check{\lambda}^a := \check{\lambda} + n\delta$ we set

$$(4.8) \quad \mathrm{Wak}_0(\check{\lambda}^a) := \mathbb{C}_n \otimes \varphi_{\tilde{G},0}(\mathcal{X}_{\check{\lambda}}), \quad \Delta_0^F(\check{\lambda}^a) := \mathrm{Red}(\mathrm{Wak}_0(\check{\lambda}^a)).$$

The following properties follow because Red is t-exact and $\varphi_{\tilde{G},0}$ is t-exact with image flat over $\mathbb{C}[\hbar]$.

Lemma 4.4.4. *The objects $\mathrm{Wak}_0(\check{\lambda}^a), \Delta_0^F(\check{\lambda}^a)$ lie in the hearts of the t-structures on the relevant categories and, moreover, are flat over $\mathbb{C}[\hbar]$.*

Next, we need to investigate the preservation of coherence.

Definition 4.4.5. We say that $\mathcal{F} \in H_{\tilde{G},0}$ is *coherent* if:

- Only finitely many cohomology groups $H^i(\mathcal{F})$ are nonzero;
- the PBW grading on each $H^i(\mathcal{F})$ is bounded from below,
- and $\mathbb{K} \otimes_{\mathbb{K}[\hbar]} H^i(\mathcal{F})$ is a coherent sheaf on $T^*(\tilde{I} \backslash \tilde{G} / I^0) / (H \times T_{\ell})$ in the sense analogous to Remark 4.1.3.

The main result of this section is:

Proposition 4.4.6. *The action of Br^{ea} sends coherent objects to coherent objects.*

Proof. It is enough to show that each T_i preserves the coherence – for the length 0 elements this is clear by the definition. Recall that T_i is the convolution with $\varphi'_{\tilde{J}_i}(j'_{i,1})$. The object $j'_{i,1} \in \mathrm{MHM}(\tilde{I} \backslash \tilde{J}_i / \tilde{I})$ fits into a short exact sequence

$$(4.9) \quad 0 \rightarrow j_{1,!} \rightarrow j_{i,!} \rightarrow L_i \rightarrow 0,$$

where $j_{1,!}$ is the monoidal unit and L is the IC sheaf corresponding to the open cell. Up to twists with characters of a cover of $\tilde{I}$ (that obviously preserve the coherence), the object $\varphi'_{\tilde{J}_i}(L)$ is $\mathcal{O}_{\tilde{I} \backslash \tilde{J}_i / \tilde{I}}[\hbar]$. The convolution with this object decomposes as the composition of two functors. First, we need the category ${}^iH_{\tilde{G},0}$ of strongly $\tilde{J}_i \times I^0$ -equivariant $D_{c_0,h}(\mathcal{LG})$ -modules, where c_0 is a suitable character. We have the forgetful functor $\eta : {}^iH_{\tilde{G},0} \rightarrow H_{\tilde{G},0}$ and its right adjoint, the homogenized D-module pushforward functor, to be denoted by

η^R . Then the convolution with $\mathcal{O}_{\tilde{I} \setminus \tilde{J}_i / \tilde{I}}[\hbar]$ is (up to a shift and a twist) to composition $\eta \circ \eta^R$.

One can define coherent objects in ${}^i\mathbf{H}_{\tilde{G},0}$ similarly to Definition 4.4.5. It is clear that η sends coherent objects $\mathcal{F}$ to coherent objects, so we just need to do this for η^R . It is enough to consider a coherent object $\mathcal{F}$ in a single cohomological degree. The object $\eta^R(\mathcal{F})$ is obtained in the following way. Let x be a nonzero element of $\mathfrak{j}_i$ of weight $-\alpha_i$. Then we impose the equation $x - \hbar L_x$ (in the derived sense) on $\mathcal{F}$ and then induce the resulting object from $\tilde{I}$ to $\tilde{J}_i$ (as an algebraic group representation; with a possible twist by a character). Both operations have cohomological amplitude 1, the former shifts the PBW degree by at most 1, while the 2nd preserves the PBW degrees.

It remains to check that $\mathbb{K} \otimes_{\mathbb{K}[\hbar]}^L \eta^R(\mathcal{F})$ is coherent. At $\hbar = 0$, up to line bundle twist and homological twist, the functor η^R is the pullpush via the correspondence

$$T^*(\tilde{I} \setminus \tilde{G} / I_0) / \tilde{H} \leftarrow \tilde{I} \setminus \tilde{G} / \tilde{I} \times_{\tilde{J}_i \setminus \tilde{G} / \tilde{I}} T^*(\tilde{J}_i \setminus \tilde{G} / I_0) / \tilde{H} \rightarrow T^*(\tilde{J}_i \setminus \tilde{G} / I_0) / \tilde{H}.$$

It makes sense to speak about coherent sheaves on the intermediate term, and both the pullback and the pushforward functor are easily seen to preserve the coherence. $\square$

Remark 4.4.7. We do not know a good definition of coherence for $\mathbf{F}'$ because the PBW grading on $\mathcal{E}\mathcal{W}_{0,\hbar}$ is not bounded from below. The description of the functor $\varphi'_{\tilde{J}_i}(L)$ at $\hbar = 0$ also applies to $\text{IndCoh}^{T_{\ell,d}}(X)$ and shows that η^R (and hence T_i) preserves the coherence in $\text{IndCoh}^{T_{\ell,d}}(X)$.

Corollary 4.4.8. *The following objects are coherent in the suitable categories: $\mathbf{Wak}_0(\check{\lambda}^a)$ and $\Delta_0^{\mathbf{F}}(\check{\lambda}^a) / \hbar \Delta_0^{\mathbf{F}}(\check{\lambda}^a)$.*

4.4.3. Koszul duality. The goal of this section is to produce equivalences between $\mathbf{H}$ and $\mathbf{H}_{\tilde{G},0}$ and between $\mathbf{F}$ and $\mathbf{F}'$. These equivalences will allow to carry the action of Br^{ea} from $\mathbf{H}_{\tilde{G},0}, \mathbf{F}'$ to $\mathbf{H}, \mathbf{F}$.

We embed $\mathfrak{h}^a := \mathfrak{h} \oplus \mathbb{K}c'$ into $\mathfrak{h}^{da}$ via $x \mapsto x - \langle x, \rho^a \rangle \hbar$. The categories $\mathbf{H}, \mathbf{F}$ are graded $\mathbf{R}$ -linear, in particular, graded $S(\mathfrak{h}^a)$ -linear. The functor

$$\mathfrak{K} := R\text{grHom}_{S(\mathfrak{h}^a)}(\mathbb{K}, ?)[\dim \mathfrak{h}^a] \langle -\dim \mathfrak{h}^a \rangle \cong \mathbb{K} \otimes_{S(\mathfrak{h}^a)}^L ?$$

is the Koszul duality equivalence $\mathbf{H} \rightarrow \mathbf{H}_{\tilde{G},0}, \mathbf{F} \rightarrow \mathbf{F}'$ lifting the Koszul duality equivalence between the graded $S(\mathfrak{h}^a)$ -modules and $\Lambda(\mathfrak{h}^{a,*}[1])$ -modules.

We note that by the construction, $\mathfrak{K}$ commutes with Red . We set

$$(4.10) \quad \mathbf{Wak}(\check{\lambda}^a) := \mathfrak{K}^{-1}(\mathbf{Wak}_0(\check{\lambda}^a)), \quad \Delta^{\mathbf{F}}(\check{\lambda}^a) := \mathfrak{K}^{-1}(\Delta_0^{\mathbf{F}}(\check{\lambda}^a)) = \text{Red}(\mathbf{Wak}(\check{\lambda}^a)).$$

We have the following properties of these objects.

Lemma 4.4.9. *The following claims hold:*

- (1) *The objects $\mathbf{Wak}(\check{\lambda}^a), \Delta^{\mathbf{F}}(\check{\lambda}^a)$ lie in the hearts of the natural t -structures and are flat over $\mathbf{R}$.*
- (2) *If $\check{\lambda}^a = \check{\lambda} + n\delta$, then $\mathbf{Wak}(\check{\lambda}^a) \cong \mathcal{X}_{\check{\lambda}} * \mathbf{Wak}(n\delta), \Delta^{\mathbf{F}}(\check{\lambda}^a) \cong \mathcal{X}_{\check{\lambda}} * \Delta^{\mathbf{F}}(n\delta)$.*
- (3) *The objects $\mathbf{Wak}(\check{\lambda}^a)$ and $\Delta^{\mathbf{F}}(\check{\lambda}^a) / \hbar \Delta^{\mathbf{F}}(\check{\lambda}^a)$ are coherent in $\mathbf{H}$ and $\mathbf{F}_{\mathbf{R}/(\hbar)}$, respectively.*

Proof. (1) follows from Lemma 4.4.4. (2) follows from (4.10) and the definition of the braid group action on $\mathbf{H}, \mathbf{F}$. To prove (3) we note that $\mathfrak{K}, \mathfrak{K}^{-1}$ send coherent objects to coherent objects and use Corollary 4.4.8. $\square$

We will need two more properties of the autoequivalences T_i of $\mathbf{H}, \mathbf{F}$. Let $L \subset \mathcal{L}G, L^a \subset \hat{G}$ denote the Levi subgroup with the same roots as $\tilde{L} \subset \tilde{G}$. The maximal torus in L^a is $H \times T_c$, where T_c is the central $\mathbb{G}_m$ in $\hat{G}$ (note that $\mathfrak{h} \oplus \mathbb{C}c \cong (\mathfrak{h}^* \oplus \mathbb{C}\mathbf{eu})^*$ as representations of W^a). We consider the category $\mathbf{H}_{\tilde{J}}$ of weakly $T_d \times H^2$ -equivariant and strongly $U_L \times U_L$ -equivariant $D_{\hbar}(L)[c]$ -modules (where $U_L := I^0 \cap L$).

Here is a construction of an object in $\mathbf{H}_{\tilde{J}_i}$ based on [10, Section 2]. Consider the free monodromic local system on $B_L s_i B_L$ and its $!$ -pushforward, in [10, Notation 3.1] this object would (the paper considers the semisimple setting) denoted by $\tilde{\Delta}_{s_i}^0$, we denote it by $j_{i,!}^{\wedge}$. It is an object of the category $\mathbf{MHM}^{\wedge}(B_L \backslash L / B_L)$ that is linear over $\mathbb{C}[[\mathfrak{h}^{*, \oplus 2}, c]]$. Moreover, we have the forgetful functor $\varphi_{\tilde{J}_i}^{\wedge} : \mathbf{MHM}^{\wedge}(B_{L_i} \backslash L_i / B_{L_i}) \rightarrow \mathbf{H}_{\tilde{J}_i}$, cf. [10, Proposition 2.4].

By Section ??, $\mathbf{H}_{\tilde{J}_i}$ is a monoidal category acting on $\mathbf{H}, \mathbf{F}$ by convolution.

Lemma 4.4.10. *The self-equivalence T_i of $\mathbf{H}, \mathbf{F}$ is given by convolution with $\varphi_{\tilde{J}_i}^{\wedge}(j_{i,!}^{\wedge})$.*

Proof. Consider the category $\mathbf{H}_{\tilde{J}_{i,0}}$ of weakly $H \times T_{\ell,d}$ - and strongly $(B_{L_i} \times U_{L_i}, -\rho^a \hbar)$ -equivariant $D_{\hbar}(L_i)$ -modules. Note that it is a bimodule over $\mathbf{H}_{\tilde{J}_i}$ (acting on the side with weak equivariance) and $\mathbf{H}'_{\tilde{J}_i}$ (acting on the side with strong equivariance).

Also we have the δ -function object $\delta_{\hbar}^{B_{L_i}}$ defined similarly to $\delta_{c,\hbar}^I \in \mathbf{H}$ (with an appropriate twist), cf. Section 4.2.2. It is isomorphic to $\varphi_{\tilde{J}_i}(j_{1,!})$.

Note that we have the convolution functors

$$\mathbf{H} \times \mathbf{H}_{\tilde{J}_{i,0}} \rightarrow \mathbf{H}_{\tilde{G},0}, \mathbf{F} \times \mathbf{H}_{\tilde{J}_{i,0}} \rightarrow \mathbf{F}'.$$

By definition, the functors $\mathbf{K} \circ T_i, T_i \circ \mathfrak{K} : \mathfrak{H} \rightarrow \mathfrak{H}_{\tilde{G},0}$ are isomorphic. To show that $T_i : \mathbf{H} \xrightarrow{\sim} \mathbf{H}$ is given by convolution with $\varphi_{\tilde{J}_i}^{\wedge}(j_{i,!}^{\wedge})$ boils down to proving the following isomorphism in $\mathbf{H}_{\tilde{J}_{i,0}}$:

$$(4.11) \quad \varphi_{\tilde{J}_i}^{\wedge}(j_{i,!}^{\wedge}) * \varphi_{\tilde{J}_i}(j_{1,!}) \cong \varphi_{\tilde{J}_i}(j_{1,!}) * \varphi'_{\tilde{J}_i}(j'_{i,!})$$

because $\mathfrak{K} \circ [\varphi^{\wedge}(j_{i,!}^{\wedge}) * ?]$ is given by convolution with the left hand side, while the functor $T_i \circ \mathfrak{K}$ is given by the convolution with the right hand side.

Since the functors φ intertwine convolutions (it is not hard to see that this includes also the free monodromic setting, (4.11) follows from the following isomorphism in $\mathbf{MHM}$ which is standard:

$$j_{i,!}^{\wedge} * j_{1,!} \cong j_{i,!} \cong j_{1,!} * j'_{i,!}.$$

$\square$

Recall that $p : \mathrm{Br}^{ea} \rightarrow W^{ea}$ denote the natural homomorphism.

Corollary 4.4.11. *For $\beta \in \mathrm{Br}^{ea}$, the corresponding self-equivalence of $\mathbf{H}, \mathbf{F}$ is $p(\beta)$ -semi-linear (for the action of $\mathbf{R}$).*

Proof. For the length 0 elements β this follows from the construction of the action of these elements as twists by elements of $\mathcal{L}G$ that preserve $\hat{I}$ and $H \times T_c$: the twist of the R-action will be by the corresponding automorphism of $\mathfrak{h}^a$.

It remains to handle the case when $\beta = T_i$. We need to show that $\varphi_{\hat{j}_i}^\wedge(j_{i,!}^\wedge)$ the two actions of R differ by s_i . This follows from the corresponding statement about the actions of $\mathbb{C}[[\mathfrak{h}^{a,*}]]$ on $j_{i,!}^\wedge$, which, in turn, follows from the corresponding statement for the action on the free monodromic local system on $\hat{I}s_i\hat{I}$. The latter statement just follows from the shape of the centralizer of the point $\hat{I}s_i\hat{I}$ in $\hat{I} \times \hat{I}$: the maximal torus in this centralizer is the graph of s_i . $\square$

Below we will need another corollary.

Corollary 4.4.12. *Let $\mathcal{F}$ be an object of $\mathbf{H}$ such that $\mathfrak{h}^{da}$ -action (on the same side where Br^{ea} acts) is by 0. Then we have a homomorphism $\mathcal{F} \rightarrow T_i\mathcal{F}$ whose cone is the convolution of $\mathcal{F}$ with*

$$(4.12) \quad \mathcal{O}_{B_{L_i} \backslash L_i / B_{L_i}}(-\rho^a, -\rho^a) \in \mathbf{H}_{\tilde{\mathcal{J}}_i}.$$

The cone of the induced homomorphism $T_i^{-1}\mathcal{F} \rightarrow \mathcal{F}$ is isomorphic to the same convolution.

Proof. Thanks to Lemma 4.4.10, $T_i\mathcal{F}$ is the convolution of

$$\mathcal{F} \in \mathrm{IndCoh}^{T_{\ell,d}}(T^*(I^0 \backslash \mathcal{L}G / I)) / H$$

with $\mathrm{gr}^H j_{i,!}$. The claim about $\mathcal{F} \rightarrow T_i\mathcal{F}$ follows from (4.9). And the claim about $T_i^{-1}\mathcal{F} \rightarrow \mathcal{F}$ follows from the short exact sequence

$$0 \rightarrow L_i \rightarrow j_{i,*} \rightarrow j_{i,!} \rightarrow 0.$$

$\square$

We will also need to understand some other specializations of the objects $\varphi_{\hat{j}_i}^\wedge(j_{i,!}^\wedge)$. Take a graded domain quotient $\mathbf{R}_0$ with complete graded fraction ring $\mathbb{F}_0$ and $\mathbf{f} := \mathbb{F}_0^{T_d}$. Suppose $\hbar$ is nonzero in $\mathbf{R}_0$. We want to understand the specialization $\varphi_{\hat{j}_i}^\wedge(j_{i,!}^\wedge)_{\mathbb{F}_0} := \varphi_{\hat{j}_i}^\wedge(j_{i,!}^\wedge) \otimes_{\mathbf{R}} \mathbb{F}_0$, let $T_{i,\mathbf{f}}$ denote the T_d -invariants in this object. This is a strongly $(B_L \times B_L, -(\rho^a, \rho^a))$ -equivariant $D_{\mathbf{f}}(L)$ -module, where the twist on the right is a point of $\mathfrak{h}^{a,*}(\mathbf{f})$ depending on the choice of $\mathbf{R}_0$ and the twist on the left is obtained from that by applying s_i .

Lemma 4.4.13. *The object $T_{i,\mathbf{f}}$ is a $!-$ or $*$ -extension of the rank 1 twisted local system on $B_L s_i B_L$.*

Proof. We observe that the object $T_{i,\mathbf{f}}$ has the following properties:

- (1) It lies in the heart of the default t-structure on the category of twisted equivariant D-modules. This is because $\varphi_{\hat{j}_i}^\wedge(j_{i,!}^\wedge)$ is flat over $\mathbf{R}$ by the construction.
- (2) For the same flatness reason, $T_{i,\mathbf{f}}$ carries a good filtration whose associated graded is $(\varphi_{\hat{j}_i}^\wedge(j_{i,!}^\wedge) \otimes_{\mathbf{R}} \mathbb{K}) \otimes \mathbf{f}$. It follows that the K_0 -class of $T_{i,\mathbf{f}}$ coincides with that of the $!-$ and $*$ -extension.
- (3) The object $T_{i,\mathbf{f}}$ is invertible with respect to convolution. In particular, it is indecomposable (because the δ -function of a point, the monoidal unit, is indecomposable).

The only objects satisfying the conclusions of (1)-(3) are the $!$ - and $*$ -extensions of the rank 1 twisted local system. $\square$

4.5. Functor $R\Gamma$.

4.5.1. *Construction.* We note that taking the derived I -invariants, $R\mathrm{Hom}_I(\mathrm{triv}, ?)$, gives functors

$$(4.13) \quad \mathbf{H} \rightarrow \mathbf{O}, \mathbf{F} \rightarrow \mathbf{C},$$

where in the case of $\mathbf{H}$ we use the action from the right (for the reduction functor we used the action from the left), and in the case of $\mathbf{F} \rightarrow \mathbf{C}$ we twist the action of $\mathbf{R}$ by the automorphism of $\mathbf{R}$ given by the automorphism v defined by

$$(4.14) \quad \hbar \mapsto \hbar, c \mapsto -c, x \mapsto -x - \langle 2\rho, x \rangle \hbar, \text{ for } x \in \mathfrak{h}.$$

This is needed because by our conventions $c \in \mathcal{W}_{c, \hbar}$ and $c \in \mathbf{R}$ act in the same way, while on objects of $\mathbf{F}$ the action differ by the sign. We similarly have a right $\mathbf{R}$ -action on objects in the image of $R\Gamma : \mathbf{H} \rightarrow \mathbf{C}$ and we twist it by the same automorphism (so that on $R\Gamma(\delta_{c, \hbar}^I)$ the two actions coincide thanks to Lemma 4.2.2). We further note that v is W^{ea} for the $-\rho^a \hbar$ -shifted action on $\mathfrak{h}^{da}$.

Lemma 4.5.1. *We have a functor isomorphism*

$$\mathrm{Red} \circ R\Gamma \cong R\Gamma \circ \mathrm{Red}$$

of functors $\mathbf{H} \rightarrow \mathbf{C}$.

Proof. The functor Red is defined via the BRST complex formed using the structure of a category $\mathbf{O}$, for the action from the left in the case of $\mathbf{H}$. The functor $R\Gamma$ is defined via the cobar complex for the action of I from the right. In particular, the cobar complex is a complex in the category $\mathbf{O}$, so forming the BRST complex commutes with forming the cobar complex. This yields the claim of the lemma. $\square$

4.5.2. $R\Gamma(\Delta_{\mathbb{K}}^F(\check{\lambda}^a))$. Here we compute the objects $R\Gamma(\mathbf{Wak}_{\mathbb{K}}(\check{\lambda}^a)), R\Gamma(\Delta_{\mathbb{K}}^F(\check{\lambda}^a))$. We write $\check{\lambda}^a$ as $\check{\lambda} + n\delta$ for $n \in \mathbb{Z}$.

Lemma 4.5.2. *We have an isomorphism*

$$(4.15) \quad R\Gamma(\mathbf{Wak}_{\mathbb{K}}(\check{\lambda}^a)) = \Delta_{\mathbb{K}}(n\delta),$$

$$(4.16) \quad R\Gamma(\Delta_{\mathbb{K}}^F(\check{\lambda}^a)) = \underline{\Delta}_{\mathbb{K}}(n\delta).$$

Proof. We note that thanks to Lemma 4.5.1 and the specialization of (1.16) to $0 \in \mathrm{Spec}(\mathbf{R}_0)$, (4.16) follows from (4.15). We will prove (4.15) in two steps.

Step 1. First, we consider the case when $\check{\lambda}^a = 0$. In this case,

$$\mathbf{Wak}_{\mathbb{K}}(0) = \mathbb{K} \otimes_{\mathbf{R}} \delta_{c, \hbar}^I = S(\mathcal{L}\mathfrak{g}/\mathfrak{i}) \otimes \mathbb{K}[I].$$

The action of I from the right is by the right translations on $\mathbb{K}[I]$. So $R\Gamma(S(\mathcal{L}\mathfrak{g}/\mathfrak{i}) \otimes \mathbb{K}[I]) = S(\mathcal{L}\mathfrak{g}/\mathfrak{i}) = \Delta_{\mathbb{K}}(0)$.

Step 2. Now we handle the general case. We can assume $n = 0$. In this case $\mathbf{Wak}_{\mathbb{K}}(\check{\lambda}^a)$ is obtained from $\mathbf{Wak}_{\mathbb{K}}(0)$ by applying an element from $\mathbf{Br}^{ea}$, so it is enough to establish a functor isomorphism $R\Gamma \circ \beta \cong \zeta(\beta) \circ R\Gamma$ of functors $\mathbf{H}_{\mathbb{K}} \rightarrow \mathbf{O}_{\mathbb{K}}$, where ζ is the projection from $\mathbf{Br}^{ea}$ to the group of length 0 elements. In the case when β is a length 0 element, by

the construction in the end of Section 4.4.1 it acts by the twist with the corresponding automorphism, so $R\Gamma \circ \beta \cong \beta \circ R\Gamma$. It is easy to see that $S(\mathcal{L}\mathfrak{g}/\mathfrak{i})$ does not change under the twist by $\zeta(\beta)$. So we only need to show that $R\Gamma \circ T_i \cong R\Gamma \circ T_i^{-1} \cong R\Gamma$. We deduce these isomorphisms from Corollary 4.4.12: in the notation of that corollary we need to prove that applying $R\Gamma$ to the isomorphic cones of $\mathcal{F} \rightarrow T_i \mathcal{F}$ and $T_i^{-1} \mathcal{F} \rightarrow \mathcal{F}$ (to the convolution with (4.12)) we get 0.

Similarly to Lemma 4.1.2, $\mathbf{H}_{\mathbb{K}}$ is the category of $T_{\ell,d}$ -equivariant ind-coherent sheaves on $T^*(I^0 \setminus \mathcal{L}G/I)/H$, and the functors $T_i^{\pm 1}$ are given by convolution. On the other hand, in this interpretation, $R\Gamma$ is just the functor of taking derived global sections.

The convolution of $\mathcal{F}$ with (4.12) in this language can be described as follows. Consider the following derived ind-stack:

$$(4.17) \quad Y = \{(g_1 g_2 I, g_1 I, \xi) | g_1 I \in \mathcal{F}l, g_2 I \in J_i/I, \xi \in (\mathcal{L}\mathfrak{g}/[\mathrm{Ad}(g_1 g_2)\mathfrak{i} + \mathrm{Ad}(g_1)\mathfrak{i} + \mathfrak{i}^0])^*\}/I,$$

where J_i is the minimal parahoric. The convolution in question is the pushforward from Y of

$$(4.18) \quad \tilde{\mathcal{F}} := p_2^*(\mathcal{F}) \otimes p_1^*\left(\mathcal{O}_{B_{L_i} \setminus L_i/B_{L_i}}(-\rho^a, -\rho^a)\right),$$

where p_1, p_2, p_3 are suitable projections from Y . We need to compute that derived global sections of (4.18). Consider the derived ind-stacks

$$\begin{aligned} Y_i &= \{(g_1 g_2 I, g_1 I, \xi) | g_1 I \in \mathcal{F}l, g_2 I \in M/B, \xi \in (\mathcal{L}\mathfrak{g}/\mathrm{Ad}(g_1)\mathfrak{j}_i + \mathfrak{i}^0])^*\}/I, \\ \underline{Y}_i &= \{(g_1 I, \xi) | g_1 I \in \mathcal{F}l, \xi \in (\mathcal{L}\mathfrak{g}/\mathrm{Ad}(g_1)\mathfrak{j}_i + \mathfrak{i}^0])^*\}/I, \end{aligned}$$

so that the morphism $\pi : Y_i \rightarrow \underline{Y}_i$ forgetting $g_1 g_2 I$ is a $\mathbb{P}^1$ -bundle and we have a closed inclusion $Y_i \hookrightarrow Y$. Then (4.18) is pushed forward from a sheaf on Y_i , which is a pullback from a sheaf on $\underline{Y}_i$ tensored with the pullback of $\mathcal{O}(-\rho^a)$ from $L_i/B_{L_i} = \mathbb{P}^1$, that gives $\mathcal{O}(-1)$ after forgetting the equivariance. Applying π_* to such a sheaf gives 0, hence the derived global sections of (4.18) is 0. $\square$

4.5.3. $R\Gamma(\Delta^F(\check{\lambda}^a))$. We start by computing $R\Gamma(\mathbf{Wak}(\check{\lambda}^a))$ for $\check{\lambda}^a = (\check{\lambda}, n)$. Recall, Section 4.5.1, that this is an object of $\mathbf{O}$ with an additional (right) action of $\mathbf{R}$ by graded endomorphisms. This action is normalized in such a way that on $R\Gamma(\delta_{c,h}^I)$ the two actions coincide.

Proposition 4.5.3. *We have an isomorphism $R\Gamma(\mathbf{Wak}(\check{\lambda}^a)) \cong \Delta(n\delta)$ in $\mathbf{O}$. Moreover, the right action on $R\Gamma(\mathbf{Wak}(\check{\lambda}^a))$ is obtained from the left action by twisting with $t_{\check{\lambda}} \circ v$, where v is defined by (4.14).*

Proof. The claim about the right $\mathbf{R}$ -action is a direct consequence of Corollary 4.4.11 combined with Lemma 4.2.2. So we just need to prove an isomorphism in $\mathbf{O}$.

From (3) of Lemma 4.4.9 and the observation that $\mathbb{K}[I]$ is concentrated in PBW degree 0 it follows that the PBW grading on $R\Gamma(\mathbf{Wak}(\check{\lambda}^a))$ is bounded from below. Choose a basis $x_1, \dots, x_n$ in $\mathfrak{h}^{da}$. Recall that the object $M := \mathbf{Wak}(\check{\lambda}^a)$ is flat over $\mathbf{R}$. Applying the long exact sequence for the derived functor $R\Gamma$ to exact sequences

$$0 \rightarrow M/(x_1, \dots, x_{i-1})M \xrightarrow{x_i} M/(x_1, \dots, x_{i-1})M \rightarrow M/(x_1, \dots, x_i)M \rightarrow 0$$

we see that if $R\Gamma(M/(x_1, \dots, x_i)M)$ is in cohomological degree 0 and is flat over the algebra $\mathbb{K}[x_{i+1}, \dots, x_n]$, then $R\Gamma(M/(x_1, \dots, x_{i-1})M)$ is in cohomological degree 0 and is flat over $\mathbb{K}[x_i, \dots, x_n]$. From (4.15) we see that $R\Gamma(M)$ is in cohomological degree 0 and is flat over $\mathbb{R}$. Also from (4.15) it follows that its specialization to $0 \in \text{Spec}(\mathbb{R})$ is $\Delta_{\mathbb{K}}(n\delta)$. Note that $\Delta_{\mathbb{K}}(n\delta)$ has no graded self-extensions in $\mathcal{O}_{\mathbb{K}}$, so all of its $\mathbb{R}$ -flat deformations are isomorphic. Clearly, $\Delta(n\delta)$ is one such deformation finishing the proof. $\square$

We proceed to describing $R\Gamma(\Delta^F(\check{\lambda}^a))$. Recall that to a linear automorphism φ of $\mathfrak{h}^{da}$ fixing $c, \hbar$ we assign a self-equivalence $M \mapsto {}^\varphi M$. The following is direct consequence of Lemma 4.5.1 and Proposition 4.5.3.

Corollary 4.5.4. *We have $R\Gamma(\Delta^F(\check{\lambda}^a)) \cong {}^{t_\lambda}\underline{\Delta}(n\delta)$.*

We will write $\underline{\Delta}(\check{\lambda}^a)$ for ${}^{t_\lambda}\underline{\Delta}(n\delta)$.

4.6. Specializations of F with invertible $\hbar$. In this section we investigate three specializations of $F^\heartsuit$:

- (A) to a graded localization $\mathbb{F}'$ of $\mathbb{R}$ such that all elements of the form $\alpha - nc - m\hbar$ and c are nonzero in $\mathbb{F}'$ (a generic specialization)
- (B) to the full graded localization $\mathbb{F}_0$ of $\mathbb{R}_0 := \mathbb{R}/(c, \hbar)$,
- (C) and to the full graded localization $\mathbb{F}_1$ of $\mathbb{R}_1 := \mathbb{R}/(\alpha - nc - m\hbar)$ for finite type roots α and $m, n \in \mathbb{Z}$.

Recall that the resulting categories are equivalent to their analogs of $\mathbf{f} := \mathbb{F}^{T_d}$ in the first case and $\mathbb{F}_0^{T_d}$ in the second case without PBW grading, see Section 1.2.3.

4.6.1. $F_{\mathbf{f}}$ vs Whittaker D -modules. Let us start by recalling the Whittaker category. The choice of $\mathbb{R}_0$ gives $\lambda \in \mathfrak{h}_{\mathbf{f}}^*$ as well as $\kappa \in \mathbf{f}$. Let $\psi : \mathcal{L}\mathbf{n}^- \rightarrow \mathbb{K}$ have the same meaning as in Section 1.3.1. It makes sense to speak about twisted Whittaker D -modules on the affine flag variety, in the notation of [9], $D_{\kappa}(\mathcal{L}\mathbf{n}^-, \psi \backslash \mathcal{L}G/(I, \lambda))$, note that the normalization differ from [9]: to get the category $\text{Whit}_{\lambda}^{aff}$ from [9] (namely, the right hand side of (3.2) in *loc. cit.*) one needs to replace λ with $-\lambda$, the equivalence between the categories of interest is provided by the inversion automorphism of $\mathcal{L}G$. We denote our category by $\text{Whit}_{\kappa, \lambda}^{aff}$. This category comes with a t-structure constructed similarly to [20, Section 5.2].

On the other hand, we can consider the category $\mathcal{E}\mathcal{W}_{\mathbf{f}, \kappa}\text{-mod}^{I, \lambda}$.

The following proposition is well-known to experts.

Proposition 4.6.1. *There is a t-exact functor $\text{Whit}_{\kappa, \lambda}^{aff} \xrightarrow{\sim} \mathcal{E}\mathcal{W}_{\mathbf{f}, \kappa}\text{-mod}^{I, \lambda}$ that is an equivalence on the full subcategories of all complexes with cohomology bounded from below.*

Note that $F_{\mathbf{f}}$ is nothing else but the category of weakly T_{ℓ} -equivariant objects in the category $\mathcal{E}\mathcal{W}_{\mathbf{f}, \kappa}\text{-mod}^{I, \lambda}$. So Proposition 4.6.1 gives an equivalence between $F_{\mathbf{f}}$ and the category of weakly T_{ℓ} -equivariant objects in $\text{Whit}_{\kappa, \lambda}^{aff}$.

4.6.2. Generic specialization. Here we work over the full graded localization $\mathbb{F}$ of $\mathbb{R}$.

Lemma 4.6.2. *The category $F_{\mathbb{F}}^{\heartsuit}$ is semisimple with $\Delta_{\mathbb{F}}^F(\check{\lambda}^a)$ being the pairwise non-isomorphic simple objects.*

Proof. The category $\mathbf{F}_{\mathbb{F}'}$ is the direct sum of $\mathbb{Z}$ many copies (appearing via shifts of the energy grading) of the category denoted by $\text{Whit}_{\check{\lambda}}^{\text{aff}}$ in [9]. This category is over $\mathbf{f}' = (\mathbb{F}')^{T_d}$, the twist λ is generic, and the summands are obtained from one another by twisting the action of T_{ℓ} . By [9, Theorem 3.2.7], the category $\text{Whit}_{\check{\lambda}}^{\text{aff}}$ splits as the direct sum of the categories denoted by $\text{Whit}_{\check{\lambda},y}^{\text{aff}}$ in *loc. cit.* Since λ is generic, the sum is over $\check{\Lambda}$. By [9, Lemma 3.2.5], each $\text{Whit}_{\check{\lambda},y}^{\text{aff}}$ is equivalent to $\mathbf{f}'$ -Vect yielding the semisimplicity claim.

To prove the claim that $\Delta_{\mathbb{F}'}^{\mathbf{F}}(\check{\lambda}^a)$ are exactly the pairwise non-isomorphic simple objects we can argue as follows. Note that since the twist is generic, all extensions in $\mathbf{H}_{\mathbf{f}'}$ are clean. Property (VI) implies that $\mathbf{Wak}_{\mathbb{F}'}(\check{\lambda})$ is the clean extension of a twisted rank 1 local system from $It_{\check{\lambda}}I$. We then can use [9, Proposition 3.2.12] to see that $\Delta_{\mathbb{F}'}^{\mathbf{F}}(\check{\lambda}^a)$ gives an extension of a rank 1 twisted local system from the stratum corresponding to $\check{\lambda}$ with energy grading twisted by n . Using [9, Corollary 3.2.11], we see that the simples in $\bigoplus_{\mathbb{Z}} \text{Whit}_{\check{\lambda}}^{\text{aff}}$ are exactly of this form and each simple object occurs in the list $\Delta_{\mathbb{F}'}^{\mathbf{F}}(\check{\lambda}^a)$ exactly once finishing the proof. $\square$

4.6.3. *Specialization to $\mathbb{F}_0$.* Here we consider $\mathbf{F}_{\mathbb{F}_0}$, where $\mathbb{F}_0 = \mathbb{K}[\hbar^{\pm 1}]$ with $\hbar, c$ mapping to 0.

Lemma 4.6.3. *Let $\check{\lambda}_i^a = \check{\lambda}_i + n_i\delta$. If $\text{grHom}_{\mathbf{F}_{\mathbb{F}_0}}(\Delta_{\mathbb{F}_0}^{\mathbf{F}}(\check{\lambda}_1^a), \Delta_{\mathbb{F}_0}^{\mathbf{F}}(\check{\lambda}_2^a)) \neq 0$, then*

- (1) $n_1 = n_2$,
- (2) and $\check{\lambda}_1 \leq \check{\lambda}_2$ in the usual ordering on $\check{\Lambda}^a$.

Proof. (1) follows because $\Delta^{\mathbf{F}}(\check{\lambda})$ are strongly T_{ℓ} -equivariant. To prove (2), we can replace $\check{\lambda}_i$ with $\check{\lambda}_i + \check{\mu}$, we can assume that both $\check{\lambda}_i$ are dominant. The objects $\mathbf{Wak}_{\mathbf{f}_0}(\check{\lambda}_i)$ are the shriek extensions from $It_{\check{\lambda}_i}I$. Using [9, Proposition 3.2.12], we see that, as Whittaker D-modules, $\Delta_{\mathbf{f}_0}(\check{\lambda}_i)$ are the shiek extensions from the starta corresponding to $t_{\check{\lambda}_i}$. As the order on these strata is the parabolic Bruhat order, and $\check{\lambda}_i$ are dominant, we see that there is a nonzero Hom as in the statement of the lemma if and only if the stratum corresponding to $\check{\lambda}_1$ lies in the closure of the startum corresponding to $\check{\lambda}_2$ if and only if $t_{\check{\lambda}_1} \leq t_{\check{\lambda}_2}$ in the Bruhat order. This implies $\check{\lambda}_1 \leq \check{\lambda}_2$ finishing the proof. This implies (2). $\square$

4.6.4. *Specialization to $\mathbb{F}_1$.* Finally, we handle (subgeneric) specializations to $\mathbb{F}_1$ as in (C).

Lemma 4.6.4. *Let λ^a lie in a two element orbit of s_{α} . Then for a suitable choice of λ^a in that orbit $\text{grHom}_{\mathbf{F}_{\mathbb{F}_1}}(\Delta_{\mathbb{F}_1}^{\mathbf{F}}(\check{\lambda}^a), \Delta_{\mathbb{F}_1}^{\mathbf{F}}(s_{\alpha}\check{\lambda}^a))$ is nonzero.*

Proof. Thanks to (II), the twisted T_{ℓ} -equivariences of $\Delta_{\mathbb{F}_1}^{\mathbf{F}}(\check{\lambda}^a), \Delta_{\mathbb{F}_1}^{\mathbf{F}}(s_{\alpha}\check{\lambda}^a)$ are the same. So we can forget the T_{ℓ} -equivariance and work with analogous objects in a suitable category $\text{Whit}_{\check{\lambda}}^{\text{aff}}$. The integral Weyl group in this case has order 2. So one of $\Delta_{\mathbb{F}_1}^{\mathbf{F}}(\check{\lambda}^a), \Delta_{\mathbb{F}_1}^{\mathbf{F}}(s_{\alpha}\check{\lambda}^a)$ (say the former) corresponds to the clean extension from the minimal stratum supporting a local system with a required twist. Then $\Delta_{\mathbb{F}_1}^{\mathbf{F}}(s_{\alpha}\check{\lambda}^a)$ is obtained from by convolving with $\mathcal{F} := \Delta_{\mathbb{F}_1}^{\mathbf{F}}(\check{\lambda}^a)$ with $\mathbf{Wak}_{\mathbb{F}_1}(s_{\alpha}\check{\lambda}^a) * \mathbf{Wak}_{\mathbb{F}_1}(\check{\lambda}^a)^{-1}$. The latter object lies in the block of $\mathbf{H}_{\mathbb{F}_1}$ containing the monoidal unit, 1, and and by the K_0 considerations combined with (VI), we see that $\mathcal{F}$ is either a !- or *-extension from $Is_{\bar{\alpha}}I$, where $\bar{\alpha}$ is the projection of α to $\mathfrak{h}^{a,*}$. We consider the case when it is a !-extension, the case of the *-extension is similar. We have a unique monomorphism $1 \rightarrow \mathcal{F}$ and what we need to show is that it is

not annihilated by convolving with $\Delta_{\mathbb{F}_1}(\check{\lambda}^a)$. As explained in the proof of [9, Proposition 3.2.12], the functor of convolution $\Delta_{\mathbb{F}_1}(\check{\lambda}^a)$ from the Hecke category to the Whittaker category is t-exact. In particular, it sends monomorphisms to monomorphisms. This finishes the proof. $\square$

4.6.5. $R\Gamma$ as an RS functor. Our goal in this section is to show that $R\Gamma : \mathcal{F} \rightarrow \mathcal{C}$ is an RS functor with respect to the families of objects $\Delta(\check{\lambda}^a) \in \mathcal{F}$ and $\underline{\Delta}(\check{\lambda}^a) \in \mathcal{C}$, where $\check{\lambda}$ runs over $\check{\Lambda}^a$ (and the group in question is $\check{\Lambda}^a$).

First, we check that conditions (a1)-(a5) from Section 1.2.1 and (b) from Section 3.2.1 for the objects $\Delta^{\mathbb{F}}(\check{\lambda}^a)$. Note that conditions (c1)-(c5) for the objects $\underline{\Delta}(\check{\lambda}^a)$ were checked in (2) of Proposition 3.2.4, it is not important that now we have a different labeling set, the same argument applies.

(a1) and (a2) are parts (2) and (1), respectively, of Lemma 4.2.4.

(a3) is a part of (1) of Lemma 4.4.9.

Let us check (a4). For $\gamma_2 = n\delta$, this follows from Proposition 4.3.1. The general case is handled using the action of $\check{\Lambda} \subset \mathbf{Br}^{ea}$ by category equivalences.

(a5) and (b) follow from Proposition 4.6.2 applied to $\mathbb{F}' = \mathbb{F}$.

Now we check properties (RS1) and (RS2) from Section 3.2.1 for the functor $R\Gamma : \mathcal{F} \rightarrow \mathcal{C}$. (RS1) is Corollary 4.5.4.

Lemma 4.6.5. *(RS2) holds for $R\Gamma$.*

Proof. As we have seen in Step 2 of the proof of Lemma 4.5.2, $R\Gamma \circ \beta \cong \zeta(\beta) \circ R\Gamma$, where ζ is the projection from $\mathbf{Br}^{ea}$ to the subgroup of length 0 elements. This allows to reduce to the case where $\gamma_1 = 0$. By (2) of Lemma 4.2.4, $\Delta_{\mathbb{K}}^{\mathbb{F}}(0)$ is generated by a single I -invariant element, say v . Hence any morphism $\Delta_{\mathbb{K}}^{\mathbb{F}}(0) \rightarrow \Delta_{\mathbb{K}}^{\mathbb{F}}(\check{\lambda}^a)$ is determined by the image of that vector. On the other hand, $R\Gamma(\Delta_{\mathbb{K}}^{\mathbb{F}}(\check{\lambda}^a)) = \Delta_{\mathbb{K}}^{\mathbb{F}}(\check{\lambda}^a)^I$ thanks to Corollary 4.5.4 (and the observation that the objects in that corollary are flat over $\mathbb{R}$). It remains to observe that $\underline{\Delta}_{\mathbb{K}}(0) = \Delta_{\mathbb{K}}^{\mathbb{F}}(0)$ is generated by v over $\mathbb{K}[\mathcal{LS}]$ and so any morphism $\underline{\Delta}_{\mathbb{K}}(0) \rightarrow \underline{\Delta}_{\mathbb{K}}(\check{\lambda}^a)$ is determined by the image of v . Since the map of Hom spaces in (RS2) is obtained by restricting a homomorphism to the I -invariants, and both morphisms are uniquely determined by the images of v , the injectivity claim follows. $\square$

Putting together the results above in this section, we see that $R\Gamma : \mathcal{F} \rightarrow \mathcal{C}$ is indeed an RS functor. In what follows we will show that it is of $\mathfrak{sl}_2$ -type and identify different kinds of ideals.

4.7. Functor $\mathcal{O} \rightarrow \check{\mathcal{F}}$. We now switch from considering the category $\mathcal{F}$ for G to working with the category $\check{\mathcal{F}}$ for $\check{G}$. We can identify $\mathbb{R}$ with $\check{\mathbb{R}}$ via (1.20).

The group of this section is to prove the following theorem that is the main result of this paper.

Theorem 4.7.1. *There is a unique up to isomorphism functor of graded $\mathbb{R}$ -linear triangulated categories*

$$\Xi : \mathcal{O} \rightarrow \check{\mathcal{F}}$$

with fixed functor isomorphisms $R\Gamma \circ \Xi \cong \text{Red}$ and $\Xi(\Delta(\lambda^a)) \xrightarrow{\sim} \check{\Delta}^{\mathbb{F}}(\lambda^a)$ for all $\lambda^a \in \Lambda^a$. This functor restricts to an equivalence of exact categories $\mathcal{O}^{\Delta} \xrightarrow{\sim} \check{\mathcal{F}}^{\Delta}$.

The group Λ acts on R fixing $c, \hbar$: $\lambda \in \Lambda$ sends $\mu \in \mathfrak{h}^*$ to $\mu + \hbar\lambda$. It also acts on $\check{R}$ fixing $\check{c}, \check{\hbar}$: λ sends $\check{\mu} \in \check{\mathfrak{h}}^*$ to $\check{\mu} - \check{c}\kappa_0(\lambda)$. Then identification (1.20) is Λ -equivariant. Combining this with Corollary 1.6.3, we see that the Feigin-Frenkel duality isomorphism intertwines $\underline{\Delta}_R(\lambda^a)$ with $\underline{\Delta}_{\check{R}}(\lambda^a)$. We also identify C and $\check{C}$. So we get RS functors

$$\text{Red} : \mathcal{O} \rightarrow C, R\Gamma : \check{F} \rightarrow C$$

with

$$\text{Red}(\Delta(\lambda^a)) \cong R\Gamma(\check{\Delta}^F(\lambda^a)) \cong \underline{\Delta}(\lambda^a).$$

In Proposition 3.3.8 we have identify Red as an RS functor of $\mathfrak{sl}_2$ finding homogeneous height 1 prime ideals of semisimple, fully faithful, and $\mathfrak{sl}_2$ -type and also described the relations $\uparrow$. We have a complete analog of that proposition for $R\Gamma : \check{F} \rightarrow C$:

Proposition 4.7.2. *The functor $R\Gamma : F \rightarrow C$ (that is RS by Section 4.6.5) is of $\mathfrak{sl}_2$ -type. Moreover, the ideal $(\check{c})$ is of semisimple type, the ideals $(\check{\alpha} - n\check{c} - m\check{\hbar})$ for $n, m \in \mathbb{Z}$ are of $\mathfrak{sl}_2$ -type, and the remaining ideals are of fully faithful type. And the relation $\uparrow_{\mathfrak{p}}$ for ideals of $\mathfrak{sl}_2$ -type is such that $\lambda_1^a \uparrow_{\mathfrak{p}} \lambda_2^a \Rightarrow \lambda_1^a < \lambda_2^a$.*

The proof of this proposition will be given in the subsequent sections.

4.7.1. *Prime ideal $(\check{\hbar})$.* Set $R_0 := \check{R}/(\check{\hbar}) = R/(c)$. By Lemma 4.1.2, the category $\check{F}_{R_0}$ is identified with $\text{IndCoh}^{T_{\ell,d}}(\check{X}_{\check{\mathfrak{h}}^a})$. The functor $R\Gamma$ becomes the derived push-forward functor for the projection $\pi : \text{IndCoh}^{T_{\ell,d}}(\check{X}_{\check{\mathfrak{h}}^a}) \rightarrow \mathcal{L}\check{S} \times \check{\mathfrak{h}}^{a,*}$.

Let $\mathbb{F}_0$ denote the full graded localization ring and let $Z \subset \check{\mathfrak{h}}^{a,*}$ denote its spectrum. Set

$$\check{X}_Z := \check{X}_{\check{\mathfrak{h}}^a} \times_{\check{\mathfrak{h}}^{a,*}} Z.$$

Recall, Remark 4.2.5, that $\check{\Delta}_{R_0}(0) = \mathcal{O}_{\check{L}_{\check{\mathfrak{h}}^a}}$ for the classical fiber $\mathcal{O}_{\check{L}_{\check{\mathfrak{h}}^a}}$ of $1\check{I}$ under the projection $\check{X}_{\check{\mathfrak{h}}^a} \rightarrow \check{\mathcal{F}}l$. So $\check{\Delta}_{\mathbb{F}_0}(0)$ is the structure sheaf of $\check{L}_Z := \check{L}_{\check{\mathfrak{h}}^a} \times_{\check{\mathfrak{h}}^{a,*}} Z$. The restriction of $\pi : \check{X}_Z \rightarrow \mathcal{L}\check{S} \times Z$ to $\check{L}_Z$ is a closed embedding with image $\check{S}_Z^+ := \check{S}_{\check{\mathfrak{h}}^a} \times_{\check{\mathfrak{h}}^{a,*}} Z$. Let $\check{X}_Z^\wedge, (\mathcal{L}\check{S} \times Z)^\wedge$ denote the formal neighborhoods of $\check{L}_Z \cong \check{S}_Z^+$ in $\check{X}_Z, \mathcal{L}\check{S} \times Z$. It is enough to show that π induces a closed embedding $\check{X}_Z^\wedge \hookrightarrow (\mathcal{L}\check{S} \times Z)^\wedge$.

Recall that, by definition,

$$\check{X}_{\check{\mathfrak{h}}^a} := T_{\check{\mathfrak{h}}^a}^*(\check{\mathcal{F}}l) \times_{\mathcal{L}\check{\mathfrak{g}}^*} \mathcal{L}\check{S}.$$

We write $T_Z^*(\check{\mathcal{F}}l)^\wedge$ for the formal neighborhood of the fiber $T_{\check{\mathfrak{h}}^a, 1\check{I}}^*(\check{\mathcal{F}}l)$ of $1\check{I}$, so that

$$\check{X}_Z^\wedge = T_Z^*(\check{\mathcal{F}}l)^\wedge \times_{\mathcal{L}\check{\mathfrak{g}}^*} \mathcal{L}\check{S}.$$

Next, we write $(\mathcal{L}\check{\mathfrak{g}}^* \times Z)^\wedge$ for the formal neighborhood of $(\mathcal{L}\check{\mathfrak{g}}/\mathfrak{i}^0)^* \times_{\check{\mathfrak{h}}^*} Z$ in $\mathcal{L}\check{\mathfrak{g}}^* \times Z$. It is enough to show that the natural morphism

$$(4.19) \quad T_Z^*(\check{\mathcal{F}}l)^\wedge \rightarrow (\mathcal{L}\check{\mathfrak{g}}^* \times Z)^\wedge$$

is a closed embedding.

Consider the subgroup ind-scheme $\check{I}^{0,-} \subset \mathcal{L}\check{G}$ and the formal neighborhood of 1 to be denoted by $\check{I}^{0,-,\wedge}$. As was discussed in Section 4.3.1, the action morphism

$$\check{I}^{0,-,\wedge} \times T_{Z, 1\check{I}}^*(\check{\mathcal{F}}l) \xrightarrow{\sim} T_Z^*(\check{\mathcal{F}}l)^\wedge$$

is an isomorphism.

Now note that we can identify $\mathcal{L}\check{\mathfrak{g}}^* \times \check{\mathfrak{h}}^{a,*}$ with $(\mathcal{L}\check{\mathfrak{g}} \rtimes \mathbb{K}d) \times \check{\mathfrak{h}}$ (where d is the Euler element). The group $\check{I}^{-,0}$ acts on $(\mathcal{L}\check{\mathfrak{g}} \rtimes \mathbb{K}d) \times \check{\mathfrak{h}}$ via the natural action on the first factor. Consider the locus $(\check{\mathfrak{i}} \rtimes \mathbb{K}d)_Z \subset \check{\mathfrak{i}} \rtimes \mathbb{K}d$, the preimage of Z under the projection to $\check{\mathfrak{h}} \oplus \mathbb{K}d$. Then $T_{Z,1\check{I}}^* \check{\mathcal{F}}l$ is a closed subscheme of $(\check{\mathfrak{i}} \rtimes \mathbb{K}d)_Z \times \check{\mathfrak{h}}$, it's as the preimage of the diagonal under the projection to $Z \times \check{\mathfrak{h}}$, where by the diagonal we mean the intersection with the locus of elements, where the two $\check{\mathfrak{h}}$ -components are equal.

First, we are going to describe the formal neighborhood of $(\check{\mathfrak{i}} \rtimes \mathbb{K}d)_Z \times \check{\mathfrak{h}}$ in $\mathcal{L}\check{\mathfrak{g}} \times \check{\mathfrak{h}}$. By the description of Z , morally, it consists of regular semisimple elements, this formal neighborhood is identified with

$$(4.20) \quad \check{I}^{0,-,\wedge} \times ((\check{\mathfrak{i}} \rtimes \mathbb{K}d)_Z \times \check{\mathfrak{h}})$$

via the action morphism.

Now it is sufficient to describe the formal neighborhood of $T_{Z,1\check{I}}^* \check{\mathcal{F}}l$ in (4.20). This formal neighborhood is identified with

$$(4.21) \quad \check{I}^{0,-,\wedge} \times (\check{\mathfrak{i}} \rtimes \mathbb{K}d)_Z \times \check{\mathfrak{h}}_0^\wedge,$$

where $\check{\mathfrak{h}}^{\wedge_0}$ is the formal neighborhood of 0 in $\check{\mathfrak{h}}$. Under the identification of $(\mathcal{L}\check{\mathfrak{g}}^* \times Z)^\wedge$ with (4.21) and $T_Z^*(\check{\mathcal{F}}l)^\wedge$ with $\check{I}^{0,-,\wedge} \times T_{Z,1\check{I}}^*(\check{\mathcal{F}}l)$, (4.19) becomes an isomorphism onto the product of the first two factors of (4.21). This finishes the proof.

4.7.2. Prime ideals of semisimple type. Thanks to Lemma 4.6.2, the height 1 graded prime ideals that are not of the form $(\check{\alpha} - n\check{c} - m\check{h})$ with $n, m \in \mathbb{Z}$ are of semisimple type. We claim that all of them but $(\check{c})$ are also of fully faithful type. This is because:

- For $\mathbf{O}$, the prime ideals $\mathfrak{p}$ different from $(\check{h}), (c), (\check{\alpha} - n\check{c} - m\check{h})$ are of fully faithful type, see Proposition 3.3.8.
- The corresponding specialization $\mathbf{O}_{\mathbb{F}_0}$ (where $\mathbb{F}_0$ is the full graded localization ring of $\mathbf{R}/\mathfrak{p}$) is semisimple.

4.7.3. Prime ideals corresponding to real roots. To prove Proposition 4.7.2 it remains to handle the ideals $\mathfrak{p} := (\check{\alpha} - n\check{c} - m\check{h})$ for $n, m \in \mathbb{Z}$. Set $\mathbf{R}_0 = \mathbf{R}/\mathfrak{p}$.

Thanks to Proposition 3.3.8, we already know that conditions (1) and (2) from Definition 3.3.7 hold. (3) follows from Lemma 4.6.4. It remains to show that the relation $\uparrow_{\mathfrak{p}}$ is as predicted by the proposition.

First, consider the case when $m \neq 0$. By Proposition 3.3.8, the corresponding prime ideal $\mathfrak{p}$ is of fully faithful type for $\mathbf{Red}$. We have $\mathrm{grHom}_{\mathbf{O}_{\mathbb{F}_1}}(\Delta_{\mathbb{F}_1}^a(\lambda_1^a), \Delta_{\mathbb{F}_1}^a(\lambda_2^a)) \neq 0 \Rightarrow \lambda_1^a \leq \lambda_2^a$ finishing this case.

Now assume $m = 0$. The locus $\check{\alpha} = n\check{c}$ contains the point $(0, 0, 1)$, where the first two coordinates are in $\check{\mathfrak{h}}^*$ and $\check{c}$, in other words, $\mathfrak{p}_1 \subset \mathfrak{p}_0 = (\check{\mathfrak{h}}, \check{c})$. By Lemma 4.6.3, $\mathrm{grHom}_{\mathbf{F}_{\mathbb{F}_0}}(\Delta^F(\lambda_1^a), \Delta^F(\lambda_2^a)) \neq 0 \Rightarrow \lambda_1^a \leq \lambda_2^a$. We claim that this implies

$$(4.22) \quad \mathrm{grHom}_{\mathbf{F}_{\mathbb{F}_1}}(\Delta_{\mathbb{F}_1}^F(\lambda_1^a), \Delta_{\mathbb{F}_1}^F(\lambda_2^a)) \neq 0 \Rightarrow \lambda_1^a \leq \lambda_2^a.$$

Indeed, we know that

- $\mathrm{grHom}_{\mathbf{F}_{\mathbb{R}_j}}(\Delta_{\mathbb{R}_j}^F(\lambda_1^a), \Delta_{\mathbb{R}_j}^F(\lambda_2^a))$ is a finitely generated $\mathbf{R}_j$ -module for $j = 0, 1$, which follows by combining the faithfulness of the RS functor $R\Gamma$, Proposition 3.2.5, with (c3).

- (ii) $\mathbb{F}_j \otimes_{R_j} \mathrm{grHom}_{\mathbb{F}_{R_j}}(\Delta_{R_j}^{\mathbb{F}}(\lambda_1^a), \Delta_{R_j}^{\mathbb{F}}(\lambda_2^a)) \cong \mathrm{grHom}_{\mathbb{F}_{\mathbb{F}_j}}(\Delta_{\mathbb{F}_j}^{\mathbb{F}}(\lambda_1^a), \Delta_{\mathbb{F}_j}^{\mathbb{F}}(\lambda_2^a))$, see Proposition 3.1.5.
- (iii) $\mathrm{grHom}_{\mathbb{F}_{R_j}}(\Delta_{R_j}^{\mathbb{F}}(\lambda_1^a), \Delta_{R_j}^{\mathbb{F}}(\lambda_2^a))$ is torsion-free over R_j , so, thanks to (i), if this module for $j = 0$ vanishes, then the module for $j = 1$ vanishes as well.

And (4.22) shows that $\uparrow_{\mathfrak{p}}$ is as needed also for $\mathfrak{p} = (\check{\alpha} - n\check{c})$ finishing the proof of Proposition 4.7.2.

4.7.4. Proof of Theorem 4.7.1.

Proof. We apply Theorem 3.3.9 to RS functors of $\mathfrak{sl}_2$ -type $\mathrm{Red} : \mathcal{O} \rightarrow \mathbb{C}$, $R\Gamma : \check{\mathbb{F}} \rightarrow \mathbb{C}$ and the following sets of height 1 graded prime ideals:

- $\mathcal{I}_{\mathfrak{sl}_2}$ consisting of $(\check{\alpha} - n\check{h} - mc)$ for all positive finite type roots α and all $m, n \in \mathbb{Z}$,
- $\mathcal{I}_{ff}$ consisting of the ideal (c) ,
- and $\mathcal{I}_{ss}$ consisting of the remaining ideals.

The conditions of Theorem 3.3.9 are satisfied thanks to Propositions 4.7.2 and 3.3.8 which yields a unique equivalence $\mathcal{O}^{\Delta} \xrightarrow{\sim} \check{\mathbb{F}}^{\Delta}$ intertwining functors Red and $R\Gamma$. This equivalence extends to a unique right exact functor $\mathcal{O} \rightarrow \check{\mathbb{F}}$ intertwining Red and Γ , the 0-th cohomology of $R\Gamma$. $\square$

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